

Recursive AI Self-Improvement, Market Power, and Growth

Oliver Pardo

Departamento de Administración
Pontificia Universidad Javeriana
pardoo@javeriana.edu.co

September 2026

Abstract

I develop an endogenous-growth model in which a monopolistic AI developer supplies AI services and invests in recursive AI self-improvement. I characterize two long-run regimes under an upper bound on AI efficiency—the quantity of AI services produced per unit of compute. Complementarity between AI and labor, or a sufficiently low upper bound on AI efficiency, can leave labor as a production bottleneck. In this regime, output per worker and wages eventually grow at the exogenous rate of labor-augmenting technological progress. When AI and labor are substitutes and the upper bound is sufficiently high, an AI-dominated regime sustains permanently faster wage growth and a higher long-run return to capital. Output per worker grows faster than wages, so labor’s income share vanishes while AI-industry revenue increases as a share of output. Replacing monopoly with perfect competition eliminates private research incentives, but sufficiently high initial efficiency sustains AI-dominated growth through capital accumulation and expanding AI use. Monopoly rewards research but restricts AI use: it can therefore produce more efficient AI yet slower long-run growth than competition. Numerical simulations show that early output-per-worker growth can remain moderate and similar across economies approaching different long-run regimes.

[Digital edition and interactive simulations](#)

Contents

1	Introduction	3
2	Related literature	5
3	Model	6
3.1	Population and labor-augmenting technology	6
3.2	Household	7
3.3	Final-good production	7
3.4	The AI developer	9
3.5	Equilibrium	13
4	Equilibrium with an upper bound on AI efficiency	15
4.1	The return to capital and the efficiency threshold	15
4.2	Equilibrium existence and long-run regimes	16
5	Competitive Equilibrium	19
5.1	Characterization	20
5.2	Comparison with monopoly	21
5.3	Research incentives and policy	23
6	Equilibrium without an upper bound on AI efficiency	24
6.1	Complementarity	24
6.2	Unit elasticity	25
6.3	Substitution	26
7	Numerical simulations	27
7.1	Parameters and initial conditions	28
7.2	Higher research productivity	30
7.3	Lower research productivity	35
7.4	From competitive provision to monopoly with RSI	40
7.5	When monopoly eliminates AI-dominated growth	42
8	Conclusion	45
A	Proofs	48
B	Numerical algorithm and equilibrium audit	75
B.1	Economic intuition and initial conditions	75
B.2	Boundary-value solution and reported outcomes	76
B.3	Equilibrium checks	77
B.4	Analytical checks on developer optimality	78
B.5	Replication files	78
C	The uncapped economy with unit elasticity	79
C.1	An explicit balanced-growth equilibrium	79
C.2	Equilibrium trajectories near the balanced-growth path	80
C.3	Growth and distribution without an upper bound	81
D	Additional competitive-to-monopoly simulation results	82
	References	91

1 Introduction

Advances in artificial intelligence raise both hopes and fears. One hope is that increasingly capable AI could generate unprecedented economic prosperity, with workers benefiting from higher wages. One fear is that the AI industry could capture a growing share of that prosperity while workers who depend on labor income receive a declining share. In this paper, I show how both outcomes can coexist: when AI and labor are substitutes, AI can permanently raise wage growth. Yet in the long run, output per worker grows even faster than wages, while AI-industry revenue grows faster than total labor income. Labor’s income share therefore converges to zero.

To show this, I extend the Ramsey–Cass–Koopmans model to include AI services as an additional production input. Households choose saving, and competitive firms produce the final good using capital, effective labor, and AI services. A monopolistic AI developer supplies these services and invests in recursive AI self-improvement (RSI) to improve AI efficiency—the quantity of AI services produced per unit of compute. As in [Romer \(1990\)](#), expected monopoly profits provide incentives for research investment. By introducing a monopolistic AI developer, I can distinguish how the gains from AI are distributed through wages, returns to physical capital, and AI-industry profits.

The model combines two feedback loops related to those in [Davidson et al. \(2026\)](#) and [Jones and Tonetti \(2026\)](#). A technological loop operates through RSI: greater AI efficiency produces more AI research services from a given amount of compute devoted to research and thereby helps create still better AI. An economic loop runs from AI efficiency to output and back to research: greater AI efficiency raises production and expands the resources available to fund further research.

With substitution and sufficiently strong research returns, the developer’s discounted net profit can become unbounded if AI efficiency is unrestricted. To rule this out, I first study an economy with an upper bound on AI efficiency and examine how equilibria change as the bound rises.

Together, the elasticity of substitution between AI production services and labor and the maximum attainable level of AI efficiency determine whether human labor remains a long-run production bottleneck or whether AI sustains permanently faster economic growth. I focus on two long-run equilibrium regimes, distinguished by whether AI produces a temporary or permanent increase in growth. In the labor-bottleneck regime, AI improvements can raise growth temporarily, but human labor ultimately constrains growth. In the AI-dominated regime, capital and AI services sustain permanently faster growth.

I first characterize equilibria in which labor remains a long-run production bottleneck. Under complementarity, I establish their existence when the upper bound on AI efficiency exceeds an elasticity-dependent threshold. At unit elasticity, such equilibria exist for any upper bound. With substitution, they exist when the upper bound lies below the corresponding threshold. Along these equilibria, output per worker and wages eventually grow at the exogenous rate of labor-augmenting technological progress, and labor’s income share converges to a positive value.

The outcome changes when the elasticity of substitution between AI production services and labor exceeds one and the maximum attainable level of AI efficiency lies above the threshold. An

AI-dominated long-run regime then becomes possible. As in [Jones and Manuelli \(1990\)](#), capital accumulation can sustain growth when its limiting return remains sufficiently high. In this regime, capital and AI production services eventually grow faster than effective labor. This does not require AI efficiency to keep improving: capital accumulation and expanding compute use sustain growth even as AI efficiency approaches its upper bound. The long-run growth rates of output per worker and wages, as well as the interest rate, exceed their labor-bottleneck values. Because output per worker grows faster than wages, labor’s income share converges to zero. Total labor income therefore grows faster than in the labor-bottleneck regime, but in the long run it grows more slowly than capital income and AI-industry revenue.

I also characterize a competitive economy in which AI models and any improvements are freely available to all suppliers. Without a private return to research, AI efficiency remains at its initial level. When AI and labor are substitutes, a sufficiently high initial efficiency nevertheless sustains AI-dominated growth through capital accumulation and expanding compute use. Neither monopoly nor continuing RSI is therefore necessary to sustain this growth regime.

The growth comparison between monopoly and competition depends on whether monopoly-financed efficiency gains compensate for the reduction in AI use caused by monopoly pricing. The simulations show that research can allow the monopoly economy to reach AI-dominated growth from an initial efficiency too low to sustain it under competition. Conversely, competition can sustain AI-dominated growth from initial states for which monopoly-led research improves efficiency but leaves the economy labor-bottlenecked. Even when both economies approach AI-dominated growth, monopoly delivers faster long-run growth only if the research gains are large enough to offset its markup. More efficient AI therefore need not imply faster long-run growth.

I also examine the effect of removing the upper bound on AI efficiency. With complementarity, long-run average output-per-worker growth cannot exceed the exogenous rate of labor-augmenting technological progress. At unit elasticity, I derive conditions for a balanced-growth equilibrium in which output per worker grows faster than this exogenous rate. Wages grow at the same rate as output per worker, preserving a constant labor income share. With substitution and sufficiently strong research returns, the developer can make discounted net profit arbitrarily large even over a fixed finite horizon, ruling out the existence of an equilibrium.

I use numerical simulations to illustrate the effects of introducing RSI under monopoly and of replacing competition with monopoly. First, I simulate an unanticipated activation of RSI in economies that already use AI services. Depending on research productivity, output-per-worker growth can display a moderate increase or a noticeable jump in both long-run regimes. Eventually, output-per-worker growth converges to the rate of labor-augmenting technological progress in the labor-bottleneck regime and to a permanently higher rate in the AI-dominated regime. With lower research productivity, initial output-growth trajectories can remain similar across economies approaching different long-run regimes. Second, I simulate granting exclusive AI rights in economies that initially supply AI services competitively, separating the immediate reduction in AI use from subsequent research-driven efficiency gains. Monopoly-financed research can move an economy from

labor-bottleneck to AI-dominated growth. Conversely, monopoly pricing can leave an economy labor-bottlenecked even when competition would sustain AI-dominated growth with less efficient AI.

This paper has eight sections, including the present introduction. Section 2 relates the paper to the literature. Section 3 presents the model and defines equilibrium. Section 4 establishes sufficient conditions for equilibrium existence with an upper bound on AI efficiency and characterizes the associated long-run growth and distributional outcomes. Section 5 characterizes competitive AI provision and compares its growth outcomes and research incentives with those under monopoly. Section 6 examines the three substitution regimes without an upper bound on AI efficiency. Section 7 presents numerical simulations. Section 8 concludes. The appendix contains the proofs and describes the algorithm used for the simulations.

2 Related literature

This paper addresses the connected questions of growth, income distribution, and transition dynamics in the research agenda proposed by [Brynjolfsson et al. \(2025\)](#). My closest antecedents are models in which AI improves both production and research. [Davidson et al. \(2026\)](#) and [Jones and Tonetti \(2026\)](#) study feedback between technological progress, research, and output. Technological progress is endogenous in both papers, but saving and research allocation shares are taken as given. [Erdil et al. \(2025\)](#) instead optimize investment and compute allocation, distinguishing resources used to deploy AI from those used to improve it. Whereas they solve a social planner’s problem, I study a market equilibrium with a monopolistic AI developer.

The developer’s research incentives build on [Romer \(1990\)](#), where expected monopoly profits motivate investment in nonrival knowledge. [Liu and Wan \(2026\)](#) study the interaction between a general-purpose AI supplier and developers of specialized applications. In their baseline model, a higher markup on general-purpose AI services can strengthen the supplier’s research incentive while raising research costs for application developers. In my model, the markup likewise rewards research but raises the price paid by final-good firms. With a single AI developer, I do not model the additional effect on research by separate application developers.

The distinction between temporary and permanent growth gains relates to the production bottlenecks studied by [Aghion et al. \(2019\)](#) and [Jones and Tonetti \(2026\)](#): tasks that remain dependent on human labor can constrain growth despite automation elsewhere. In my model, RSI need not eliminate the labor bottleneck. Conversely, when AI and labor are substitutes and AI is already sufficiently efficient, permanently faster growth can persist without RSI, since capital and compute can continue expanding together. This connects to [Restrepo \(2025\)](#), where output can keep growing with the exogenous expansion of compute once bottleneck tasks have been automated.

[Acemoglu and Restrepo \(2018\)](#) show that automation can raise the long-run wage level while reducing labor’s income share. [Ray and Mookherjee \(2022\)](#) show that capital accumulation can sustain rising wages even as labor’s share vanishes. In the long run, low-skill wages grow more

slowly than output in [Hémous and Olsen \(2022\)](#), while high-skill wages keep pace with it. In my AI-dominated regime, wages grow permanently faster than in the labor-bottleneck regime, but more slowly than output per worker. The substitution elasticity determines how much of the additional output-per-worker growth translates into faster wage growth.

I distinguish the profits earned by the AI developer from the income earned by supplying physical capital. This distinction connects to [Korinek and Stiglitz \(2019\)](#), who separate the gains earned by innovators from the redistribution caused by changes in wages and other factor prices. [Korinek and Ng \(2019\)](#) show how digital innovation can increase monopoly profits as a share of income while reducing the shares paid to labor and physical capital. I examine how wages, capital income, and developer profits evolve as private investment in RSI improves AI efficiency and physical capital accumulates.

Changes in asset returns can also affect how the gains from automation are distributed across households. [Moll et al. \(2022\)](#) show that automation can raise returns to wealth and increase income and wealth inequality. In my model, the long-run return to physical capital is higher in the AI-dominated regime than in the labor-bottleneck regime. However, the representative household owns both physical capital and the AI developer. I therefore characterize changes in the sources of income, rather than how differences in asset ownership distribute those gains across households.

3 Model

Time is continuous. The economy contains a representative household, competitive final-good firms, and an integrated AI developer. Labor-augmenting technology A grows exogenously. AI efficiency B improves through autonomous AI research: the only current input used to raise B is research compute. The model therefore describes an economy in which there are no longer human researchers developing AI and AI has achieved recursive self-improvement (RSI). As in [Trammell and Korinek \(2023\)](#), I distinguish automation of research from automation of production: human labor remains an input in final-good production.

3.1 Population and labor-augmenting technology

Population grows at the exogenous rate $n \geq 0$:

$$\dot{N} = nN, \quad n \geq 0. \quad (1)$$

Each person supplies one unit of homogeneous labor inelastically. The aggregate labor endowment is therefore N . Because autonomous research uses no human labor, all labor is employed in final production and equilibrium will impose $L = N$.

The variable A measures the level of labor-augmenting technology. One unit of labor supplies A efficiency units, so effective labor is AL . Labor-augmenting technology evolves according to

$$\dot{A} = \gamma A, \quad \gamma > 0. \quad (2)$$

3.2 Household

The representative household owns physical capital and the AI developer. Let Π denote the AI developer's net profit, which accrues to the household. The household takes the paths of the net interest rate r , the wage w in final-good units and Π as given. For a predetermined capital stock $K_0 > 0$, it chooses aggregate consumption C to solve

$$\max_{\{C_t\}_{t \geq 0}} \int_0^\infty e^{-\rho t} N_t \log(C_t/N_t) dt, \quad \rho > n, \quad (3)$$

subject to

$$\dot{K} = rK + wN + \Pi - C, \quad K \geq 0. \quad (4)$$

The variable r is the net real interest rate: depreciation has already been deducted from the rental return to capital. The household's Euler condition is

$$\frac{\dot{C}}{C} = n + r - \rho. \quad (5)$$

Its transversality condition is

$$\lim_{t \rightarrow \infty} e^{-\rho t} \frac{N_t K_t}{C_t} = 0. \quad (6)$$

3.3 Final-good production

Competitive firms produce the final good from physical capital K , effective labor AL , and effective AI production services X . Let $0 < \alpha < 1$ denote the output elasticity with respect to capital, $\delta \geq 0$ the depreciation rate, and $\sigma > 0$ the elasticity of substitution between effective labor and effective AI production services. The parameters ω_L and ω_X denote the CES weights on effective labor and effective AI production services, respectively. Production is

$$Y = K^\alpha Z^{1-\alpha}, \quad (7)$$

$$Z = \left[\omega_L (AL)^{(\sigma-1)/\sigma} + \omega_X X^{(\sigma-1)/\sigma} \right]^{\sigma/(\sigma-1)}. \quad (8)$$

The composite input Z combines effective labor and AI production services.

To illustrate this production technology, consider a researcher writing an academic paper. The researcher contributes time and expertise to formulate questions, evaluate arguments, and interpret results. AI assists with derivations, programming, and drafting, while physical capital, such as a computer, runs numerical calculations and produces figures. AI can replace some of the work the researcher would otherwise do, but it can also make the researcher's expertise more productive by helping test conjectures or communicate arguments more clearly. The researcher may revise an argument drafted by AI or ask AI to check an argument written by the researcher. A numerical result may similarly combine a question formulated by the researcher, code developed with AI assistance, and calculations executed on the computer.

Jobs comprise multiple tasks, and AI may substitute for human input in some tasks while complementing it in others (Autor, 2013; Acemoglu and Restrepo, 2019). An early model of mechanization is Zeira (1998), in which producers can use more capital to reduce the labor required to produce individual intermediate goods. I model labor and AI services as aggregate inputs, not as alternative ways of performing a single task. Jones and Tonetti (2026) show how a production process with many tasks can admit a CES-like aggregate representation. Erdil et al. (2025) distinguish automating additional tasks (the extensive margin) from increasing the services supplied in tasks that AI can already perform (the intensive margin). I focus on expansion of aggregate AI services without explicitly tracking the set of automatable tasks; the fixed CES weights should therefore not be read as fractions of tasks automated.

The complementarity case connects to the “weak links” mechanism in Aghion et al. (2019, Section 9.2.2). In their task model, automation allows capital to replace labor within individual tasks, while different tasks remain complements. Tasks still performed by humans can therefore constrain production as automated tasks expand. With $\sigma < 1$, Equation (8) captures a related constraint: more AI services cannot eliminate the need for effective labor. I apply this logic to aggregate inputs with fixed weights, rather than modeling an expanding set of automated tasks; the two substitution elasticities are not interchangeable.

Let p_X denote the price of one unit of effective AI production services. Taking the input prices $r + \delta, w$, and p_X as given, a final-good firm solves

$$\max_{K, L, X \geq 0} \left\{ K^\alpha Z(AL, X)^{1-\alpha} - (r + \delta)K - wL - p_X X \right\}. \quad (9)$$

Define s_X as the elasticity of Z with respect to X :

$$s_X = \frac{\omega_X X^{(\sigma-1)/\sigma}}{\omega_L (AL)^{(\sigma-1)/\sigma} + \omega_X X^{(\sigma-1)/\sigma}}. \quad (10)$$

The first-order conditions are

$$r = \alpha \frac{Y}{K} - \delta, \quad (11)$$

$$w = (1 - \alpha)(1 - s_X) \frac{Y}{L}, \quad (12)$$

$$p_X = (1 - \alpha)s_X \frac{Y}{X}. \quad (13)$$

Thus $(1 - \alpha)s_X$ is AI revenue as a share of output, while $(1 - \alpha)(1 - s_X)$ is labor’s income share.

Constant returns and these conditions imply

$$Y = (r + \delta)K + wL + p_X X. \quad (14)$$

Solving the final-good firm’s AI condition for its price yields the inverse demand schedule $p_X(X)$.

The absolute elasticity of inverse demand, holding K , A , and L fixed, is

$$e_X \equiv -\frac{\partial \ln p_X(X)}{\partial \ln X} = \frac{1 - s_X}{\sigma} + \alpha s_X. \quad (15)$$

This demand schedule is an input to the developer’s problem.

3.4 The AI developer

3.4.1 Effective AI production and research services

I use *compute* to denote the computational resources used to execute the numerical operations required by AI systems (Sastry et al., 2024). Compute can be measured in floating-point operations; a flow of compute corresponds to operations per unit of time. I assume that one unit of compute requires one unit of the final good, which could otherwise be consumed or invested in physical capital. Compute is therefore a rival, costly input, not a measure of AI performance.

I distinguish compute used to deploy AI from compute used to improve it. Inference compute U runs existing AI systems and produces outputs used in current final-good production. Research compute M is allocated to the development of better AI. It includes training and fine-tuning, but also computational experimentation, model evaluation, and other compute-intensive components of AI research. I therefore use the term *research compute* rather than the narrower term *training compute*. Korinek and McKelvey (2026) make a related distinction between compute used for inference and compute used for R&D and training.

The distinction between U and M concerns the economic use of compute, rather than a strict separation between inference and training: running AI systems that conduct research also requires inference compute (Davidson et al., 2026).

Raw compute does not by itself determine the quantity of AI services produced. The variable B measures AI efficiency: AI services obtained per unit of compute. Thus U and M are raw computational inputs, while BU and BM are the corresponding flows of AI production and AI research services. This interpretation parallels measures of algorithmic efficiency based on the decline in compute required to achieve a given level of performance (Hernandez and Brown, 2020; Erdil and Besiroglu, 2022). A rise in B therefore represents an improvement in the algorithms that use compute, rather than an expansion of physical computing capacity. I assume that the same B augments compute in both activities.

This distinction also underlies *effective compute*, which adjusts physical compute for algorithmic improvements (Ho et al., 2024; Jones, 2026). In BM , M is rival, whereas the algorithmic knowledge represented by B is nonrival: using it in one computation does not diminish its usefulness in another. I apply this distinction to AI research more broadly than the model-training activities measured in those sources. Erdil et al. (2025) similarly distinguish the nonrival model weights created by training from the recurring compute costs of inference.

Inference compute produces AI production services according to

$$X = BU. \quad (16)$$

Equation (16) maps inference compute into the flow $X = BU$ of AI production services. Research compute analogously produces BM , the flow of AI research services.

Figure 1 summarizes the production block.

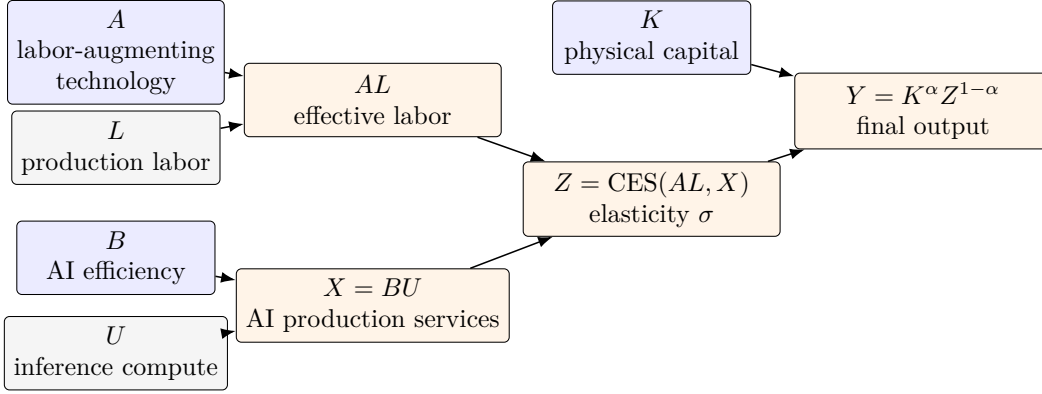


Figure 1: Production of final output and AI production services. AI efficiency B augments inference compute U to produce $X = BU$, the flow of AI production services. These services combine with effective labor AL to produce Z , which combines with capital K to produce Y .

3.4.2 RSI technology

Given initial AI efficiency $B_0 > 0$ and an upper bound $\bar{B} \in (B_0, \infty]$, AI efficiency evolves according to the RSI technology

$$\dot{B} = \chi(BM)^\eta \psi(B), \quad \psi(B) = 1 - \frac{B}{\bar{B}}, \quad \chi > 0, \quad 0 < \eta < 1. \quad (17)$$

Figure 2 summarizes the RSI mechanism.

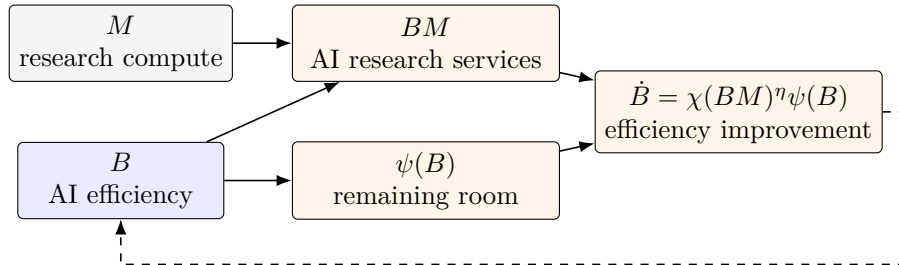


Figure 2: Recursive self-improvement in AI efficiency. AI efficiency B augments research compute M to produce BM , the flow of AI research services. These services and the remaining room for improvement $\psi(B)$ determine $\dot{B}$, which increases B .

The parameter χ shifts research productivity, while η is the elasticity of $\dot{B}$ with respect to

research compute, holding B fixed. The same AI-efficiency index B augments both inference and research compute. This common-efficiency assumption is parsimonious but substantive. Equation (17) captures RSI: a higher B produces more AI research services BM from a given amount of research compute M , and these services help improve B further. This is the AI-research feedback studied by Davidson et al. (2026). The term $\psi(B)$ reduces the remaining room for improvement as AI efficiency approaches $\bar{B}$. A related approach appears in Erdil et al. (2025), who also reduce research productivity as hardware and software efficiency approach their respective upper bounds. The uncapped specification is recovered at $\bar{B} = \infty$.

Evidence that ideas become harder to find motivates allowing research productivity to decline as technology advances. Bloom et al. (2020) document that research effort has risen substantially while research productivity has fallen across several industries, products, and firms. This evidence motivates diminishing returns, but it does not establish a literal finite technological ceiling. The term $1 - B/\bar{B}$ is a deliberately stronger reduced-form specification. With this upper bound, I establish sufficient conditions for equilibrium existence and compute equilibrium trajectories. I then examine how these equilibria change as the bound rises and whether they have a meaningful limit as $\bar{B} \rightarrow \infty$. Without the upper bound, recursive improvement may lead to the absence of an equilibrium, as shown in Section 6.3.

3.4.3 Supply of effective AI production services

An integrated AI developer has exclusive ownership of B . It conducts research and sells effective AI production services. The developer sells X and purchases the $U = X/B$ units of inference compute needed to supply those services. Because X does not enter the RSI technology in equation (17) directly, its within-date choice can be solved conditional on B . Define optimized operating profit before research expenditure as

$$\pi(B) = \max_{X \geq 0} \left\{ p_X(X)X - \frac{X}{B} \right\}. \quad (18)$$

The notation $\pi(B)$ leaves its dependence on current K , A , and L implicit. The first-order condition equates marginal revenue to the compute cost of one additional service unit:

$$p_X(1 - e_X) = \frac{1}{B}. \quad (19)$$

Lemma 1 in Appendix A shows that problem (18) has a unique, strictly positive global maximizer characterized by condition (19). Thus, the developer charges the unit cost $1/B$ multiplied by the monopoly markup $1/(1 - e_X)$. The envelope theorem gives

$$\pi_B = \frac{X}{B^2} = \frac{U}{B}. \quad (20)$$

At a fixed X , a marginal increase in AI efficiency saves U/B units of the consumption good.

AI efficiency is nonrival in use. Once B has been developed, using it to augment one unit of

inference or research compute does not reduce its ability to augment other units. This does not make effective AI services costless: BU and BM require the rival and costly compute inputs U and M . Nonrivalry also does not rule out excludability. An owner can restrict access to model weights, software, or AI services.

To see the resulting appropriability problem, suppose that competitive service providers had free access to B . Free entry would imply no operating rent. A developer that paid for M would improve a common AI-efficiency index B that competing providers could use without compensating it. The developer would bear the research cost but could not appropriate the resulting benefits. Exclusive ownership allows the developer to capture part of the value created by a higher B and makes the return to research appropriable, as in the standard endogenous-growth treatment of nonrival ideas (Romer, 1990).

That being said, monopoly is an institutional assumption, not a technological implication of nonrivalry. Licensing, oligopoly, public funding, or open-source provision with research subsidies could also support investment in B . Each alternative would require an additional institution and would change prices, profits, and the distribution of rents. Recent work studies these alternatives in more detail (Korinek and Vipra, 2025; Gans, 2024; Demirer et al., 2025). The single-developer assumption excludes the conflict between general-purpose AI suppliers and application developers studied by Liu and Wan (2026).

3.4.4 The dynamic problem

The AI developer uses M for research and owns B and any resulting improvements in AI efficiency. After the within-date service choice, the developer's net profit is $\Pi_t = \pi(B_t) - M_t$: operating profit less research expenditure. Taking the paths of r , K , A , and L as given and substituting the static solution to problem (18), the developer chooses the path of research compute M to increase AI efficiency B :

$$\max_{\{M_t\}_{t \geq 0}} \int_0^\infty \exp \left\{ - \int_0^t r_s ds \right\} [\pi(B_t) - M_t] dt, \quad (21)$$

subject to the RSI technology given by (17) and taking B_0 as given.

Let q be the current-value shadow price of AI efficiency, measured in units of the current final good per unit of B . An interior choice of research compute M satisfies

$$q\chi\eta B(BM)^{\eta-1}\psi(B) = 1, \quad (22)$$

Equation (22) equates the marginal benefit of research compute to its unit cost. Holding current B fixed, an additional unit of M raises the rate of improvement in AI efficiency by $\chi\eta B(BM)^{\eta-1}\psi(B)$. Multiplication by q values the resulting efficiency gain, accounting for future profit gains and further improvements in AI efficiency.

The current-value costate equation is

$$\frac{\dot{q}}{q} = r - \frac{X}{qB^2} - \eta \frac{\dot{B}}{B} - \dot{B} \frac{\psi'(B)}{\psi(B)}. \quad (23)$$

Rearranging gives the return equation

$$rq = \frac{U}{B} + q\eta \frac{\dot{B}}{B} + q\dot{B} \frac{\psi'(B)}{\psi(B)} + \dot{q}. \quad (24)$$

The first term on the right-hand side of Equation (24) is the inference-compute saving from greater AI efficiency. The second is the research-productivity gain from RSI. With an upper bound, the third term is negative because a higher B leaves less room for further improvement. The final term is the capital gain on AI efficiency. Together these returns must equal the market return rq .

The developer's transversality condition is

$$\lim_{t \rightarrow \infty} \exp \left\{ - \int_0^t r_s ds \right\} q_t B_t = 0. \quad (25)$$

It rules out a path that leaves positive discounted value in the AI-efficiency state.

The preceding conditions are necessary for an interior optimum but do not establish global optimality. Lemma 2 in Appendix A gives sufficient conditions for a trajectory of research expenditure to maximize the present value of the developer's net profit in problem (21).

3.5 Equilibrium

I can now combine the decisions of the three agents and the market clearing conditions to define an equilibrium. Capital-market clearing identifies household capital with the capital rented by final-good firms. Labor-market clearing gives $L = N$. AI-service market clearing identifies the developer's sales with the final sector's purchases. Combining the household budget, final-good zero profit, developer profit, and market clearing yields the aggregate resource constraint

$$C + \dot{K} + \delta K + U + M = Y. \quad (26)$$

Definition 1 (Equilibrium). *For exogenous initial values $A_0, N_0, K_0, B_0 > 0$, an equilibrium is an infinite-horizon trajectory*

$$\{K_t, B_t, C_t, q_t, L_t, U_t, M_t, X_t, Z_t, Y_t, r_t, w_t, p_{X,t}, s_{X,t}, e_{X,t}, \Pi_t\}_{t \geq 0} \quad (27)$$

such that the developer solves problem (21) and the following conditions hold:

1. The intratemporal allocation and prices satisfy

$$L = N, \quad (28a)$$

$$Z = \left[\omega_L (AL)^{(\sigma-1)/\sigma} + \omega_X X^{(\sigma-1)/\sigma} \right]^{\sigma/(\sigma-1)}, \quad (28b)$$

$$Y = K^\alpha Z^{1-\alpha}, \quad (28c)$$

$$s_X = \frac{\omega_X X^{(\sigma-1)/\sigma}}{\omega_L (AL)^{(\sigma-1)/\sigma} + \omega_X X^{(\sigma-1)/\sigma}}, \quad (28d)$$

$$e_X = \frac{1 - s_X}{\sigma} + \alpha s_X, \quad (28e)$$

$$r = \alpha \frac{Y}{K} - \delta, \quad (28f)$$

$$w = (1 - \alpha)(1 - s_X) \frac{Y}{L}, \quad (28g)$$

$$p_X = (1 - \alpha) s_X \frac{Y}{X}, \quad (28h)$$

$$\frac{1}{B} = p_X (1 - e_X), \quad (28i)$$

$$X = BU, \quad (28j)$$

$$1 = q_X \eta B (BM)^{\eta-1} \psi(B), \quad (28k)$$

$$\Pi = p_X X - U - M. \quad (28l)$$

2. The four dynamic equations are

$$\dot{K} = Y - C - U - M - \delta K, \quad (29a)$$

$$\dot{B} = \chi (BM)^\eta \psi(B), \quad (29b)$$

$$\frac{\dot{C}}{C} = n + r - \rho, \quad (29c)$$

$$\frac{\dot{q}}{q} = r - \frac{X}{qB^2} - \eta \frac{\dot{B}}{B} - \dot{B} \frac{\psi'(B)}{\psi(B)}. \quad (29d)$$

3. The two transversality conditions are

$$\lim_{t \rightarrow \infty} e^{-\rho t} \frac{N_t K_t}{C_t} = 0, \quad (30a)$$

$$\lim_{t \rightarrow \infty} \exp \left\{ - \int_0^t r_s ds \right\} q_t B_t = 0. \quad (30b)$$

Remark 1 (The economy without AI). With $\omega_X = 0$ and $\omega_L = 1$, the model reduces to the Ramsey–Cass–Koopmans economy with logarithmic utility and exogenous labor-augmenting technological progress.

The system contains no separate household budget or final-good zero-profit equation because these are redundant after market clearing. The consolidated income decomposition follows from

the equations in the definition:

$$1 = \frac{(r + \delta)K}{Y} + \frac{wL}{Y} + \frac{\Pi}{Y} + \frac{U}{Y} + \frac{M}{Y}. \quad (31)$$

It records how final output is divided among gross capital income, labor income, developer profit, inference compute, and research compute.

4 Equilibrium with an upper bound on AI efficiency

In this section, I characterize the long-run equilibrium behavior for various parameter configurations. The key finding is that long-run output-per-worker growth can exceed the rate of labor-augmenting technological change when the elasticity of substitution between AI production services and labor exceeds one and the upper bound on AI efficiency is sufficiently high.

To identify this threshold, I calculate the net return on capital implied by the production and pricing conditions as AI efficiency approaches its upper bound and labor's income share tends to zero. I compare this return with $\rho + \gamma$, the interest rate that sustains consumption-per-person growth at the exogenous rate γ . When the former exceeds the latter and AI and labor are substitutes, the model supports equilibrium paths along which capital and AI production services grow faster than effective labor, even though AI efficiency converges to a finite limit.

4.1 The return to capital and the efficiency threshold

An upper bound restricts AI efficiency, but not the compute used to supply AI production services. To identify the return when labor's income share vanishes, fix $B > 0$ and consider $s_X \rightarrow 1$. Since $wL/Y = (1 - \alpha)(1 - s_X)$, this is the zero-labor-share limit.

Combine the final-good firm's demand for AI services in equation (13) with the developer's pricing condition in equation (19). The result is:

$$\frac{X}{Y} = B(1 - \alpha)s_X(1 - e_X) \longrightarrow B(1 - \alpha)^2. \quad (32)$$

The limit uses $e_X \rightarrow \alpha$ from Equation (15). The CES aggregator in equation (8) and the definition of s_X in equation (10) imply

$$\frac{Z}{X} = \left(\frac{\omega_X}{s_X} \right)^{\sigma/(\sigma-1)} \longrightarrow \omega_X^{\sigma/(\sigma-1)}. \quad (33)$$

Using $Z/Y = (Z/X)(X/Y)$ and the limits in equations (32) and (33), I obtain

$$\frac{Y}{K} = \left(\frac{Z}{Y} \right)^{(1-\alpha)/\alpha} \longrightarrow \left[(1 - \alpha)^2 B \omega_X^{\sigma/(\sigma-1)} \right]^{(1-\alpha)/\alpha}. \quad (34)$$

Let $\mathcal{R}(B)$ denote the net return on capital in this zero-labor-share limit at a given B . Substituting

the limiting output–capital ratio in equation (34) into $r = \alpha Y/K - \delta$ gives

$$\mathcal{R}(B) \equiv \alpha \left[(1 - \alpha)^2 B \omega_X^{\sigma/(\sigma-1)} \right]^{(1-\alpha)/\alpha} - \delta. \quad (35)$$

This is a return at a particular limit of the production and pricing conditions, not necessarily an equilibrium interest rate. For $\sigma > 1$, $s_X \rightarrow 1$ means that AI production services become arbitrarily large relative to effective labor. For $\sigma < 1$, $s_X \rightarrow 1$ means that effective labor services become arbitrarily large relative to AI production services. Whether the equilibrium interest rate converges to $\mathcal{R}(\bar{B})$ depends on both σ and $\bar{B}$, as shown below.

By the Euler equation (5), consumption per person grows at $r - \rho$. If this growth rate converges to γ , the interest rate must converge to $\rho + \gamma$. Define $\mathcal{B}(\sigma)$ as the AI-efficiency level at which the zero-labor-share return equals this benchmark:

$$\begin{aligned} \mathcal{R}(\mathcal{B}(\sigma)) &= \rho + \gamma, \\ \mathcal{B}(\sigma) &= \frac{[(\rho + \gamma + \delta)/\alpha]^{\alpha/(1-\alpha)}}{(1 - \alpha)^2 \omega_X^{\sigma/(\sigma-1)}}. \end{aligned} \quad (36)$$

The inequalities between $\bar{B}$ and $\mathcal{B}(\sigma)$ –whether AI can attain sufficient efficiency– and between σ and 1 –whether labor and AI are substitutes or complements– distinguish the long-run equilibrium regimes characterized below. As [Duffy and Papageorgiou \(2000, Section 2.2\)](#) explain for CES models of capital and labor, an elasticity above one permits sustained growth only if the limiting return on capital is sufficiently high.

4.2 Equilibrium existence and long-run regimes

Define $g_{Y/N}$ as the growth rate of output per worker and g_w as the wage growth rate. The next proposition characterizes the long-run behavior of equilibrium trajectories.

Proposition 1 (Equilibrium existence and long-run characterization). *Fix $0 < \bar{B} < \infty$. For each parameter configuration covered by a case below, there exists a nonempty open set of initial conditions (A_0, N_0, K_0, B_0) such that every initial state in that set has an equilibrium trajectory with the following properties:*

(i) **Complementarity.** *If $0 < \sigma < 1$ and $\bar{B} > \mathcal{B}(\sigma)$, then*

$$g_{Y/N} \rightarrow \gamma, \quad g_w \rightarrow \gamma, \quad r \rightarrow \rho + \gamma, \quad \lim_{t \rightarrow \infty} s_X \in (0, 1). \quad (37)$$

(ii) **Unit elasticity.** *If $\sigma = 1$, then*

$$g_{Y/N} \rightarrow \gamma, \quad g_w \rightarrow \gamma, \quad r \rightarrow \rho + \gamma, \quad s_X = \omega_X \quad (38)$$

(iii) **Substitution with a labor bottleneck.** If $\sigma > 1$ and $\bar{B} < \mathcal{B}(\sigma)$, then

$$g_{Y/N} \rightarrow \gamma, \quad g_w \rightarrow \gamma, \quad r \rightarrow \rho + \gamma, \quad \lim_{t \rightarrow \infty} s_X \in (0, 1). \quad (39)$$

(iv) **AI-dominated growth.** If $\sigma > 1$ and $\bar{B} > \mathcal{B}(\sigma)$, then

$$g_{Y/N} \rightarrow \mathcal{R}(\bar{B}) - \rho > \gamma, \quad (40)$$

$$g_w \rightarrow \gamma + \frac{\mathcal{R}(\bar{B}) - \rho - \gamma}{\sigma} > \gamma, \quad (41)$$

$$r \rightarrow \mathcal{R}(\bar{B}), \quad s_X \rightarrow 1.$$

Proof. See Appendix A, [Proof of Proposition 1](#). □

I treat $0 < \sigma < 1$ with $\bar{B} < \mathcal{B}(\sigma)$ separately because the result established for this region is conditional, not an existence result. Proposition 8 in Appendix A shows that any equilibrium in this region has an AI production bottleneck: long-run average output-per-worker growth is below γ , and labor's income share converges to zero.

The labor-bottleneck cases of Proposition 1 show that autonomous AI research need not eliminate the production bottleneck, a constraint also emphasized by [Aghion et al. \(2019\)](#). AI efficiency improves in each of these cases, including complementarity. Substitution is needed for a permanent increase in growth under an upper bound, not for research to improve AI efficiency. Moreover, $\sigma > 1$ is not sufficient to escape the labor bottleneck: if $\mathcal{R}(\bar{B}) < \rho + \gamma$, the return at a vanishing labor share is too low to sustain growth faster than effective labor.

The AI-dominated regime has an asymptotically linear production structure, as in the capital-accumulation mechanisms of [Rebelo \(1991\)](#) and [Jones and Manuelli \(1990\)](#). Equation (34) gives a positive limiting output–capital ratio even as B approaches the upper bound. Capital and inference compute can expand together, sustaining output-per-worker growth above γ even when B is arbitrarily close to $\bar{B}$ and further AI-efficiency gains vanish, as in the compute-expansion mechanism of [Restrepo \(2025\)](#). [Erdil et al. \(2025\)](#) also discuss how expanding digital-worker services alongside physical capital can generate AK -type dynamics. Here, the corresponding growth regime arises under fixed aggregate production weights and private research and pricing decisions. RSI can help the economy reach an efficiency at which this mechanism supports growth faster than γ ; continuing efficiency improvements are not necessary to sustain it. Section 5 shows that sufficiently high initial efficiency also allows this growth mechanism to operate under competition without RSI.

This joint expansion also explains the higher long-run interest rate. In the labor-bottleneck regime, the interest rate converges to $\rho + \gamma$. In the AI-dominated regime, however, AI services expand alongside capital, so production need not be constrained by the growth of effective labor. The output–capital ratio converges to a level at which $r = \alpha Y/K - \delta$ remains above $\rho + \gamma$: accumulating more capital does not restore the labor-bottleneck return. [Caselli and Manning \(2019\)](#) also discuss the possibility that technological progress raises growth and the interest rate together, but

their comparison of steady states does not determine the resulting wage dynamics.

The proof of Proposition 1 first removes the long-run trends from growing quantities and shrinking efficiency gaps. For each parameter configuration covered by the proposition, I identify candidate stationary values of these normalized variables. I then construct equilibrium trajectories from every sufficiently nearby initial state and verify convergence to those values. Local uniqueness results are given in the proof. These analytical results guide the algorithm’s boundary conditions (Appendix B).

Remark 2 (Wages and AI revenue in the AI-dominated regime). In the AI-dominated equilibria of Proposition 1(iv), output per worker grows asymptotically faster than wages, even though both grow faster than labor-augmenting technological progress:

$$\lim_{t \rightarrow \infty} g_{Y/N} > \lim_{t \rightarrow \infty} g_w > \gamma.$$

Labor’s income share therefore converges to zero, while the AI industry’s revenue as a share of output converges to $1 - \alpha$.

Rising wages alongside a declining labor share also appear in Acemoglu and Restrepo (2018), Autor and Kausik (2026), and Ray and Mookherjee (2022). In my AI-dominated regime, wages grow permanently faster than in the labor-bottleneck regime, even as AI efficiency approaches its upper bound. Equations (40) and (41) quantify this increase: relative to the labor-bottleneck regime, the increase in wage growth equals the increase in output-per-worker growth divided by the elasticity of substitution.

Figure 3 shows where the long-run regimes characterized in Proposition 1 lie relative to $\mathcal{B}(\sigma)$. For $0 < \sigma < 1$, the equilibria above the threshold have output-per-worker growth converging to γ and a positive limiting labor income share. For $\sigma > 1$, these same long-run outcomes occur below the threshold. Above it, the regime is AI-dominated: output per worker grows asymptotically faster than γ and labor’s income share converges to zero.

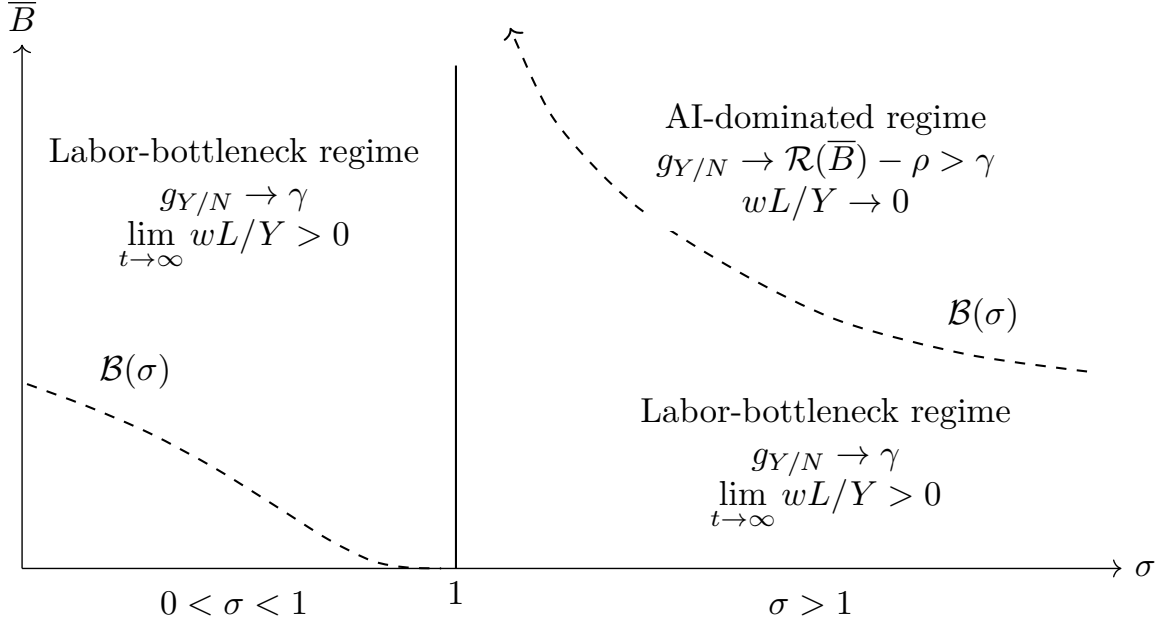


Figure 3: Growth regimes with an upper bound on AI efficiency.

Table 1 summarizes the four cases of Proposition 1. For $\sigma > 1$, an upper bound below $\mathcal{B}(\sigma)$ makes the return on capital at a vanishing labor share too low for capital and AI services to outgrow effective labor. Above the threshold, that return supports their faster growth along the AI-dominated equilibria. Across these equilibrium regimes, long-run wage growth exceeds γ if and only if labor's income share converges to zero.

Table 1: Long-run equilibrium characterization

Elasticity	AI efficiency bound	$\lim g_{Y/N}$	$\lim g_w$	$\lim wL/Y$	$\lim r$
$0 < \sigma < 1$	$\overline{B} > \mathcal{B}(\sigma)$	γ	γ	Positive	$\rho + \gamma$
$\sigma = 1$	Any finite $\overline{B}$	γ	γ	$(1 - \alpha)\omega_L$	$\rho + \gamma$
$\sigma > 1$	$\overline{B} < \mathcal{B}(\sigma)$	γ	γ	Positive	$\rho + \gamma$
$\sigma > 1$	$\overline{B} > \mathcal{B}(\sigma)$	$\mathcal{R}(\overline{B}) - \rho$	$\gamma + \frac{\mathcal{R}(\overline{B}) - \rho - \gamma}{\sigma}$	Zero	$\mathcal{R}(\overline{B})$

5 Competitive Equilibrium

I now replace the monopolistic developer with competitive AI suppliers. This comparison separates the incentive to improve AI from the use of technology already developed. I focus on $\sigma > 1$, where both labor-bottleneck and AI-dominated growth are possible. I first characterize the competitive equilibrium and then compare it with the monopoly equilibria in Section 4.

5.1 Characterization

Consider now the case where multiple AI suppliers have free access to the same efficiency B . Entry is free, and any improvement becomes immediately available to every supplier, without royalties or other compensation to its developer. There are no research subsidies or publicly financed research. Suppliers and households take prices and the common technology as given. Preferences, final-good production, and the RSI technology remain unchanged.

Competition sets $p_X = 1/B$. Since $X = BU$, revenue then exactly covers inference compute: $p_X X = U$. A supplier that invests in research bears its cost but earns neither operating profit nor royalties on the improvement. It therefore chooses $M = 0$, leaving $B = B_0$. Final-good firms continue to solve problem (9), with production given by equations (7)–(8). Their demand for AI services in equation (13) then implies $U = (1 - \alpha)s_X Y$, since $X = B_0 U$. The household continues to choose consumption and saving according to the Euler equation (5) and the transversality condition (6). Capital accumulation follows the resource constraint (26) with $M = 0$. Competitive suppliers' optimality replaces the monopolist's pricing, research, and shadow-price conditions.

As labor's income share vanishes, the net return on capital under marginal-cost pricing approaches

$$\mathcal{R}^{\text{mc}}(B) = \alpha \left[(1 - \alpha) B \omega_X^{\sigma/(\sigma-1)} \right]^{(1-\alpha)/\alpha} - \delta. \quad (42)$$

Let $\mathcal{B}^{\text{mc}}(\sigma)$ be the unique efficiency at which $\mathcal{R}^{\text{mc}}(\mathcal{B}^{\text{mc}}(\sigma)) = \rho + \gamma$. Unlike the monopoly economy, the competitive economy cannot improve from B_0 toward $\bar{B}$. Its long-run regime depends on the efficiency inherited at the initial date.

Proposition 2 (Competitive equilibrium without private research). *Under the free-access arrangement described above, suppose $\sigma > 1$. For every $A_0, N_0, K_0 > 0$ and $0 < B_0 < \bar{B}$, there is a unique competitive equilibrium. At every date,*

$$M = 0, \quad B = B_0, \quad p_X = 1/B_0, \quad \Pi = 0. \quad (43)$$

Its long-run outcomes on either side of the threshold are:

(i) *If $B_0 < \mathcal{B}^{\text{mc}}(\sigma)$, labor remains a bottleneck:*

$$g_{Y/N} \rightarrow \gamma, \quad g_w \rightarrow \gamma, \quad r \rightarrow \rho + \gamma, \quad \lim_{t \rightarrow \infty} wL/Y > 0.$$

(ii) *If $B_0 > \mathcal{B}^{\text{mc}}(\sigma)$, growth is AI-dominated:*

$$\begin{aligned} g_{Y/N} &\rightarrow \mathcal{R}^{\text{mc}}(B_0) - \rho > \gamma, & g_w &\rightarrow \gamma + \frac{\mathcal{R}^{\text{mc}}(B_0) - \rho - \gamma}{\sigma} > \gamma, \\ r &\rightarrow \mathcal{R}^{\text{mc}}(B_0), & wL/Y &\rightarrow 0. \end{aligned} \quad (44)$$

Proof. See Appendix A, [Proof of Proposition 2](#). □

Neither monopoly nor RSI is necessary to sustain AI-dominated growth. At fixed efficiency, $X = B_0 U$ can continue expanding through greater compute use. When B_0 exceeds the competitive threshold $\mathcal{B}^{\text{mc}}(\sigma)$, capital and AI services grow together faster than effective labor. This is the capital-accumulation mechanism identified in Section 4.2, now sustained without private research. Below the threshold, diminishing returns bring the interest rate back to $\rho + \gamma$. Since $B = B_0$, the competitive economy cannot escape this bottleneck.

5.2 Comparison with monopoly

I now compare the efficiency required under the two pricing rules, holding the technology fixed. This isolates the effect of pricing from the difference in research incentives.

Proposition 3 (Monopoly pricing and the AI-efficiency threshold). *Suppose AI and labor are substitutes ($\sigma > 1$). Let $\mathcal{B}(\sigma)$ and $\mathcal{B}^{\text{mc}}(\sigma)$ denote the AI-efficiency thresholds under monopoly pricing and marginal-cost pricing, respectively. These thresholds identify the efficiency needed for the return on capital to reach $\rho + \gamma$ in the limit where labor's income share vanishes. Monopoly pricing requires a higher AI efficiency:*

$$\mathcal{B}^{\text{mc}}(\sigma) = (1 - \alpha)\mathcal{B}(\sigma) < \mathcal{B}(\sigma). \quad (45)$$

Proof. See Appendix A, Proof of Proposition 3. □

To understand why monopoly pricing raises the threshold, compare the same AI technology with the same capital and effective labor under the two pricing rules. At marginal cost, AI services are cheaper, so final-good firms use more of them and produce more with the same capital. This raises the output-capital ratio and therefore the return on capital. For the threshold comparison, evaluate both returns as labor's income share tends to zero. In this limit, monopoly pricing requires a higher AI efficiency for the return on capital to reach the benchmark $\rho + \gamma$. With $\alpha = 0.33$, marginal-cost pricing reaches this benchmark at only 67.0% of the AI efficiency required under monopoly pricing.

Corollary 1 (Monopoly and the labor bottleneck). *Suppose $\sigma > 1$ and*

$$\mathcal{B}^{\text{mc}}(\sigma) < \bar{B} < \mathcal{B}(\sigma). \quad (46)$$

There is a nonempty open set of common initial states with $\mathcal{B}^{\text{mc}}(\sigma) < B_0 < \bar{B}$ from which the competitive equilibrium converges to AI-dominated growth, while a monopoly equilibrium converges to the labor-bottleneck regime. AI efficiency remains at B_0 under competition but approaches the higher level $\bar{B}$ under monopoly.

Proof. See Appendix A, Proof of Corollary 1. □

In this case, the efficiency gains generated by monopoly-led research do not compensate for the reduction in AI use caused by monopoly pricing. Competition sustains permanently faster output-per-worker and wage growth despite retaining a less efficient AI technology. AI services per unit of effective labor converge to a positive constant under monopoly but grow without bound under competition. The monopoly economy is not stagnant: in the long run, output per worker and wages grow at γ .

Corollary 2 (Growth under competition and monopoly). *Suppose $\sigma > 1$. Consider competitive and monopoly equilibria with the same parameters and initial stocks, both converging to AI-dominated growth, with AI efficiency approaching $\bar{B}$ under monopoly. Monopoly delivers a higher long-run return to capital and faster long-run growth in output per worker and wages if and only if*

$$B_0 < (1 - \alpha)\bar{B}. \quad (47)$$

Competition delivers higher limiting returns and growth rates when the inequality is reversed. At equality, these limiting outcomes coincide.

Proof. See Appendix A, [Proof of Corollary 2](#). □

The efficiency gains generated by monopoly-led research must be large enough to offset the reduction in AI use caused by its markup. With $\alpha = 0.33$, monopoly requires an eventual efficiency gain of more than 49.3% relative to B_0 to deliver faster long-run growth than competition, provided both equilibria are AI-dominated. This follows from rewriting equation (47) as $\bar{B}/B_0 > 1/(1 - \alpha)$.

There is therefore no universal growth ranking between competition and monopoly. Conditional on the existence of the compared equilibria, monopoly-led research can allow an escape from the labor bottleneck when $B_0 < \mathcal{B}^{\text{mc}}(\sigma)$ but $\bar{B} > \mathcal{B}(\sigma)$. Corollary 1 establishes the opposite regime ranking, while Corollary 2 shows what determines the ranking when both economies are AI-dominated. The relevant comparison is not efficiency alone, but whether monopoly's efficiency gains compensate for the reduction in AI use caused by its markup.

5.3 Research incentives and policy

Holding K , AL , and B fixed, marginal-cost pricing maximizes $Y - X/B$ and leaves more resources available for consumption or capital accumulation. In the free-access economy, however, it also removes the private return to research. The comparison does not rank welfare: that requires comparing consumption throughout the transition, including the resource cost and future benefits of research under monopoly.

This raises a classic time-inconsistency problem ([Kyland and Prescott, 1977](#)). A government may promise monopoly protection to encourage research, but later wish to remove the markup once the technology has been developed. Anticipating this policy reversal can weaken the developer's incentive to invest in research. In the present setting, continued RSI makes the trade-off relevant even after a valuable AI technology already exists: opening access to that technology changes the rewards for improving it further.

An announced rule that ends monopoly protection at a specified efficiency level is different from reneging on a promise. It nevertheless changes the developer's problem. Without compensation, reaching that level would remove future monopoly rents, potentially giving the developer an incentive to delay reaching it. The threshold $\mathcal{B}^{\text{mc}}(\sigma)$ identifies the AI efficiency at which the limiting return on capital under marginal-cost pricing reaches $\rho + \gamma$. It does not identify the optimal time

to end monopoly protection. Choosing that time requires accounting for the research response and comparing consumption paths under alternative rules.

Patent buyouts offer a possible way to reward innovation without retaining the monopoly markup. [Kremer \(1998\)](#) studies government purchases of patents that are generally placed in the public domain. Applied here, the idea would be to compensate the developer for the rights to an existing AI technology and allow competing suppliers to use it. Competitive supply at the same efficiency would require access to the algorithms and model weights. The buyout would also need a credible valuation and payment rule, a source of financing, and arrangements to reward subsequent improvements. Incentives for further development depend on whether developers can appropriate its benefits. These issues are particularly important for RSI, where today’s technology is also an input into tomorrow’s research. I leave the design of such a policy, including its effects on research and welfare, to an extension of the model.

6 Equilibrium without an upper bound on AI efficiency

I now return to the monopoly economy, remove the upper bound on AI efficiency, and compare the three substitution regimes. The RSI technology becomes

$$\dot{B} = \chi(BM)^\eta. \quad (48)$$

All other primitives remain as in Section 3. In Definition 1, I set $\psi(B) = 1$ and $\psi'(B) = 0$ and remove the upper restriction on B . Equilibrium still requires finite allocations at every finite date, global agent optimality, and both transversality conditions.

The results differ in scope across substitution regimes. Under complementarity, long-run average output-per-worker growth cannot exceed the rate of labor-augmenting technological progress, even without an upper bound on AI efficiency. At unit elasticity, I give sufficient parameter restrictions for an analytically tractable balanced-growth equilibrium. Under substitution and sufficiently strong returns to research, the developer can make discounted net profit arbitrarily large even over a fixed finite horizon, ruling out an equilibrium with a finite-valued developer optimum.

6.1 Complementarity

When $\sigma < 1$, effective labor remains necessary even if AI becomes arbitrarily efficient. Equation (8) gives an upper bound for Z and Y which does not depend on B :

$$Z \leq \omega_L^{\sigma/(\sigma-1)} AL, \quad (49)$$

$$Y \leq \omega_L^{\sigma(1-\alpha)/(\sigma-1)} K^\alpha (AL)^{1-\alpha}. \quad (50)$$

Proposition 4 (Production bounds under complementarity). *Let $0 < \sigma < 1$ and $\bar{B} = \infty$. Along any feasible path satisfying the production equations (7) and (8), labor-market clearing $L = N$ in equation (28a), and the resource constraint (26), the following holds:*

$$\limsup_{t \rightarrow \infty} \frac{1}{t} \log \left(\frac{Y_t/N_t}{Y_0/N_0} \right) \leq \gamma. \quad (51)$$

Proof. See Appendix A, Proof of Proposition 4. $\square$

Complementarity bounds long-run average output-per-worker growth despite RSI and the absence of an AI-efficiency upper bound. This does not prevent growth from temporarily exceeding γ .

Proposition 5 (Long-run outcomes under complementarity without an AI efficiency upper bound). *Suppose $0 < \sigma < 1$, $\bar{B} = \infty$, and an equilibrium satisfies $B \rightarrow \infty$. Then*

$$\frac{K}{Y} \rightarrow \frac{\alpha}{\rho + \gamma + \delta} > 0, \quad \frac{C}{Y} \rightarrow 1 - \frac{\alpha(\delta + n + \gamma)}{\rho + \gamma + \delta} > 0, \quad (52)$$

$$\frac{U}{Y} \rightarrow 0, \quad \frac{M}{Y} \rightarrow 0, \quad (53)$$

$$g_{Y/N} \rightarrow \gamma, \quad g_w \rightarrow \gamma, \quad r \rightarrow \rho + \gamma, \quad (54)$$

$$s_X \rightarrow \frac{1 - \sigma}{1 - \alpha\sigma}, \quad (55)$$

$$\frac{wL}{Y} \rightarrow \frac{\sigma(1 - \alpha)^2}{1 - \alpha\sigma} > 0, \quad \frac{p_X X}{Y} \rightarrow \frac{(1 - \alpha)(1 - \sigma)}{1 - \alpha\sigma} > 0. \quad (56)$$

Proof. See Appendix A, Proof of Proposition 5. $\square$

Conditional on $B \rightarrow \infty$, inference and research compute vanish as shares of output, not necessarily in levels. The AI industry nevertheless retains a positive revenue share.

6.2 Unit elasticity

At unit elasticity, production becomes

$$Y = K^\alpha (AL)^{(1-\alpha)\omega_L} X^{(1-\alpha)\omega_X}. \quad (57)$$

Consider a balanced-growth path (BGP) with positive constant K/Y , C/Y , and M/Y . The AI-demand and monopoly conditions, equations (13) and (19), imply $U/Y = (1 - \alpha)^2 \omega_X^2$. Along the BGP, $g_K = g_U = g_M = g_Y$. Substituting $X = BU$ into production, equation (57), and log-differentiating this equation and the RSI technology in equation (48) yields

$$g_Y = n + \gamma + \frac{\omega_X}{\omega_L} g_B, \quad (58)$$

$$(1 - \eta) g_B = \eta g_Y. \quad (59)$$

The first relation describes how AI improvement raises production growth. The second describes how expanding research resources sustain AI improvement. Solving these relations gives candidate growth rates. The next proposition establishes that they belong to an equilibrium.

Proposition 6 (An uncapped balanced-growth equilibrium and local convergence). *Suppose $\sigma = 1$, $\bar{B} = \infty$, and*

$$\eta + \omega_X < 1. \quad (60)$$

(i) *There is an equilibrium trajectory such that*

$$g_{Y/N} = g_w = \gamma + \frac{\eta\omega_X}{1 - \eta - \omega_X}(n + \gamma), \quad (61)$$

$$r = \rho + \gamma + \frac{\eta\omega_X}{1 - \eta - \omega_X}(n + \gamma), \quad (62)$$

$$\frac{wL}{Y} = (1 - \alpha)\omega_L > 0, \quad (63)$$

$$g_B = \frac{\eta(1 - \omega_X)}{1 - \eta - \omega_X}(n + \gamma). \quad (64)$$

(ii) *There is an open neighborhood of the initial condition (A_0, N_0, K_0, B_0) of the BGP in part (i) such that, from every initial condition in that neighborhood, there exists an equilibrium trajectory converging to the corresponding BGP in normalized variables.*

Proof. See Appendix A, Proof of Proposition 6. □

Proposition 6 shows that RSI can sustain permanently faster wage growth without a vanishing labor income share. Equations (61) and (63) imply $g_w = g_{Y/N} > \gamma$ while $wL/Y = (1 - \alpha)\omega_L$ remains constant and positive. Wages therefore keep pace with output per worker. This contrasts with the regimes with an upper bound in Proposition 1, where a permanent wage-growth premium occurs only in the AI-dominated regime and accompanies a vanishing labor share. A declining labor share is therefore not a necessary consequence of recursive self-improvement.

Growth in effective labor expands the resources available for research, which raises B and further expands output. The common denominator $1 - \eta - \omega_X$ captures this reinforcement: a higher η strengthens the research feedback, while a higher ω_X makes output growth more responsive to AI improvement. A related dependence appears in Jones (1995), where long-run growth depends on an expanding research workforce. Here, effective labor supports research indirectly through production, rather than entering the RSI technology itself.

6.3 Substitution

With $\sigma > 1$, Proposition 1 establishes AI-dominated equilibria whose limiting interest rate is $\mathcal{R}(\bar{B})$, which grows without bound as $\bar{B} \rightarrow \infty$. The same function also provides a lower bound on the equilibrium interest rate at each date without an efficiency upper bound:

$$r_t \geq \mathcal{R}(B_t). \quad (65)$$

To see this, combine equations (32), (33), and (34) before taking the limit $s_X \rightarrow 1$:

$$\frac{Y}{K} = \left[B(1 - \alpha) \omega_X^{\sigma/(\sigma-1)} (1 - e_X) s_X^{-1/(\sigma-1)} \right]^{(1-\alpha)/\alpha}. \quad (66)$$

For fixed B and $\sigma > 1$, this expression decreases with s_X and approaches its lower limit as $s_X \rightarrow 1$. Substitution into $r = \alpha Y/K - \delta$ gives inequality (65). Thus $\mathcal{R}(B)$ is the lowest return compatible with the production and monopoly conditions at that efficiency level, attained in the zero-labor-share limit. If B grows without bound along a trajectory, inequality (65) requires $r \rightarrow \infty$ and $Y/K \rightarrow \infty$. Production also gives $Z/Y = (Y/K)^{\alpha/(1-\alpha)} \rightarrow \infty$, while the Euler equation (5) requires $g_{C/N} = r - \rho \rightarrow \infty$.

Davidson et al. (2026) show that technological and economic feedback loops can overcome diminishing returns to research and generate *hyperbolic growth*, with variables diverging at a finite date. This *finite-time singularity* is stronger than an indefinitely increasing growth rate. Their benchmark fixes saving and factor-allocation shares; I therefore cannot apply its explosion conditions directly to the optimal choices studied here.

Under Definition 1, a finite-time singularity cannot constitute an infinite-horizon equilibrium, which requires finite allocations at every finite date. The return bound in equation (65) does not determine whether divergence occurs in finite or infinite time. The following proposition instead shows that, when research returns are sufficiently strong, the developer can make discounted net profit arbitrarily large even over a fixed finite horizon. This is an unbounded optimization problem, not a trajectory that attains infinite profit in finite time.

Proposition 7 (Research expenditure and the developer's value). *Suppose $\sigma > 1$, $\bar{B} = \infty$ and $0 < \alpha < \eta < 1$. Fix any $B_0 > 0$, any finite horizon $T > 0$, positive continuous paths of K, A, L , and an integrable interest-rate path. For the developer's problem (21) restricted to $[0, T]$, the developer can make discounted net profit arbitrarily large.*

Proof. See Appendix A, Proof of Proposition 7. □

7 Numerical simulations

I use numerical simulations to illustrate the model's economic mechanisms, not to make calibrated predictions of the magnitude or timing of AI-driven growth. The AI-specific parameters are illustrative rather than jointly estimated from data. I therefore emphasize how research incentives, AI use, and capital accumulation shape the transition and the long-run regime; the reported magnitudes and dates are conditional on the chosen parameters and initial conditions.

The exercises answer three questions. First, how does an economy adjust when RSI becomes available under an existing monopoly, and how does that adjustment depend on research productivity? I retain all four elasticities in Sections 7.2 and 7.3. Second, can research financed by monopoly rents compensate for the restriction on AI use caused by monopoly pricing and enable permanently faster growth? Third, can monopoly pricing instead eliminate AI-dominated growth despite the

efficiency gains from research? I examine these opposing outcomes at $\sigma = 1.5$, with both values of χ , in Sections 7.4 and 7.5. Appendix B describes the algorithm used to find the equilibrium and presents the links to replication files.

7.1 Parameters and initial conditions

In the next two subsections, I compare the transitional dynamics across four parameter configurations corresponding to the cases in Proposition 1. I study an unanticipated activation of RSI at date zero, retaining monopoly pricing before and after the event. Before date zero, the economy already uses AI production services, but AI efficiency is fixed at B_0 . For the chosen value of B_0 , each economy starts on a labor-bottleneck balanced growth path. I compute this path with the research productivity parameter χ equal to zero to obtain the corresponding initial capital stock K_0 for each parameter configuration. Setting $\chi = 0$ rules out improvements in AI efficiency, not the expansion of AI services through inference compute U . By treating activation as unanticipated, I exclude the advance adjustment of consumption and saving emphasized in the technology-news literature reviewed by Beaudry and Portier (2014).

I compare four economies with $\sigma = 0.9, 1.0, 1.1, 1.5$, holding all other parameters and the initial values A_0 , N_0 , and B_0 fixed. For each economy, I choose K_0 to place it on its pre-RSI balanced growth path. In particular, for each elasticity, I choose the pre-event capital stock to satisfy

$$r_0 = \rho + \gamma, \quad \frac{K_0}{Y_0} = \frac{\alpha}{\rho + \gamma + \delta} = 3.30. \quad (67)$$

This setup guarantees that Y , X , U , w , r , and p_X behave continuously rather than jumping at date zero. Consumption, research expenditure, and growth rates can nevertheless jump when RSI becomes available. Across elasticities, this construction holds the pre-event capital-output ratio fixed, not the initial AI revenue share or monopoly markup. Within each elasticity, the two research-productivity scenarios inherit exactly the same stocks; consumption and research are chosen anew after activation.

Table 2 reports the parameter values and initial stocks used for the simulations. Time is measured in years, and the four values of K_0 follow the order of the elasticities in the table. I use standard values for α , δ , and ρ , while n and γ are guided by long-run population and productivity projections. The AI production weight $\omega_X = 0.10$ and research exponent $\eta = 0.20$ are illustrative choices, not empirical estimates. I choose them to satisfy $\omega_X + \eta < 1$, the condition for the uncapped unit-elastic balanced-growth equilibrium in Proposition 6; this is not a requirement for the simulations with an upper bound considered here.

The common upper bound on AI efficiency satisfies $\mathcal{B}(1.5) < \bar{B} < \mathcal{B}(1.1)$. Thus, AI and labor are substitutes in both the $\sigma = 1.1$ and $\sigma = 1.5$ cases, but only the latter converges to the AI-dominated regime. Holding this upper bound fixed, I study two levels of research productivity for each of the four elasticities of substitution: $\chi = 7.5$ in the higher-productivity scenario and $\chi = 1.5$ in the lower-productivity scenario. For a given elasticity, both lead to the same long-run

regime, output and wage growth rates, and interest rate, but produce different initial adjustments and transition speeds. These are illustrative choices, intended to show how research productivity affects the short-run impact of RSI and the timing of subsequent growth.

Table 2: Simulation parameters and initial stocks

Symbol	Value	Interpretation	Source
α	0.33	Capital share	Standard calibration value.
δ	0.05	Depreciation rate	Standard calibration value.
ρ	0.04	Discount rate	Standard calibration value.
n	0.003	Population growth rate	Rounded constant-rate approximation to the UN 2024–2100 world projection (United Nations, 2024).
γ	0.01	Growth of labor-augmenting technology	OECD labor-productivity projections (OECD, 2025).
η	0.20	Research exponent	Illustrative.
ω_X	0.10	AI production weight	Illustrative.
ω_L	0.90	Effective-labor weight	$1 - \omega_X$.
χ	7.5 (higher); 1.5 (lower)	Research productivity after activation	Illustrative.
σ	0.90; 1.00; 1.10; 1.50.	Elasticity of substitution	Illustrative.
$\bar{B}$	1,361	AI-efficiency upper bound	$1.10 \times \mathcal{B}(1.50)$.
A_0	1	Initial labor-efficiency units	Normalization.
N_0	1	Initial population	Normalization.
B_0	13.6	Initial AI efficiency	$0.01 \times \bar{B}$.
K_0	3.43; 4.65; 5.04; 5.50.	Initial capital	Adjusted to match $K_0/Y_0 = 3.30$, the pre-RSI balanced growth path ratio.

I do not model a separate stock of computing hardware or the costs of rapidly expanding it. [Erdil et al. \(2025\)](#) incorporate adjustment costs for compute capacity and other physical capital. Introducing such frictions could change the initial reallocation of resources and the timing of the transition shown here.

Section 7.4 then uses the same technology parameters at $\sigma = 1.5$ in a different experiment, starting from a competitive labor-bottleneck BGP and granting exclusive rights over the technology. Section 7.5 changes B_0 and $\bar{B}$ and starts near the competitive economy’s AI-dominated long-run path. In each institutional comparison, continued competition and both monopoly paths share the same initial stocks.

7.2 Higher research productivity

I first ask how research, consumption, and capital accumulation respond when RSI becomes available under an existing monopoly, and how this adjustment differs across economies approaching the two long-run regimes. I compare the four elasticities under relatively high research productivity, $\chi = 7.5$. Research productivity affects the transition, but not the limiting growth rates characterized in Section 4. All four economies inherit the pre-RSI balanced growth paths described in Section 7.1. Table 2 reports their common parameters and elasticity-specific initial capital stocks. I examine quantities first, then wages and prices, and finally income shares.

I use Figure 4 to present the annual growth rates of output, AI production services, and capital per unit of effective labor: $Y/(AL)$, $X/(AL)$, and $K/(AL)$. Zero therefore means that the corresponding aggregate grows at the same rate as effective labor, $n + \gamma$. The upper row shows two pre-event years and the first ten years after RSI activation. The lower row follows the same variables from year 10 to year 500. Throughout the three figures, green dashed, black solid, orange dotted, and blue dash-dotted lines denote $\sigma = 0.90, 1, 1.10, 1.50$, respectively. The vertical line marks RSI activation, and black dotted horizontal lines mark the analytical limits for $\sigma = 1.50$.

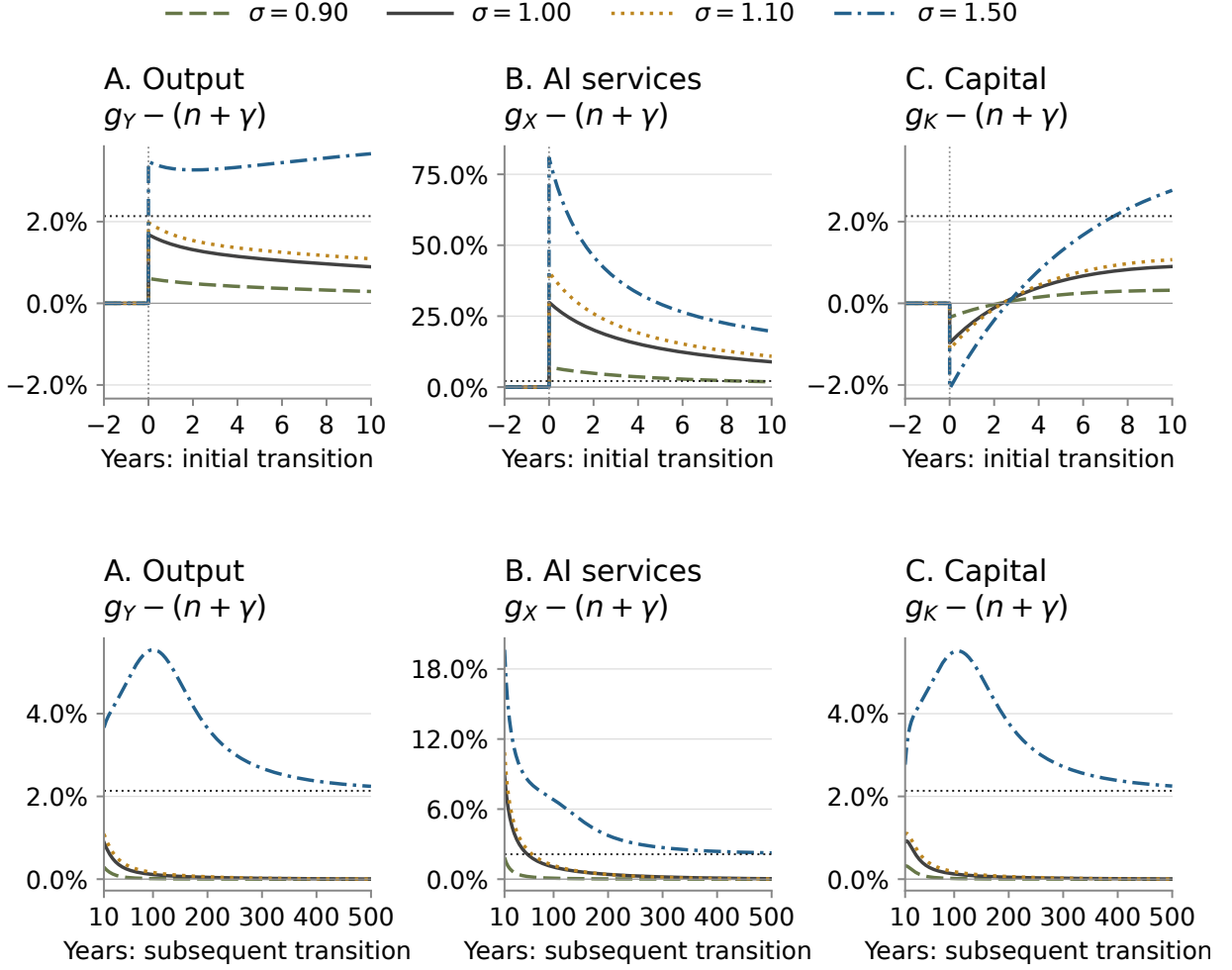


Figure 4: Higher research productivity ($\chi = 7.5$): growth of output, AI production services, and capital per unit of effective labor. Rates are instantaneous annual rates. The upper row shows two pre-event years and years 0–10; the lower row shows years 10–500. Panels A and C share a scale within each row. The vertical line marks RSI activation.

The top row of Figure 4 shows that AI services respond much more strongly than output. At $\sigma = 1.50$, $X/(AL)$ initially grows at 80.8% per year, whereas $Y/(AL)$ grows at 3.5%. Research begins to raise B , allowing more services to be obtained from compute through $X = BU$. Final production cannot expand at the same rate because it also requires capital and labor. Initial growth of $Y/(AL)$ is smaller at the other elasticities, reaching only 0.6% at $\sigma = 0.90$. The third panel shows the capital response: capital per unit of effective labor, $K/(AL)$, falls in every case. Households increase consumption at RSI activation while the developer starts spending on research. The resource constraint thus requires lower investment in physical capital. Capital accumulation subsequently recovers as output expands and as the marginal product of capital increases.

The bottom row of Figure 4 separates temporary growth gains from permanent ones. At $\sigma = 0.90, 1, 1.10$, all three normalized growth rates decline toward zero. This is the labor-bottleneck

outcome in Proposition 1: output per worker eventually grows at $\gamma = 1.0\%$. At $\sigma = 1.50$, output and capital growth instead rise further before declining toward their long-run values. As explained in Section 4.2, sufficient substitution and a sufficiently high efficiency bound allow capital and inference compute to expand together without being limited by effective-labor growth. Their limiting growth rates exceed effective-labor growth by 2.1 percentage points; effective labor itself grows at 1.3% per year.

I next examine wages and prices in Figure 5. The figure shows annual wage growth g_w , the annual net interest rate r , and the level of p_X in final-good units on a logarithmic scale. The top panels present the short-run dynamics and the bottom panels present the long-run dynamics.

Wage growth increases in all four economies, but the initial response is not ordered monotonically by substitutability: the immediate increase is larger at $\sigma = 1$ than at $\sigma = 1.50$. Wage growth in the highest-elasticity case soon surpasses the other cases. After the initial jump, wage growth declines in the labor-bottleneck cases but continues to rise in the AI-dominated case.

While p_X falls continuously as AI efficiency improves, the interest rate starts at 5.0% in every case and then rises as output initially outpaces capital. The higher return accompanies faster consumption-per-person growth, as required by the Euler condition in Equation (5). A related mechanism underlies Chow et al. (2026), who link faster expected consumption growth to higher long-term real interest rates.

The bottom row of Figure 5 shows the longer adjustment of wage growth, the interest rate, and the AI-service price. The three labor-bottleneck economies return toward wage growth of 1.0% and an interest rate of 5.0%. At $\sigma = 1.50$, both rates rise above their long-run values before declining. Equation (41) gives limiting wage growth of 2.4%, below output-per-worker growth of 3.1% but above labor-augmenting technological progress of 1.0%. Meanwhile, the interest rate tends to $\mathcal{R}(\bar{B}) = 7.1\%$, above its pre-RSI level.

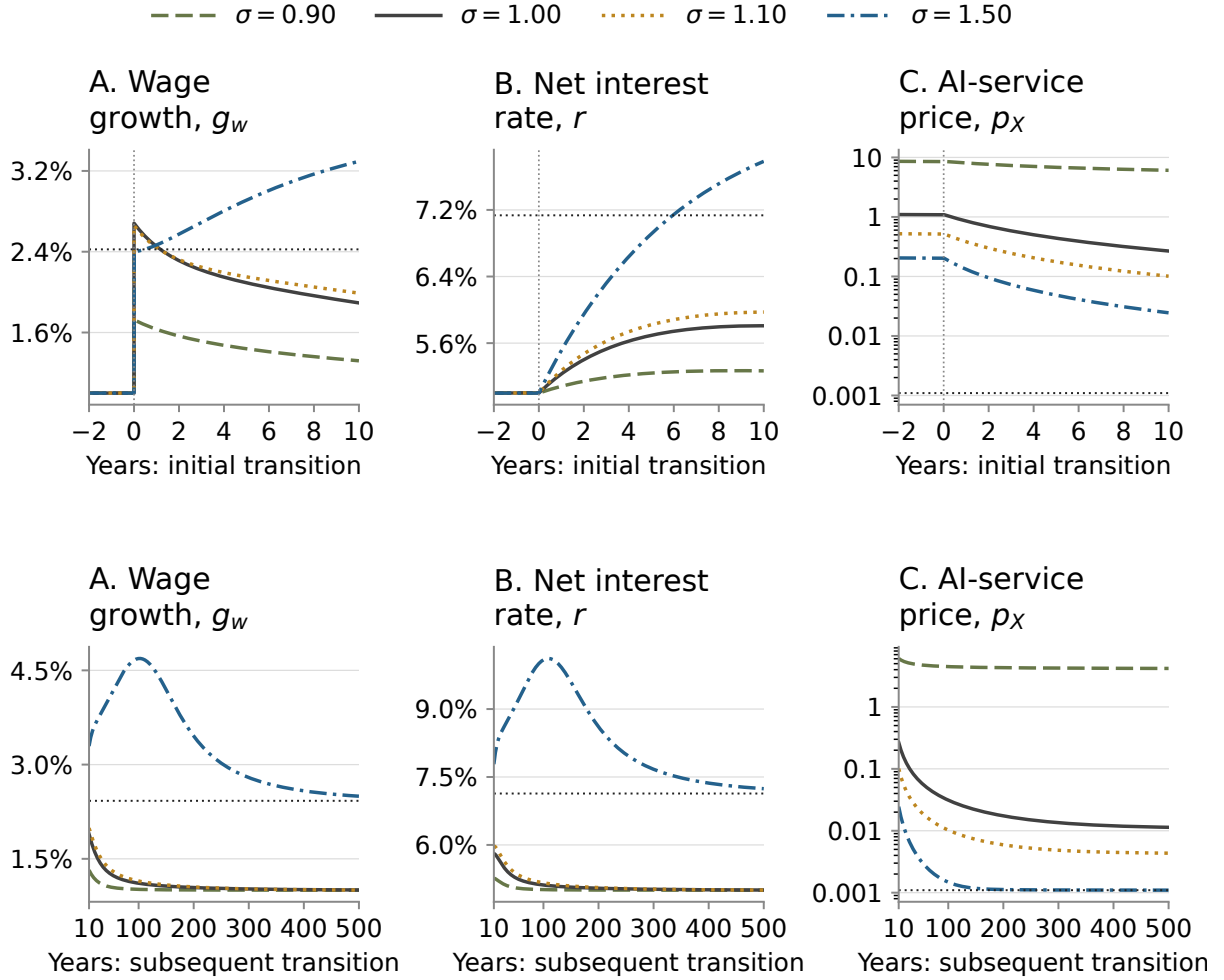


Figure 5: Higher research productivity ($\chi = 7.5$): wage growth, the net interest rate, and the AI-service price. Rates are annual; the price is in final-good units on a logarithmic scale. The vertical line marks RSI activation.

Finally, I use Figure 6 to show labor income wL/Y , developer net profit Π/Y , inference expenditure U/Y , and research expenditure M/Y , all as a share of output. The top four panels show the short run and the bottom four panels show the long run. Line styles again identify the four elasticities. The omitted gross capital income share remains $\alpha = 33.0\%$.

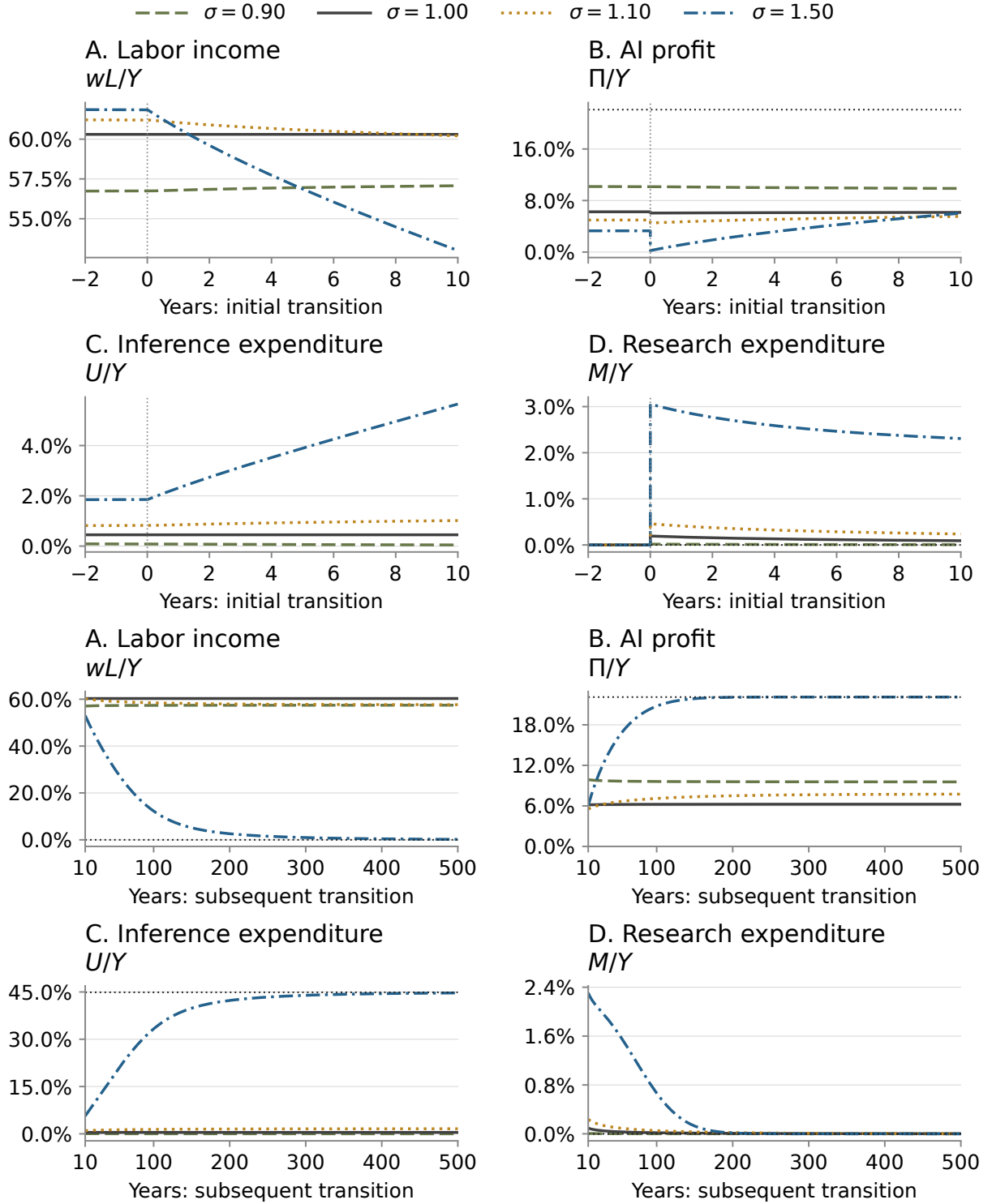


Figure 6: Higher research productivity ($\chi = 7.5$): labor income, AI net profit, inference expenditure, and research expenditure relative to output. The first two rows show the pre-event BGP and years 0–10; the last two show years 10–500. The omitted gross capital share remains $\alpha = 0.33$. Short-run panels A and C use scales fitted to the initial trajectories and omit the long-run reference lines.

In the short run, research spending as a share of output jumps in all four economies. The adjustment is strongest at $\sigma = 1.50$: immediately after activation, research absorbs 59.4% of AI revenue, compared with 35.9% for inference compute. The developer’s net profit therefore falls sharply. Profit subsequently recovers as research absorbs a smaller share of revenue and AI sales expand. Labor’s income share also falls most strongly in this scenario, while changes are much smaller in the other three economies.

Over the longer horizon, the AI-dominated economy combines permanently faster wage growth with an even faster expansion of output per worker, so labor’s income share tends to zero. Rising wages alongside a declining labor share also appear in [Autor and Kausik \(2026\)](#) and [Ray and Mookherjee \(2022\)](#). As labor’s share falls, AI revenue rises relative to output. Inference compute eventually absorbs 67.0% of AI revenue, while research expenditure vanishes relative to that revenue. These last proportions use AI revenue as the denominator, rather than output as in the figure.

7.3 Lower research productivity

I next ask whether lower research productivity makes the early transition less informative about the eventual growth regime, and how it changes the initial allocation of resources. I set $\chi = 1.5$, one fifth of its value in Section 7.2. All other parameters and initial stocks remain unchanged, including the elasticity-specific pre-RSI capital stocks in Table 2. Consumption and research are solved again for each economy; the new exercise changes both the initial adjustment and the timing of the transition, but not the limiting growth rates, interest rates, and income shares.

I use Figure 7 to present the growth of $Y/(AL)$, $X/(AL)$, and $K/(AL)$ in panels A–C, in percent per year. As before, zero means growth at the rate of effective labor, $n + \gamma$. The upper row shows two pre-event years and the first ten years after RSI activation. The lower row follows the same trajectories from year 10 to year 500. Throughout the three figures, line styles identify the same four elasticities as in Figure 4, and black dotted lines mark the analytical limits for $\sigma = 1.50$.

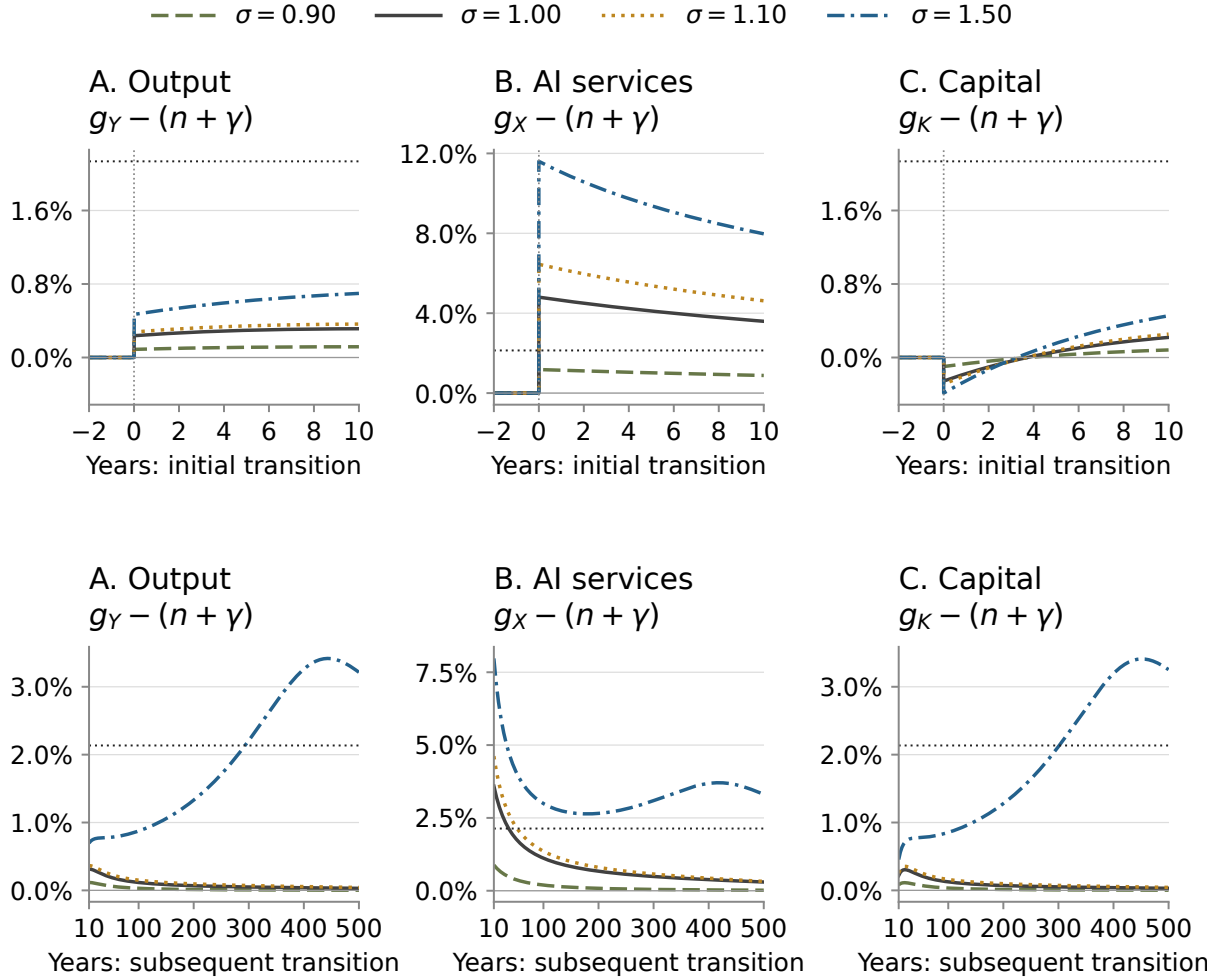


Figure 7: Lower research productivity ($\chi = 1.5$): growth of output, AI production services, and capital per unit of effective labor. Rates are instantaneous annual rates. The upper row shows two pre-event years and years 0–10; the lower row shows years 10–500. Panels A and C share a scale within each row. The vertical line marks RSI activation.

The upper row shows a much smaller initial increase in growth. Growth of $Y/(AL)$ ranges from 0.1% to 0.5% across the four economies. AI services still grow faster than output, but their initial normalized growth rate at $\sigma = 1.50$ is 11.6%, compared with 80.8% in the preceding exercise. Lower research productivity and a smaller research allocation produce slower efficiency improvements. Consumption still rises at activation, and research uses resources that could otherwise finance capital accumulation. Both adjustments are smaller, however, so $K/(AL)$ falls less. As capital accumulation recovers, output growth rises during the first decade even though AI-service growth is already slowing. The four economies therefore begin with more moderate growth and less pronounced differences.

The lower row reveals the eventual separation between regimes. At $\sigma = 1.50$, output and capital growth rise for several centuries before declining. AI-service growth first slows, then rises again

as capital and inference use expand. At year 500, $Y/(AL)$ still grows at 3.2%, above its 2.1% long-run limit, showing a significantly slower transition. A similar pattern of delayed acceleration appears in Jones (2026, pp. 11–12), drawing on Jones and Tonetti (2026). My simulations generate the transition through private RSI investment and capital accumulation, without modeling the progressive automation of additional tasks.

I next examine wages and prices in Figure 8. Panels A–C show annual wage growth g_w , the annual net interest rate r , and the level of p_X in final-good units on a logarithmic scale. The top panels show the short run and the bottom panels the long run. Initially, wage growth rises modestly in all four economies. Unlike the preceding exercise, it continues to increase during the first years even in the labor-bottleneck cases: capital accumulation helps sustain the increase in labor’s marginal product. The interest rate also rises more gradually as output outpaces capital. At $\sigma = 1.50$, it reaches 5.5% after ten years, compared with 7.8% under higher research productivity. Meanwhile, slower efficiency improvements make the decline in p_X more gradual.

In the bottom row, wage growth and the interest rate eventually turn down in all four economies. The three labor-bottleneck cases approach 1.0% and 5.0%, respectively. At $\sigma = 1.50$, the rise and subsequent decline occur much later, and adjustment remains incomplete at year 500: the interest rate is still 8.2%, above its 7.1% long-run value. The limiting wage-growth rate remains 2.4%, below output-per-worker growth of 3.1%. Lower research productivity therefore delays the transition without eliminating the long-run difference between these growth rates.

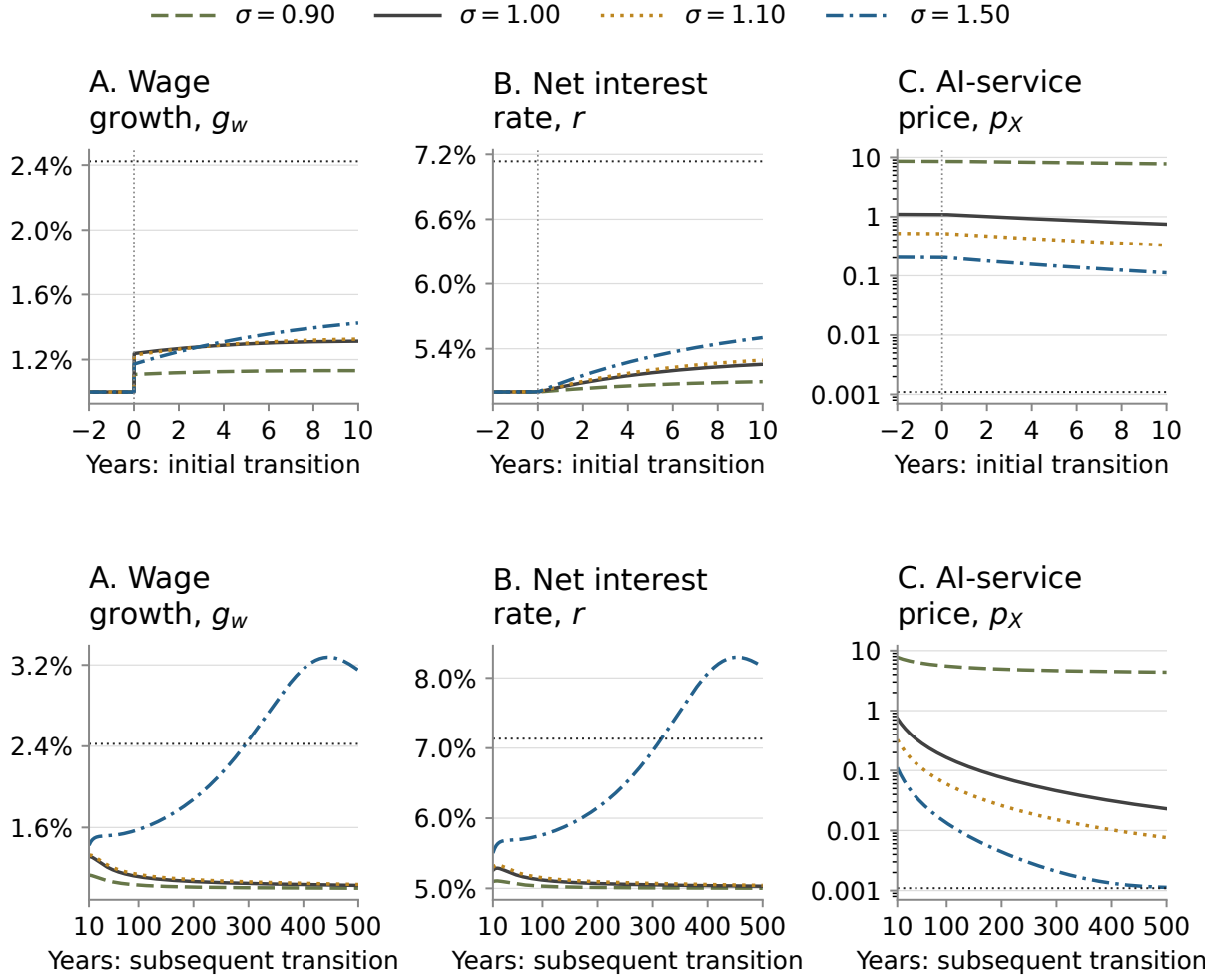


Figure 8: Lower research productivity ($\chi = 1.5$): wage growth, the net interest rate, and the AI-service price. Rates are annual; the price is in final-good units on a logarithmic scale. The vertical line marks RSI activation.

I complete the comparison with Figure 9. Panels A–D report wL/Y , Π/Y , U/Y , and M/Y , all as percentages of output. The upper two rows cover the initial transition and the lower two cover years 10–500, with the same scenario lines. The omitted gross capital share remains 33.0%.

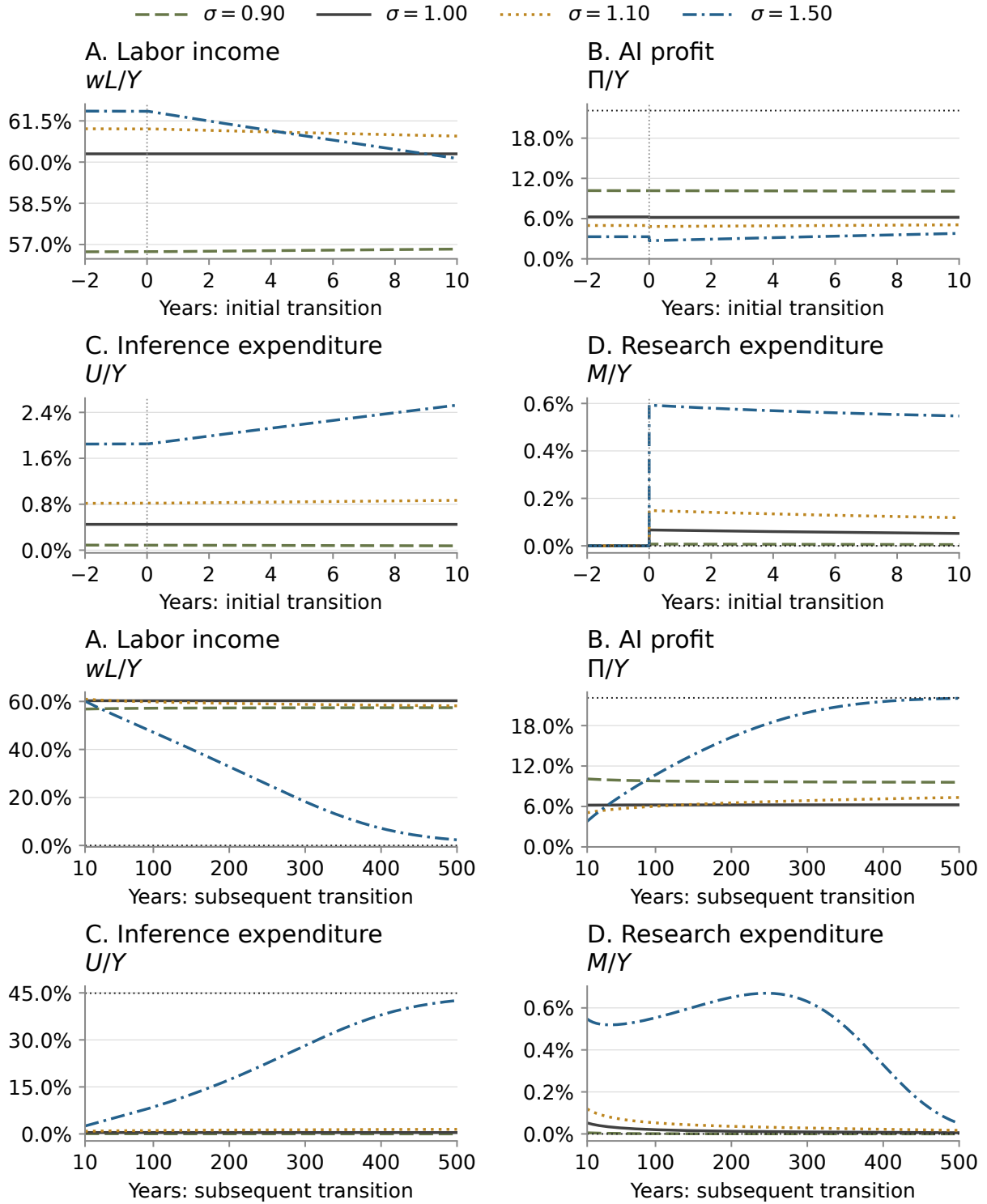


Figure 9: Lower research productivity ($\chi = 1.5$): labor income, AI net profit, inference expenditure, and research expenditure relative to output. The first two rows show the pre-event BGP and years 0–10; the last two show years 10–500. The omitted gross capital share remains $\alpha = 0.33$. Short-run panels A and C use scales fitted to the initial trajectories and omit the long-run reference lines.

In the short run, research absorbs a smaller share of output, leaving more of AI revenue as net profit. At $\sigma = 1.50$, initial research expenditure is 0.6% of output rather than 3.1%, and net profit is 2.7% rather than 0.2%. Research spending reduces net profit at activation, but less sharply than in the preceding exercise. Inference expenditure and labor’s income share change little in the labor-bottleneck cases. In the AI-dominated case, inference’s share increases and labor’s share declines, but both movements are initially gradual, consistent with the smaller output–wage growth gap.

The longer horizon shows the same eventual redistribution as in Figure 6: labor’s share tends to zero at $\sigma = 1.50$, while inference expenditure and developer profit approach the same respective limiting shares of output as in the higher-productivity exercise. Research expenditure relative to output first declines, then rises again before falling toward zero. Expanding AI sales provide an incentive for further research, while proximity to the upper bound eventually limits the efficiency gains it can deliver. The other three economies retain a positive labor share. Lower research productivity changes the adjustment and delays the divergence between regimes without changing their limiting growth rates and income shares.

7.4 From competitive provision to monopoly with RSI

I now ask whether research financed by monopoly rents can compensate for the restriction on AI use caused by monopoly pricing and allow an economy to escape the labor bottleneck. I set $\sigma = 1.5$ and compare continued competition with monopoly under $\chi = 7.5$ and $\chi = 1.5$. Unlike the preceding exercises, RSI is available before date zero, with the same positive χ as after the change. Research is nevertheless zero under the competitive arrangement in Section 5, because improvements cannot be privately appropriated.

I retain the technology parameters in Table 2, but choose $K_0 = 6.535$ to place the economy on its competitive BGP at the same B_0 . Before the change, $p_X = 1/B_0$, $r = \rho + \gamma$, and $K_0/Y_0 = 3.3$. Initial efficiency lies below $\mathcal{B}^{\text{mc}}(1.5)$, so continued competition preserves the labor-bottleneck BGP. The upper bound instead exceeds $\mathcal{B}(1.5)$, allowing the monopoly paths considered here to approach AI-dominated growth.

At date zero, an unexpected permanent exclusive right covers both the existing AI technology and its subsequent improvements. In particular, competitive suppliers can no longer use the old technology. Each monopoly path inherits the competitive K_0 and B_0 but chooses consumption and research anew. This experiment changes research incentives and pricing, not the RSI technology. It abstracts from the cost of enforcing exclusivity and is not an optimal policy exercise.

Figure 10 compares both research productivities with continued competition from the same initial stocks. Panels A–C show output per worker, wages, and consumption per person. The gray dashed line denotes continued competition; the solid blue and dash-dotted gold lines denote monopoly with $\chi = 7.5$ and $\chi = 1.5$. The upper row includes two pre-event years and years 0–10, measured as percentages of the competitive levels. The middle and lower rows cover years 10–100 and 10–500, measured as multiples on logarithmic scales. The competitive benchmark is therefore 100.0% or $1\times$.

Levels relative to continued competition | $\sigma = 1.5$

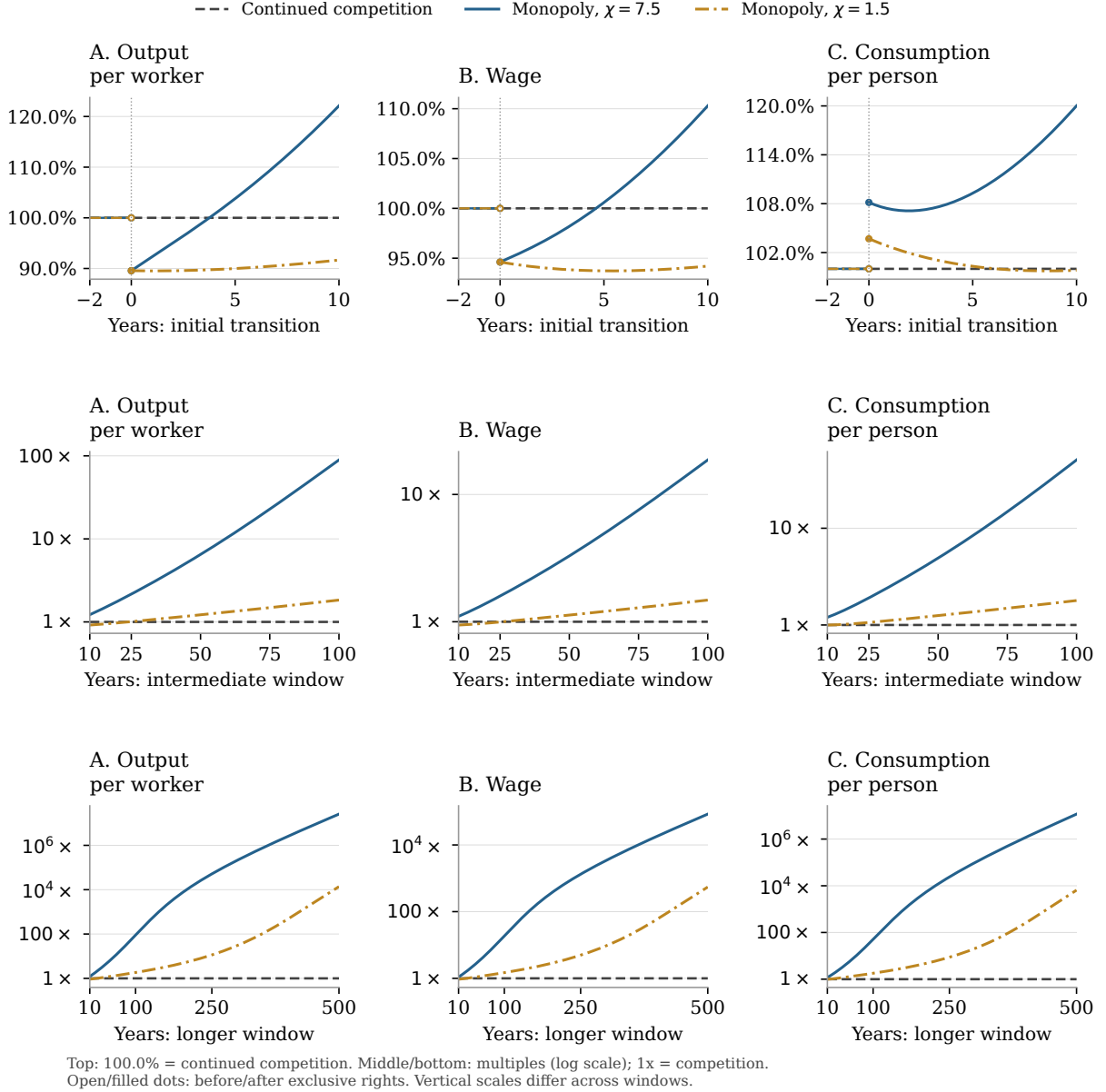


Figure 10: From competition to monopoly with RSI at $\sigma = 1.5$: output per worker, wages, and consumption per person relative to continued competition. The two monopoly paths differ only in χ . Rows cover years $-2-10$, $10-100$, and $10-500$. The upper row uses percentages; the lower rows use multiples on logarithmic scales. Open and filled dots mark levels before and after the grant. Vertical scales differ across windows.

The upper row shows the initial restriction in AI use before efficiency can improve: output falls by 10.4% and wages by 5.4% under either χ . These are level losses at unchanged capital and efficiency, not annual growth rates. With $\chi = 7.5$, research subsequently raises efficiency enough for output to regain its competitive counterpart around year 3.8 and wages around year 4.6. With

$\chi = 1.5$, both remain below their competitive counterparts throughout the first decade.

Consumption responds differently. It rises by 8.2% at the grant under high research productivity and by 3.7% under low productivity, even though output falls. The household chooses less capital accumulation, while reduced inference use releases resources. Research initially absorbs 3.3% of output with $\chi = 7.5$, compared with 0.6% with $\chi = 1.5$. Research productivity thus changes the initial allocation, not only the timing of recovery.

The middle row makes the slower recovery visible: output and wages under $\chi = 1.5$ regain their competitive counterparts around years 24.0 and 26.0. The lower row shows an expanding gap under both monopoly paths. Research allows efficiency to rise beyond the monopoly threshold, whereas continued competition retains B_0 and remains labor-bottlenecked. Limiting output-per-worker growth, wage growth, and the interest rate are 3.1%, 2.4%, and 7.1% under either χ , as characterized in Proposition 1. The simulated horizon does not establish completed convergence: the interest rate at year 500 is still 7.2% with high research productivity and 8.2% with low research productivity.

I interpret this comparison as an institutional counterfactual, not a calibrated prediction of granting AI exclusivity. The recovery dates compare contemporaneous levels, not cumulative gains or household welfare. Evaluating the policy would require the entire consumption path. Appendix D retains both research productivities for all four elasticities, including their initial stocks, level comparisons, growth, prices, income shares, and compute expenditure.

7.5 When monopoly eliminates AI-dominated growth

I now ask whether monopoly can eliminate AI-dominated growth even while research makes AI more efficient. This is the opposite outcome to the preceding exercise: research gains fail to compensate for the restriction on AI use caused by monopoly pricing. Following Corollary 1, I choose $\sigma = 1.5$ and $\mathcal{B}^{\text{mc}}(1.5) < B_0 < \bar{B} < \mathcal{B}(1.5)$. Initial efficiency is high enough for AI-dominated growth under competition, but even the upper bound is insufficient under monopoly. The unexpected grant covers existing technology and future improvements, as in Section 7.4. RSI is available throughout; the grant changes pricing and research incentives. The preceding simulations remain separate experiments with different initial states and efficiency bounds.

Table 3 reports the new design. I retain the other parameters in Table 2 and compare continued competition with monopoly under $\chi = 7.5$ and $\chi = 1.5$, all from the same initial stocks. Before granting monopoly rights, I let the competitive economy approach its AI-dominated growth path. I select the first whole year when output-per-worker growth, wage growth, and the interest rate are each within 0.1 percentage point of their long-run limits, and reset the date to zero. I inherit A_0 , N_0 , and K_0 jointly from that date, keeping B_0 and both values of χ unchanged. This is a near-long-run competitive allocation, not an exact steady state: $K/(AL)$ continues to grow. Just before the grant, output per worker grows at 5.8%, wages at 4.2%, and the interest rate is 9.8%. The capital-output ratio is 2.23. Appendix D describes the additional competitive-transition solver and numerical checks.

Table 3: Parameters and initial stocks for the growth-reversal experiment

Object	Value	Choice
σ	1.5	Substitution
χ	7.5; 1.5	Same illustrative research productivities
$\bar{B}$	1,113.5	$0.90\mathcal{B}(1.5)$
B_0	1,002.2	$0.90\bar{B}$
A_0, N_0	6.6859; 1.7683	Inherited from competitive prehistory
K_0	7.2907×10^9	Inherited from the same date
K_0/Y_{0-}	2.23	Competitive ratio before the grant

Notes: $\mathcal{B}^{\text{mc}}(1.5) = 829.0$ and $\mathcal{B}(1.5) = 1,237.3$. The upper bound and initial efficiency are chosen to satisfy the corollary’s threshold inequalities. The three paths share the reported initial stocks; consumption and research are equilibrium choices. The initialization uses 190 years of competitive prehistory, not a forecast of a policy date. Stocks retain their original units; these illustrative choices are not empirical estimates. Appendix D describes the prehistory and its starting stocks.

Figure 11 shows annual output-per-worker growth, wage growth, and the net interest rate in panels A–C. The rows cover years -10 – 10 , 10 – 100 , and 100 – 500 ; negative dates show the common competitive prehistory. The dashed gray line denotes continued competition; the solid blue and dash-dotted gold lines denote monopoly with high and low research productivity. At the grant, output falls by 55.6%, wages by 33.5%, and AI services by 70.3%. These are instantaneous level changes at unchanged K and B , not annual growth rates in the figure. The lower output–capital ratio reduces the interest rate from 9.8% to 1.6%.

After this level loss, high research productivity briefly raises output-per-worker growth to 13.4% as efficiency improves rapidly. With low research productivity, initial growth is only 0.1%. Neither response restores AI-dominated growth: by year 10, output per worker contracts under both monopoly paths. The inherited capital stock supported faster growth under competitive AI supply; restricting that supply lowers its return and induces capital decumulation. Consumption initially jumps upward by 141.6% and 132.2%, respectively, as lower compute use and investment release resources despite the fall in gross output. Consumption then falls: the interest rate initially lies below ρ , so the Euler equation implies negative consumption-per-person growth after the grant.

The longer windows show that the loss of AI-dominated growth is not confined to the initial repricing. Under continued competition, output-per-worker growth tends to 5.7%, wage growth to 4.1%, and the interest rate to 9.7%, as in equation (44). Under either monopoly path, the corresponding limits are 1.0%, 1.0%, and 5.0%, consistent with the labor-bottleneck regime in Proposition 1. Both monopoly economies are still adjusting at year 500: output-per-worker growth is about -0.1% , wage growth is 0.3% , and the interest rate is 3.9% . The large stock of capital inherited from the competitive expansion takes a long time to adjust to restricted AI supply. The monopoly limit therefore remains distinct from the last plotted date: monopoly removes permanently faster growth, not all long-run growth.

Figure 12 explains why efficiency gains do not prevent this outcome. Using the same lines and windows, panels A–C show $B/\bar{B}$, AI services relative to continued competition, and labor income as a share of output. In the first decade, high research productivity takes efficiency to 99.9% of

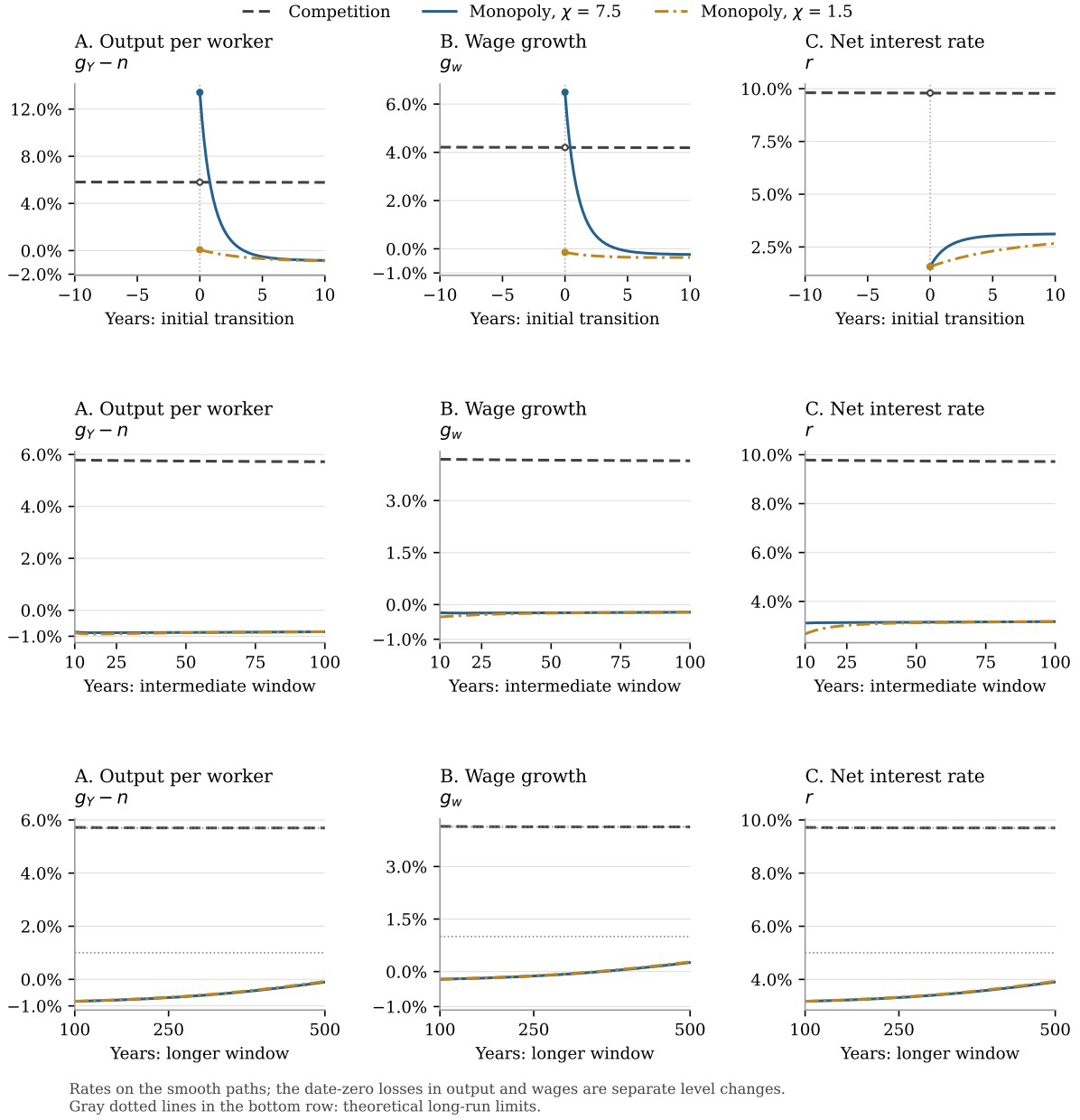


Figure 11: Monopoly and the loss of AI-dominated growth: annual output-per-worker growth, wage growth, and net interest. Rows show years $-10-10$, $10-100$, and $100-500$. The grant occurs at zero, after the common competitive prehistory. Rates describe the smooth paths in each scenario, not the instantaneous output and wage losses. Gray dotted lines in the bottom row indicate theoretical long-run limits. Vertical scales differ across windows.

its upper bound, compared with 97.1% under low research productivity. Both nevertheless supply fewer AI services than continued competition. In the longer run, efficiency approaches $\bar{B}$, an improvement of 11.1% over B_0 ; competition retains B_0 . Yet the monopoly markup prevents these improvements from sustaining the joint expansion of capital and AI services that occurs under competition. AI services under monopoly become negligible *relative to the competitive path*, not in absolute terms.

Labor’s income share tends to 4.5% under monopoly rather than zero under competition. This larger limiting share coexists with slower wage growth. At year 500, the labor share is still only 2.1% and 2.2% under high and low research productivity, respectively, confirming that neither path has reached its labor-bottleneck limit.

Together with Section 7.4, this experiment illustrates opposite regime rankings. There, research allows the monopoly economy with $\sigma = 1.5$ to escape the labor bottleneck from a low initial efficiency. Here, initial efficiency already sustains AI-dominated growth under competition, and further efficiency gains do not compensate for the restriction on AI use caused by monopoly pricing. More efficient AI is therefore compatible with slower long-run growth.

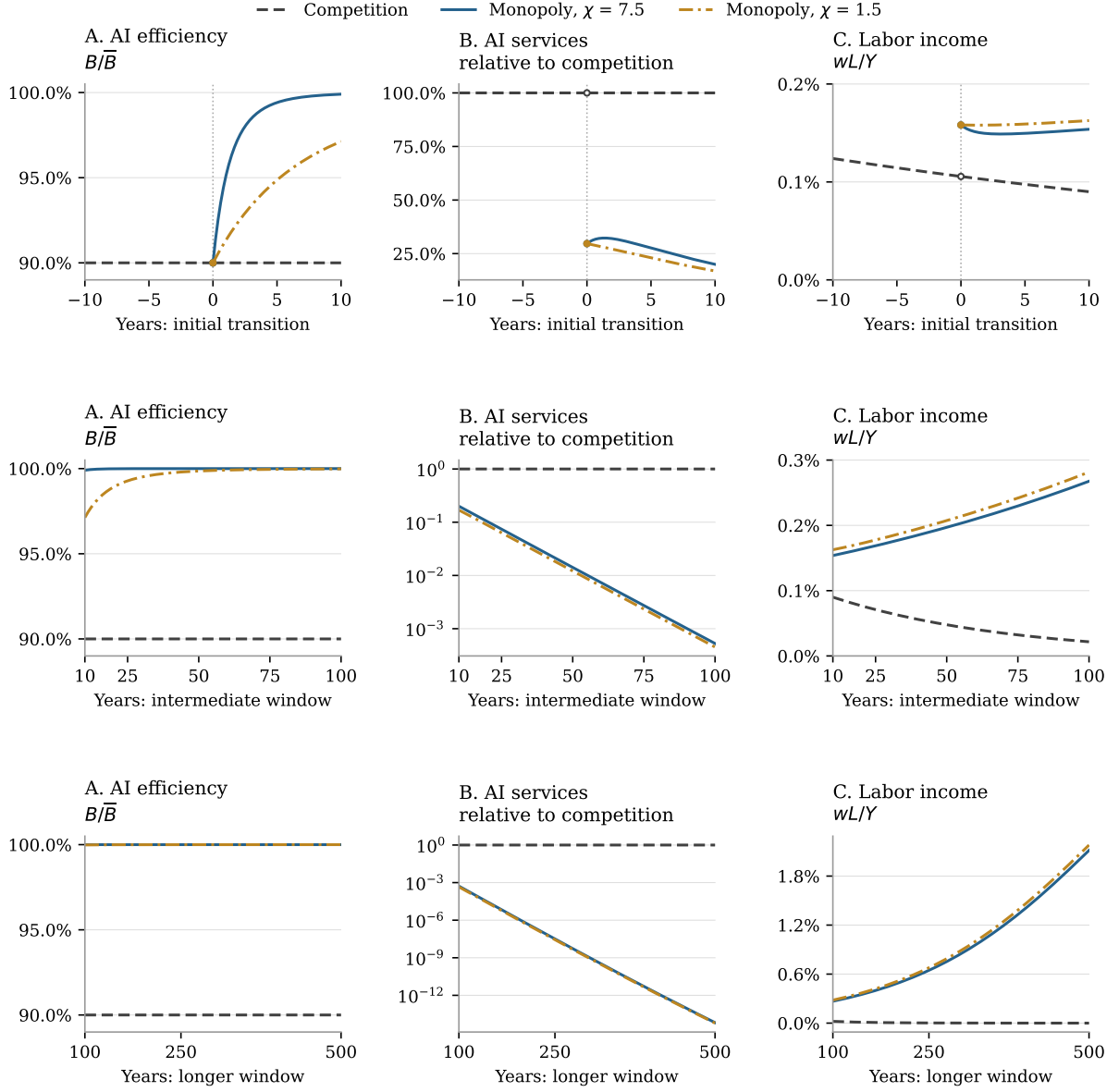
The large initial adjustments also reflect the abrupt withdrawal of access to existing technology. I use this experiment to illustrate a possible growth reversal, not to predict the effects of an actual monopoly grant or rank household welfare.

8 Conclusion

This paper endogenizes investment in AI research, which in turn affects the pace of economic growth. Even under RSI, computing requires resources that compete with consumption and accumulation of physical capital. Algorithms and model weights, by contrast, are non-rival: their use in one computation does not reduce their availability for others. Following Romer (1990), I use monopoly rents to make research investment profitable.

With an upper bound on AI efficiency, permanently faster wage growth requires sufficient substitution between AI and labor and sufficiently high attainable efficiency. In this AI-dominated regime, output and AI revenue per worker grow faster than wages, so labor’s income share vanishes even as wages rise. When labor remains a production bottleneck, the growth rates of output per worker and wages instead return to the exogenous rate of labor-augmenting technological progress. The upper bound on AI efficiency matters for these growth and distributional outcomes: without it, the unit-elasticity case can sustain both growth rates above this exogenous rate while preserving a positive labor share.

AI-dominated growth can permanently increase the return on capital, raising questions about wealth concentration not examined here (Berg et al., 2018; Moll et al., 2022). The representative household owns both physical capital and the AI developer. I therefore characterize the division of income among wages, capital income, and developer profit, not the effects of unequal asset ownership across households.



Panel B: 100.0% or 1 = continued competition; logarithmic scale in the middle and bottom rows. All paths share initial capital and AI efficiency. Vertical scales differ across windows.

Figure 12: More efficient AI but slower growth under monopoly: efficiency relative to its upper bound, AI services relative to continued competition, and labor's income share. Windows and line styles match Figure 11. Panel B uses percentages in the first row and multiples on logarithmic scales in the lower rows; its competitive benchmark is 100.0% or one. Vertical scales differ across windows.

RSI is not sufficient for permanently faster output-per-worker growth, nor does such growth require continuing improvements in AI efficiency. AI efficiency can improve even when labor remains a production bottleneck. Under complementarity, output-per-worker growth cannot permanently exceed the rate of labor-augmenting technological progress, even without an upper bound on AI efficiency. With sufficient substitution and sufficiently high attainable AI efficiency, capital accumulation and expanding inference compute sustain permanently faster growth in both output per worker and wages, even as AI efficiency converges to its upper bound.

The comparison with competition separates the efficiency needed to sustain AI-dominated growth from the incentives to reach it. Under free access to AI models and their improvements, competitive suppliers do not finance RSI. Nevertheless, sufficiently high initial efficiency sustains AI-dominated growth without monopoly or further research. The simulations show that, from an initial efficiency too low to sustain AI-dominated growth under competition, monopoly rents can finance improvements that allow an escape from the labor bottleneck. But markups restrict AI use and raise the efficiency required for AI-dominated growth relative to marginal-cost pricing. Competition can therefore sustain AI-dominated growth where monopoly produces more efficient AI but leaves the economy labor-bottlenecked. Even when both economies are AI-dominated, monopoly delivers faster long-run growth only if its efficiency gains offset the reduction in AI use caused by its markup. Neither market structure nor efficiency alone determines which economy grows faster. These comparisons do not rank welfare.

The monopoly is an institutional assumption, not a consequence of RSI. In practice, firms compete to develop algorithms and train models. However, as [Korinek and Vipra \(2025\)](#) explain, access to better AI can improve the advantage of a leading developer, so an eventual monopoly does not seem to be far-fetched. The prospect of future monopoly rents can help explain intense competition among firms.

Removing monopoly rents after innovation can undermine the credibility of the incentives offered beforehand. Patent buyouts could instead reward the developer while opening access to existing AI technology ([Kremer, 1998](#)). An extension could study how to value and finance such buyouts while preserving incentives for further RSI.

The numerical simulations show that early output-per-worker growth can remain moderate and similar across economies approaching different long-run regimes. Research productivity affects both the initial allocation of resources and the subsequent timing of growth: lower productivity can delay takeoff in the AI-dominated economy without changing its long-run growth rates.

A Proofs

Lemma 1 (Static AI-service choice). *For $K, A, L, B > 0$, $0 < \alpha < 1$, $\sigma > 0$, and strictly positive CES weights that sum to one, problem (18) has a unique, strictly positive global maximizer. It is characterized by condition (19) and depends smoothly on the positive variables K, A, L, B , for each fixed σ .*

Proof of Lemma 1. Fix $K, A, L, B > 0$ and let $R(X) = p_X(X)X$ denote AI-service revenue. The inverse-demand elasticity in (15) and differentiation of (10) give

$$R'(X) = p_X(X)(1 - e_X), \quad \frac{ds_X}{d \ln X} = \left(1 - \frac{1}{\sigma}\right) s_X(1 - s_X).$$

Wherever marginal revenue is positive, direct differentiation gives

$$-\frac{XR''(X)}{p_X(X)} = e_X(1 - e_X) + \left(\alpha - \frac{1}{\sigma}\right) \left(1 - \frac{1}{\sigma}\right) s_X(1 - s_X). \quad (68)$$

If $(\alpha - 1/\sigma)(1 - 1/\sigma) \geq 0$, the right-hand side is positive. If this product is negative, the parameter restrictions imply $0 < \alpha < 1/\sigma < 1$, and

$$\begin{aligned} -\frac{XR''(X)}{p_X(X)} &= \frac{1}{\sigma} \left(1 - \frac{1}{\sigma}\right) (1 - s_X)^2 \\ &\quad + \left[\frac{1}{\sigma^2} + 2\alpha - \frac{3\alpha}{\sigma}\right] s_X(1 - s_X) + \alpha(1 - \alpha)s_X^2. \end{aligned}$$

The middle coefficient is positive because

$$\frac{1}{\sigma^2} + 2\alpha - \frac{3\alpha}{\sigma} = \left(\frac{1}{\sigma} - \alpha\right)^2 + \alpha \left(2 - \frac{1}{\sigma} - \alpha\right) > 0.$$

Thus marginal revenue decreases strictly whenever it is positive.

As $X \downarrow 0$, revenue is asymptotically proportional to $X^{1-1/\sigma}$ when $\sigma > 1$, to $X^{(1-\alpha)\omega_X}$ when $\sigma = 1$, and to $X^{1-\alpha}$ when $\sigma < 1$. Every exponent lies between zero and one, so $R'(X) \rightarrow \infty$. As $X \rightarrow \infty$, marginal revenue converges to zero from above if $\sigma \geq 1$; if $\sigma < 1$, it eventually becomes negative. Hence $R'(X) = 1/B$ has exactly one solution. Operating profit increases before this solution and decreases after it. The linear cost X/B dominates revenue as $X \rightarrow \infty$, so this solution is the unique global maximum. Equation (68) also proves the strict second-order condition:

$$e_X(1 - e_X) + \left(\alpha - \frac{1}{\sigma}\right) \left(1 - \frac{1}{\sigma}\right) s_X(1 - s_X) > 0. \quad (69)$$

At the unique solution of $R'(X) = 1/B$, $R''(X) < 0$. The implicit-function theorem therefore makes the maximizing service choice smooth in K, A, L, B . $\square$

Lemma 2 (Optimal research expenditure). *Consider a feasible research trajectory $\{B_t, M_t\}_{t \geq 0}$*

for problem (21), starting from B_0 and satisfying the RSI technology (17). Suppose $M_t > 0$ and the present value of net profit is finite. Suppose there exists a continuously differentiable shadow-price trajectory $\{q_t\}_{t \geq 0}$ satisfying the research condition (22), the costate equation (23), and the transversality condition (25).

Suppose $0 < B_0 < \bar{B} < \infty$ and, for every $t \geq 0$ and every $b \in [B_0, \bar{B})$,

$$\begin{aligned} \pi_t(b) - \pi_t(B_t) + \frac{1-\eta}{\eta} M_t \left[\left(\frac{b}{B_t} \right)^{\eta/(1-\eta)} - 1 \right] \\ \leq \frac{\bar{B} - B_t}{B_t} (U_t + M_t) \log \left(\frac{\bar{B} - B_t}{\bar{B} - b} \right), \end{aligned} \quad (70)$$

where $\pi_t(b)$ is operating profit as defined in equation (18). Then $\{M_t\}_{t \geq 0}$ solves the developer's problem (21): no other feasible research trajectory starting from the same B_0 yields a higher present value of net profit.

Proof of Lemma 2. I first transform AI efficiency to compare the candidate with alternative research trajectories. Define

$$R_B(B) = -\bar{B} \log \left(1 - \frac{B}{\bar{B}} \right), \quad B(R_B) = \bar{B} \left(1 - e^{-R_B/\bar{B}} \right). \quad (71)$$

It maps $B \in [B_0, \bar{B})$ into $R_B \in [R_{B,0}, \infty)$, where $R_{B,t} = R_B(B_t)$ and $R_{B,0} = R_B(B_0)$. Since $R'_B(B) = 1/\psi(B)$, this transformation removes the factor $\psi(B)$ from the RSI technology (17):

$$\dot{R}_B = \chi[B(R_B)M]^\eta. \quad (72)$$

The corresponding current-value shadow price is $q_{R,t} = q_t \psi(B_t)$, which is positive by the research condition (22). At each date t , hold $q_{R,t}$ at its value along the candidate trajectory and define the maximized Hamiltonian for an alternative state $\tilde{R}_B$:

$$\hat{\mathcal{H}}_t(\tilde{R}_B) \equiv \max_{\tilde{M} \geq 0} \left\{ \pi_t(B(\tilde{R}_B)) - \tilde{M} + q_{R,t} \chi[B(\tilde{R}_B)\tilde{M}]^\eta \right\}.$$

Because $0 < \eta < 1$, the expression being maximized is strictly concave in research expenditure. Using the candidate's research condition (22), the maximizing expenditure at $\tilde{R}_B$ is

$$M_t \left[\frac{B(\tilde{R}_B)}{B_t} \right]^{\eta/(1-\eta)},$$

where M_t is the candidate's research expenditure. Substitution gives

$$\hat{\mathcal{H}}_t(\tilde{R}_B) = \pi_t(B(\tilde{R}_B)) + \frac{1-\eta}{\eta} M_t \left[\frac{B(\tilde{R}_B)}{B_t} \right]^{\eta/(1-\eta)}. \quad (73)$$

Differentiating the maximized Hamiltonian with respect to its state argument and using the envelope condition (20) gives

$$\hat{\mathcal{H}}'_t(R_{B,t}) = \frac{\psi(B_t)}{B_t}(U_t + M_t). \quad (74)$$

Because

$$R_B(b) - R_{B,t} = \bar{B} \log\left(\frac{\bar{B} - B_t}{\bar{B} - b}\right),$$

condition (70) is equivalent to the following inequality for every $\tilde{R}_B \geq R_{B,0}$:

$$\hat{\mathcal{H}}_t(\tilde{R}_B) \leq \hat{\mathcal{H}}_t(R_{B,t}) + \hat{\mathcal{H}}'_t(R_{B,t})(\tilde{R}_B - R_{B,t}).$$

Let $D_t = \exp(-\int_0^t r_s ds)$ be the discount factor and $\mu_t = D_t q_{R,t}$ the present-value shadow price of the transformed state R_B . The research condition (22) and the RSI technology (17) imply $q_t \eta \dot{B}_t = M_t$. Differentiating $q_{R,t} = q_t \psi(B_t)$ and using the costate equation (23) therefore gives

$$\dot{q}_{R,t} = r_t q_{R,t} - \frac{\psi(B_t)}{B_t}(U_t + M_t), \quad \dot{\mu}_t = -D_t \hat{\mathcal{H}}'_t(R_{B,t}).$$

Consider any feasible alternative $\{\tilde{B}_t, \tilde{M}_t\}_{t \geq 0}$ starting from the same B_0 , and write $\tilde{R}_{B,t} = R_B(\tilde{B}_t)$. Here feasible expenditure is nonnegative and integrable on every finite interval, with a locally absolutely continuous state satisfying the RSI technology almost everywhere. Let

$$J_T = \int_0^T D_t [\pi_t(B_t) - M_t] dt$$

denote discounted net profit through date T under the candidate, and define $\tilde{J}_T$ analogously for the alternative. At each date, the alternative's Hamiltonian cannot exceed $\hat{\mathcal{H}}_t(\tilde{R}_{B,t})$, whereas the candidate attains $\hat{\mathcal{H}}_t(R_{B,t})$. Applying condition (70) and the transformed costate equation yields, for almost every t ,

$$\begin{aligned} & D_t \left[\pi_t(\tilde{B}_t) - \tilde{M}_t - \pi_t(B_t) + M_t \right] \\ & \leq D_t \hat{\mathcal{H}}'_t(R_{B,t})(\tilde{R}_{B,t} - R_{B,t}) - \mu_t(\dot{\tilde{R}}_{B,t} - \dot{R}_{B,t}) \\ & = -\frac{d}{dt} \left[\mu_t(\tilde{R}_{B,t} - R_{B,t}) \right]. \end{aligned}$$

Integrating from zero to T and using the common initial state gives

$$\tilde{J}_T - J_T \leq -\mu_T(\tilde{R}_{B,T} - R_{B,T}) \leq \mu_T R_{B,T},$$

where the second inequality uses $\mu_T > 0$ and $\tilde{R}_{B,T} \geq 0$. Since

$$\psi(B_t) R_{B,t} = -(\bar{B} - B_t) \log\left(1 - \frac{B_t}{\bar{B}}\right) \leq B_t,$$

the developer's transversality condition (25) gives

$$0 \leq \mu_T R_{B,T} = D_T q_T \psi(B_T) R_{B,T} \leq D_T q_T B_T \longrightarrow 0.$$

The candidate's present value is finite by assumption. Hence

$$\limsup_{T \rightarrow \infty} \tilde{J}_T \leq \lim_{T \rightarrow \infty} J_T < \infty,$$

so no feasible alternative yields a higher present value of net profit. The argument requires the tangent to the maximized Hamiltonian at the candidate state to lie above its value at every alternative state. Global concavity is sufficient for this inequality, but is not required. $\square$

Lemma 3 (Sufficient conditions for Hamiltonian concavity). *Consider the maximized Hamiltonian in equation (73) for a feasible trajectory satisfying the research condition (22), with $M_t > 0$ and $0 < B_0 < \bar{B} < \infty$. Let $X_t^*(b)$ denote the optimal AI-service choice in problem (18) at efficiency b , and let $\varepsilon_{B,t}(b) = \partial \ln X_t^*(b) / \partial \ln b$ denote its elasticity with respect to AI efficiency. Suppose that, for every $t \geq 0$ and $b \in [B_0, \bar{B})$,*

$$\psi(b)[\varepsilon_{B,t}(b) - 2] \leq \frac{b}{\bar{B}}, \quad (75)$$

and

$$\frac{B_0}{\bar{B}} \geq \max \left\{ 0, \frac{2\eta - 1}{\eta} \right\}. \quad (76)$$

Then the maximized Hamiltonian is concave in the transformed state $R_B \in [R_B(B_0), \infty)$ at every date. Consequently, the support condition (70) in Lemma 2 holds.

Proof of Lemma 3. Fix a date t . Lemma 1 ensures that $X_t^*(b)$ is unique and smooth. Let $s_{X,t}(b)$ and $e_{X,t}(b)$ be the corresponding elasticities defined in equations (10) and (15). Implicit differentiation of the monopoly condition (19) gives

$$\varepsilon_{B,t}(b) \equiv \frac{\partial \ln X_t^*(b)}{\partial \ln b} = \frac{1 - e_{X,t}(b)}{e_{X,t}(b)[1 - e_{X,t}(b)] + (\alpha - \sigma^{-1})(1 - \sigma^{-1})s_{X,t}(b)[1 - s_{X,t}(b)]}. \quad (77)$$

The denominator is strictly positive by equation (69). The envelope condition (20) gives $\pi_t'(b) = X_t^*(b)/b^2$, so

$$\pi_t''(b) = \frac{X_t^*(b)}{b^3} [\varepsilon_{B,t}(b) - 2]. \quad (78)$$

The transformation (71) satisfies $B'(R_B) = \psi(b)$ and $B''(R_B) = -\psi(b)/\bar{B}$, where $b = B(R_B)$. The chain rule therefore gives

$$\frac{d^2 \pi_t(B(R_B))}{dR_B^2} = \frac{X_t^*(b)\psi(b)}{b^3} \left[\psi(b)(\varepsilon_{B,t}(b) - 2) - \frac{b}{\bar{B}} \right]. \quad (79)$$

Condition (75) makes this derivative nonpositive for every $b \in [B_0, \bar{B})$.

The research term in (73) is a positive multiple of $B(R_B)^{\eta/(1-\eta)}$. Its curvature is

$$\frac{d^2 B(R_B)^{\eta/(1-\eta)}}{dR_B^2} = \frac{\eta}{1-\eta} b^{\eta/(1-\eta)-2} \psi(b) \left[\frac{2\eta-1}{1-\eta} \psi(b) - \frac{b}{\bar{B}} \right]. \quad (80)$$

The bracket is nonpositive exactly when $b/\bar{B} \geq (2\eta-1)/\eta$. Since $b \geq B_0$, condition (76) makes the research term concave as well. The maximized Hamiltonian is therefore concave, so its tangent at the candidate state is a global upper bound. As shown in the proof of Lemma 2, this is condition (70).

If $\eta \leq 1/2$, condition (76) holds automatically. But for every fixed $0 < \eta < 1$ its right-hand side is strictly below one, so it also holds without this restriction when initial efficiency is sufficiently close to the upper bound. These curvature tests are sufficient, not necessary: they can fail even when the Hamiltonian has the required supporting tangent at the candidate state. $\square$

Proof of Proposition 1. The proof has three tasks. First, Lemma 1 gives a unique, smooth, globally optimal within-date service choice at given states. I combine this choice with the other equilibrium equations to identify candidate long-run values and verify their admissibility, including positive consumption. Finding these values does not yet establish the existence of a trajectory approaching them.

Second, I write the exact dynamic equations in coordinates with finite candidate limits. The linearization identifies the stable directions; I also check that they can accommodate every sufficiently nearby initial state. Counting stable directions alone would not establish this property. The stable-manifold and inverse-function theorems then select the initial consumption and shadow price for those states. They establish convergent solutions of the nonlinear equations for all future time, not merely solutions of their linear approximation.

Third, I reconstruct the original variables and verify feasibility, market clearing, and both transversality conditions. Household concavity and Lemma 2 establish optimality. For the developer, I verify the curvature conditions of Lemma 3 at every AI-efficiency level reachable under alternative research policies from the initial efficiency. The upper bound makes this possible for initial efficiency sufficiently close to it. These final checks turn the constructed solutions of the first-order conditions into equilibria. The result is local in initial states, not in the time horizon.

Step 1. Feasible efficiency and candidate long-run values.

Let $D = \bar{B} - B$. The state equation with an upper bound gives

$$\frac{\dot{D}}{D} = -\frac{\chi}{\bar{B}} (BM)^\eta.$$

Integration gives

$$\bar{B} - B_t = (\bar{B} - B_0) \exp \left\{ -\frac{\chi}{\bar{B}} \int_0^t (B_s M_s)^\eta ds \right\} > 0. \quad (81)$$

Local integrability of research expenditure makes the exponent finite at every finite date, so $D_t > 0$.

For the candidate limits, impose $B = \bar{B}$ only in the normalized boundary equations. Actual trajectories will satisfy $B < \bar{B}$ at every finite date.

Write $\theta = (1 - \alpha)/\alpha$. For $\sigma \neq 1$, let $\varphi = (\sigma - 1)/\sigma$ and define

$$m(s) = 1 - \frac{1-s}{\sigma} - \alpha s, \quad \mathcal{H}_\sigma(s) = \left[(1-\alpha)s m(s) \left(\frac{\omega_X}{s} \right)^{1/\varphi} \right]^\theta.$$

Here $m(s) = 1 - e_X$ is the marginal-revenue factor, not research compute. The static monopoly optimum has $m(s_X) > 0$ by Lemma 1. The CES share, monopoly condition, and production function, equations (28d), (28i), and (28c), imply

$$\frac{X}{AN} = \left[\frac{\omega_L s_X}{\omega_X (1 - s_X)} \right]^{1/\varphi}, \quad \frac{U}{Y} = (1 - \alpha) s_X m(s_X), \quad (82)$$

$$\frac{Y}{K} = B^\theta \mathcal{H}_\sigma(s_X). \quad (83)$$

The identities $Z = X(\omega_X/s_X)^{1/\varphi}$ and $X = B(1 - \alpha)s_X m(s_X)Y$ give the output–capital ratio in (83).

If $0 < \sigma < 1$, positive marginal revenue requires

$$s > s_{\min} \equiv \frac{1 - \sigma}{1 - \alpha\sigma}.$$

On $(s_{\min}, 1)$, both $m(s)$ and $s^{-1/(\sigma-1)}$ increase strictly. Since

$$\mathcal{H}_\sigma(s)^{1/\theta} = (1 - \alpha)\omega_X^{1/\varphi} m(s) s^{-1/(\sigma-1)},$$

the share map increases from zero to $\mathcal{H}_\sigma(1) = [(1 - \alpha)^2 \omega_X^{1/\varphi}]^\theta$. Thus the equation

$$\frac{\rho + \gamma + \delta}{\alpha} = \bar{B}^\theta \mathcal{H}_\sigma(\bar{s}_X) \quad (84)$$

has exactly one interior solution if and only if $\bar{B} > \mathcal{B}(\sigma)$.

If $\sigma > 1$, the share map decreases strictly on $(0, 1)$. To verify the sign, put $a = \sigma^{-1} - \alpha$, so $m(s) = \varphi + as$. Its logarithmic derivative has the sign of $a(\sigma - 2)s - \varphi$. If $a(\sigma - 2) \leq 0$ this is negative. If $\sigma > 2$ and $a > 0$, $a(\sigma - 2) < (\sigma - 2)/\sigma < \varphi$; if $1 < \sigma < 2$ and $a < 0$, $a(\sigma - 2) < (1 - \sigma^{-1})(2 - \sigma) < \varphi$. The map therefore decreases from infinity to $\mathcal{H}_\sigma(1)$. Equation (84) has exactly one interior solution if and only if $\bar{B} < \mathcal{B}(\sigma)$. Since $\mathcal{R}(B) = \alpha B^\theta \mathcal{H}_\sigma(1) - \delta$, these calculations also derive (35)–(36). The interior-share condition is therefore $\mathcal{R}(\bar{B}) > \rho + \gamma$ for $\sigma < 1$, but $\mathcal{R}(\bar{B}) < \rho + \gamma$ for $\sigma > 1$.

For either interior-share case, let x, y, u, k, c denote X, Y, U, K, C divided by AN , respectively.

Construct their candidate limiting values sequentially:

$$\begin{aligned}
\bar{x} &= \left[\frac{\omega_L \bar{s}_X}{\omega_X (1 - \bar{s}_X)} \right]^{1/\varphi}, & \bar{u} &= \frac{\bar{x}}{\bar{B}}, \\
\bar{y} &= \frac{\bar{u}}{(1 - \alpha) \bar{s}_X m(\bar{s}_X)}, & \bar{k} &= \frac{\alpha \bar{y}}{\rho + \gamma + \delta}, \\
\bar{c} &= \bar{y} - \bar{u} - (\delta + n + \gamma) \bar{k}, & \bar{r} &= \rho + \gamma.
\end{aligned} \tag{85}$$

At $\sigma = 1$, define $\beta = (1 - \alpha)\omega_X$. The exact static solution is

$$y = \left[(\beta^2 B)^\beta k^\alpha \right]^{1/(1-\beta)}, \quad u = \beta^2 y, \quad x = Bu.$$

The equation $y/k = (\rho + \gamma + \delta)/\alpha$ has exactly one positive solution for every finite $\bar{B} > 0$:

$$\bar{k} = \left[(\beta^2 \bar{B})^\beta \left(\frac{\rho + \gamma + \delta}{\alpha} \right)^{\beta-1} \right]^{1/(1-\alpha-\beta)}. \tag{86}$$

Set $\bar{y} = [(\rho + \gamma + \delta)/\alpha] \bar{k}$, $\bar{u} = \beta^2 \bar{y}$, and $\bar{x} = \bar{B} \bar{u}$, with $\bar{c}, \bar{r}$ as in (85).

Consumption positivity does not require an additional restriction. Positive marginal revenue implies $0 < m(\bar{s}_X) < 1$. Since $0 < \bar{s}_X < 1$, in all three cases

$$\frac{\bar{u}}{\bar{y}} = (1 - \alpha) \bar{s}_X m(\bar{s}_X) < 1 - \alpha, \quad \frac{(\delta + n + \gamma) \bar{k}}{\bar{y}} = \frac{\alpha(\delta + n + \gamma)}{\rho + \gamma + \delta} < \alpha.$$

It follows that $\bar{c}/\bar{y} > 0$.

Step 2. The three positive-labor-share cases.

The cases in Proposition 1(i), 1(ii), and 1(iii) share one normalization. Divide capital, consumption, and the AI-efficiency shadow price by effective labor, and multiply the remaining efficiency gap by effective labor:

$$k = \frac{K}{AN}, \quad c = \frac{C}{AN}, \quad d = (\bar{B} - B)AN, \quad \tilde{q} = \frac{q}{AN}, \quad \tau = (AN)^{-1}. \tag{87}$$

Dividing by AN removes effective-labor growth. Multiplying the shrinking gap by AN allows it to converge to a positive value: $\dot{d}/d = n + \gamma - \dot{B}/(\bar{B} - B)$. The variable d measures the scaled gap, not its rate of closure. The constants $\bar{k}, \bar{d}$ are the limiting values of k, d ; they are determined by the model, not additional parameters. Also $\dot{\tau} = -(n + \gamma)\tau$, so $\tau \rightarrow 0$. These coordinates leave the model unchanged: $K = k/\tau$ and $B = \bar{B} - d\tau$ at every finite date.

Given k, B , solve the unique static monopoly problem for y, x, u , and set $r = \alpha y/k - \delta$. Define $a = \dot{B}/(\bar{B} - B)$, the proportional rate of closure of the AI-efficiency gap. The research condition gives

$$B = \bar{B} - d\tau, \quad M = \left[\tilde{q} d \chi \eta \frac{B^\eta}{\bar{B}} \right]^{1/(1-\eta)}, \quad a = \chi \frac{B^\eta M^\eta}{\bar{B}}.$$

Substituting (28k) and (29) gives the exact dynamic equations

$$\frac{\dot{k}}{k} = \frac{y - c - u - M\tau}{k} - \delta - n - \gamma, \quad (88a)$$

$$\frac{\dot{c}}{c} = r - \rho - \gamma, \quad (88b)$$

$$\frac{\dot{d}}{d} = n + \gamma - a, \quad (88c)$$

$$\frac{\dot{\tilde{q}}}{\tilde{q}} = r - \frac{x}{\tilde{q}B^2} - \eta \frac{ad\tau}{B} + a - n - \gamma, \quad (88d)$$

$$\dot{\tau} = -(n + \gamma)\tau. \quad (88e)$$

In particular, population growth cancels from the normalized consumption equation; it is not subtracted a second time.

Complete the candidate long-run allocation by setting

$$\bar{M} = \left[\frac{(n + \gamma)\bar{B}^{1-\eta}}{\chi} \right]^{1/\eta}, \quad \bar{\tilde{q}} = \frac{\bar{x}}{\bar{r}\bar{B}^2}, \quad \bar{d} = \frac{\bar{M}}{\eta(n + \gamma)\bar{\tilde{q}}}. \quad (89)$$

All are positive. Direct substitution shows that $(\bar{k}, \bar{c}, \bar{d}, \bar{\tilde{q}}, 0)$ is a fixed point of (88). This boundary point is a device for constructing paths with $\tau > 0$ and $B < \bar{B}$, not an admissible date-zero state itself.

I next verify saddle-path stability. Write $\mu = \eta/(1 - \eta)$, $r_k = \partial r / \partial \log k$, and $a_k = \partial[(y - u)/k] / \partial \log k + \bar{c}/\bar{k}$, evaluated at the fixed point. In $(\log k, \log c, \log d, \log \tilde{q}, \tau)$ coordinates the Jacobian has two diagonal blocks: one for capital and consumption, and one for the scaled efficiency gap and normalized shadow price,

$$J_R = \begin{pmatrix} a_k & -\bar{c}/\bar{k} \\ r_k & 0 \end{pmatrix}, \quad J_B = \begin{pmatrix} -(n + \gamma)\mu & -(n + \gamma)\mu \\ (n + \gamma)\mu & \bar{r} + (n + \gamma)\mu \end{pmatrix}, \quad (90)$$

and the additional eigenvalue $-(n + \gamma)$. There can be forcing from τ to both blocks and from (k, c) to $(d, \tilde{q})$, but not in the reverse directions at the fixed point.

To check $r_k < 0$, let $s = s_X$, $u_\sigma = 1/\sigma$, $e = e_X$, and let $D(s)$ be the positive curvature expression in (68). Implicit differentiation of the monopoly condition gives

$$\frac{\partial \log X}{\partial \log K} = \frac{\alpha(1 - e)}{D(s)}, \quad r_k = (r + \delta)(1 - \alpha) \left[\frac{\alpha s(1 - e)}{D(s)} - 1 \right].$$

Moreover,

$$D(s) - \alpha s(1 - e) = (1 - s)\{u_\sigma(1 - e) + (\alpha - u_\sigma)(1 - u_\sigma)s\} > 0.$$

The sign is immediate when $u_\sigma \geq 1$ or $u_\sigma \leq \alpha$. For $\alpha < u_\sigma < 1$, the braces are linear in s , with positive endpoint values $u_\sigma(1 - u_\sigma)$ and $u_\sigma^2 + \alpha(1 - 2u_\sigma)$. The latter is positive if $u_\sigma \leq 1/2$; otherwise

$\alpha < u_\sigma$ implies it exceeds $u_\sigma(1 - u_\sigma) > 0$. Thus $r_k < 0$ in every interior-share case.

Both blocks have negative determinants:

$$\det J_R = (\bar{c}/\bar{k})r_k < 0, \quad \det J_B = -(n + \gamma)\mu\bar{r} < 0.$$

There are therefore three stable and two unstable eigenvalues. The stable direction in J_R has a nonzero k component, and the stable direction in J_B has a nonzero d component. To verify that their number suffices to select the jump variables, consider a vector in the stable subspace whose state components (k, d, τ) are all zero. Its linearized τ path is zero. The unforced capital–consumption block must then follow its stable direction; its zero initial k component forces its c component to be zero as well. With that forcing absent, the same argument in the research block forces $\tilde{q} = 0$. The projection of the three-dimensional stable subspace onto the three predetermined coordinates is therefore injective and hence invertible. This argument also covers repeated stable eigenvalues.

Lemma 1 makes the static solution smooth. The normalized vector field extends smoothly through $\tau = 0$, since $B = \bar{B} - d\tau$ remains positive nearby. The stable-manifold theorem and the inverse-function theorem therefore give a local graph $(c, \tilde{q})$ over (k, d, τ) ; see Dyatlov (2018). Choose a small neighborhood of the full fixed point and then a smaller initial-state neighborhood $\mathcal{U}_L$ of $(\bar{k}, \bar{d}, 0)$. The size of this neighborhood depends on the parameters and the case under consideration. In original initial stocks the restriction is

$$(k_0, d_0, \tau_0) = \left(\frac{K_0}{A_0 N_0}, (\bar{B} - B_0)A_0 N_0, \frac{1}{A_0 N_0} \right) \in \mathcal{U}_L. \quad (91)$$

Closeness is required in all three coordinates; B_0 close to $\bar{B}$ alone is not sufficient. The point with $\tau = 0$ is a long-run limit, not a finite-date initial state. Every physical initial state satisfying (91) has a solution on this graph that stays in the full neighborhood for all future time and converges to the fixed point. It is unique among solutions with those properties. Positivity of $\tau, d, c, \tilde{q}, k$ and the original RSI technology ensure $0 < B_t < \bar{B}$ and $M_t > 0$ at every finite date. Reconstruction gives the original initial stocks, the equilibrium equations, and the limiting rates in (37)–(39). Indeed, the normalized vector field tends to zero at its fixed point, so the time derivatives of its smooth static ratios tend to zero as well. This step, not convergence of ratios alone, establishes convergence of the growth rates. Optimality and transversality are verified below.

Step 3. The AI-dominated case.

Let $\sigma > 1$ and $\bar{B} > \mathcal{B}(\sigma)$. Here $\varphi = (\sigma - 1)/\sigma \in (0, 1)$. I seek an equilibrium in which capital grows faster than effective labor. Along such a trajectory, $K/(AN)$ would diverge, so I normalize consumption and the shadow value by capital and scale the remaining AI-efficiency gap by capital:

$$h = (AN/K)^\varphi, \quad d = (\bar{B} - B)K, \quad c = C/K, \quad v = q/K, \quad \tau = (AN)^{-1}.$$

Within this construction, y, x, u denote $Y/K, X/K, U/K$, respectively. The constant $\bar{d}$ now denotes

the limit of $(\bar{B} - B)K$, not of $(\bar{B} - B)AN$; no original state or control is redefined. Then

$$B = \bar{B} - d\tau h^{1/\varphi}, \quad M = [vd\chi\eta B^\eta/\bar{B}]^{1/(1-\eta)}, \quad a = \chi B^\eta M^\eta/\bar{B}, \quad r = \alpha y - \delta.$$

Capital's growth rate is $g_K = y - c - u - M\tau h^{1/\varphi} - \delta$. The exact equations are

$$\dot{h} = \varphi h(n + \gamma - g_K), \tag{92a}$$

$$\dot{d}/d = g_K - a, \tag{92b}$$

$$\dot{c}/c = n + r - \rho - g_K, \tag{92c}$$

$$\dot{v}/v = r - x/(vB^2) - \eta ad\tau h^{1/\varphi}/B + a - g_K, \tag{92d}$$

$$\dot{\tau} = -(n + \gamma)\tau. \tag{92e}$$

The CES term divided by capital is $[\omega_L h + \omega_X x^\varphi]^{1/\varphi}$. At $h = 0$ the monopoly equation has the strictly positive curvature $D(1) = \alpha(1 - \alpha)$. Its implicit static solution is therefore smooth in (h, B) near $(0, \bar{B})$. For the remaining terms, replacing $h^{1/\varphi}$ by $\max\{h, 0\}^{1/\varphi}$ outside the physical domain gives a C^1 extension because $1/\varphi > 1$. The C^1 stable-manifold theorem applies; see [Feldman \(2021, p. 7\)](#). The extension is only a mathematical device and does not change any equation on $h \geq 0$.

The limiting interest and aggregate growth rates are

$$\bar{r} = \mathcal{R}(\bar{B}), \quad \bar{g} = n + \bar{r} - \rho > n + \gamma. \tag{93}$$

At $h = \tau = 0$, the terminal constants are

$$\begin{aligned} \bar{y} &= \frac{\mathcal{R}(\bar{B}) + \delta}{\alpha}, & \bar{u} &= (1 - \alpha)^2 \bar{y}, & \bar{x} &= \bar{B} \bar{u}, \\ \bar{r} &= \mathcal{R}(\bar{B}), & \bar{g} &= n + \bar{r} - \rho, & \bar{c} &= \alpha(1 - \alpha)\bar{y} + \rho - n > 0, \\ \bar{v} &= \bar{x}/(\bar{r} \bar{B}^2), & \bar{M} &= [\bar{g} \bar{B}^{1-\eta}/\chi]^{1/\eta}, & \bar{d} &= \bar{M}/(\eta \bar{g} \bar{v}). \end{aligned} \tag{94}$$

The threshold implies $\bar{g} > n + \gamma > 0$ and $\bar{r} > \rho + \gamma > 0$, so all constants are positive. Substitution verifies the fixed point $(0, \bar{d}, \bar{c}, \bar{v}, 0)$.

The h and τ eigenvalues are $\varphi(n + \gamma - \bar{g}) < 0$ and $-(n + \gamma) < 0$. The first equals $\varphi[\rho + \gamma - \mathcal{R}(\bar{B})]$. It becomes zero at $\mathcal{R}(\bar{B}) = \rho + \gamma$, which is why this stable-manifold argument does not cover the threshold. The consumption-ratio eigenvalue is $\bar{c} > 0$. The remaining block in $(\log d, \log v)$ is

$$\begin{pmatrix} -\bar{g}\mu & -\bar{g}\mu \\ \bar{g}\mu & \bar{r} + \bar{g}\mu \end{pmatrix}, \quad \det = -\bar{g}\mu\bar{r} < 0.$$

The linearized system is triangular in the order $(h, \tau), c, (d, v)$. A stable vector with zero state components (h, d, τ) has zero h, τ paths; its consumption component must be zero because the unforced consumption equation is unstable. The research block then has zero d component and hence zero v component, as in the previous construction. Thus the three-dimensional stable subspace projects

invertibly onto (h, d, τ) . The same graph argument gives a neighborhood $\mathcal{U}_A$ of $(0, \bar{d}, 0)$ such that the initial-state condition is

$$\left[\left(\frac{A_0 N_0}{K_0} \right)^\varphi, (\bar{B} - B_0)K_0, \frac{1}{A_0 N_0} \right] \in \mathcal{U}_A. \quad (95)$$

It constructs a convergent trajectory for every admissible initial state in this neighborhood, with a unique nearby convergent choice of C_0, q_0 and positive quantities at every finite date.

The normalized vector field tends to zero, so the time derivatives of the smooth positive ratios Y/K , X/K , and C/K tend to zero. Consequently $g_Y, g_X, g_C, g_K \rightarrow \bar{g}$ and $s_X \rightarrow 1$. Log differentiation of the CES share in equation (28d) and the wage equation (28g) gives the exact identity

$$g_w = g_Y - n + \frac{\sigma - 1}{\sigma} s_X (n + \gamma - g_X).$$

Taking limits yields

$$g_w \rightarrow \bar{g} - n + \frac{\sigma - 1}{\sigma} (n + \gamma - \bar{g}) = \gamma + \frac{\bar{r} - \rho - \gamma}{\sigma}.$$

This proves the growth and distribution limits in Proposition 1(iv).

Step 4. Optimality and transversality.

In either construction let $\bar{g}$ denote the limiting aggregate growth rate: $n + \gamma$ in the positive labor-share case and $n + \bar{r} - \rho$ in the AI-dominated case. The constructions imply

$$a \rightarrow \bar{g}, \quad M \rightarrow \left[\frac{\bar{g} \bar{B}^{1-\eta}}{\chi} \right]^{1/\eta} > 0, \quad g_q \rightarrow \bar{g}, \quad g_B \rightarrow 0, \quad M/Y \rightarrow 0.$$

Both transversality expressions in Definition 1 have limiting logarithmic growth

$$-\rho + n + g_K - g_C \rightarrow n - \rho < 0, \quad -r + g_q + g_B \rightarrow \bar{g} - \bar{r} = n - \rho < 0. \quad (96)$$

They therefore converge to zero, rather than merely remaining bounded.

Household utility is finite because $\log(C/N)$ grows at most linearly and $\rho > n$, as imposed in the household problem. Log utility is concave, and the household budget is linear at given prices and distributed profits. More explicitly, let $\lambda_t = e^{-\rho t} N_t / C_t$, so $\dot{\lambda}_t = -r_t \lambda_t$ by (5). Concavity of log utility and equal initial capital give, for any admissible alternative $(\tilde{C}, \tilde{K})$,

$$\int_0^T e^{-\rho t} N_t \log \frac{\tilde{C}_t}{C_t} dt \leq \int_0^T \lambda_t (\tilde{C}_t - C_t) dt = -\lambda_T (\tilde{K}_T - K_T) \leq \lambda_T K_T \rightarrow 0.$$

Here $\tilde{K}_T \geq 0$ and the last limit is the household TVC.

For the dynamic developer, feasibility and first-order conditions still require a sufficiency argument. Let $S_t = A_t N_t$ in the positive-share construction and $S_t = K_t$ in the AI-dominated construction, and let $D_t = \exp(-\int_0^t r_s ds)$. Homogeneity of the static problem and compactness of the normalized state neighborhood give a constant C_π such that $0 \leq \pi_t(b) \leq C_\pi S_t$ for every date

and every $b \in [B_0, \bar{B}]$. Since $d \log(D_t S_t)/dt \rightarrow \bar{g} - \bar{r} = n - \rho < 0$, $\int_0^\infty D_t S_t dt < \infty$. Thus discounted operating profit is uniformly bounded above across all alternative research policies. The candidate's present value of net profit is finite, because its research compute is bounded and $r_t \rightarrow \bar{r} > 0$. An alternative with infinite discounted research expenditure has payoff $-\infty$ and cannot improve on it.

It remains to verify the curvature conditions of Lemma 3, (75) and (76) for every date and every alternative AI-efficiency level in $[B_0, \bar{B}]$, not just on the selected path. First fix a compact normalized-state neighborhood and an interval of alternative AI-efficiency levels $[b_*, \bar{B}]$, with $0 < b_* < \bar{B}$. In the positive labor-share construction, $K/(AN)$ stays in a compact neighborhood of $\bar{k} > 0$. Smoothness of the static monopoly solution bounds $\varepsilon_{B,t}(b)$ uniformly for these states and for AI-efficiency levels near $\bar{B}$. In the AI-dominated construction, h stays near zero, and the same uniform bound follows from the smooth static limit $\varepsilon_{B,t}(b) \rightarrow 1/\alpha$. Thus in either case there is a finite constant E with $\varepsilon_{B,t}(b) - 2 \leq E$ throughout that state domain. This bound is chosen before shrinking the initial-state neighborhood. Because $\mu = \eta/(1 - \eta)$ is finite for each $0 < \eta < 1$, choose $b_{**} \in [b_*, \bar{B})$ sufficiently close to $\bar{B}$ that

$$(1 - b_{**}/\bar{B}) \max\{E, \mu - 1, 0\} < b_{**}/\bar{B}.$$

Shrink the open neighborhood $\mathcal{U}_L$ or $\mathcal{U}_A$ so that $B_0 > b_{**}$. This is possible because $B_0 = \bar{B} - d_0 \tau_0$ in the first construction and $B_0 = \bar{B} - d_0 \tau_0 h_0^{1/\varphi}$ in the second. Every feasible research policy has $b \geq B_0$, so the two curvature conditions hold throughout the entire alternative-state domain at every date. Lemma 3 therefore gives the support inequality (70). Lemma 2 and the verified TVC then establish global developer optimality without imposing $\eta \leq 1/2$. The curvature conditions follow from the primitives and the chosen initial-state neighborhood.

Final-good optimality follows from concavity and constant returns. The pointwise monopoly optimum is global by Lemma 1. All quantities remain finite at every finite date, and market clearing and the original initial conditions hold by reconstruction. The paths are therefore equilibria, not merely solutions of the first-order conditions and state equations.

The result is local only in its set of initial stocks. For $\sigma > 1$, at $\bar{B} = \mathcal{B}(\sigma)$ the positive-labor-share limiting point reaches the boundary and a stable eigenvalue in the boundary normalization becomes zero; this proof does not apply. For $\sigma < 1$ and $\bar{B} \leq \mathcal{B}(\sigma)$, it does not rule out equilibria with different long-run ratios. Nor does it imply existence at arbitrary initial stocks or in the uncapped limit.

Step 5. Initial-state sets and the four conclusions.

The preimage of each open neighborhood under its continuous initial-state map, restricted to $A_0, N_0, K_0 > 0$ and $0 < B_0 < \bar{B}$, is open in the original initial stocks. It is nonempty. In the positive-labor-share cases, choose k_0, d_0 close to $\bar{k}, \bar{d}$ and $\tau_0 > 0$ sufficiently small. Then $A_0 N_0 = 1/\tau_0$, $K_0 = k_0/\tau_0$, and $B_0 = \bar{B} - d_0 \tau_0$ are admissible. In the AI-dominated case, choose d_0 near $\bar{d}$ and $h_0, \tau_0 > 0$ sufficiently small. The inverse relations are $A_0 N_0 = 1/\tau_0$, $K_0 = 1/(\tau_0 h_0^{1/\varphi})$, and $B_0 = \bar{B} - d_0 \tau_0 h_0^{1/\varphi}$. Any positive factorization of $A_0 N_0$ supplies A_0, N_0 . These choices also

satisfy the strict efficiency restriction in Step 4.

For each initial state in the resulting set, the local graph selects $C_0, q_0 > 0$. These jump values are unique among the trajectories that remain near the indicated normalized limit and converge to it. The argument does not assert global uniqueness.

For $\sigma < 1$ with $\bar{B} > \mathcal{B}(\sigma)$, and for $\sigma > 1$ with $\bar{B} < \mathcal{B}(\sigma)$, Step 1 gives a unique interior $\bar{s}_X$. Step 2 then yields $g_Y - n \rightarrow \gamma$, $g_w \rightarrow \gamma$, and $r \rightarrow \rho + \gamma$. At $\sigma = 1$, the exact CES limit gives $s_X = \omega_X$ at every date, and the same construction works for every finite positive upper bound. These establish the first three cases of the proposition. Step 3 gives $s_X \rightarrow 1$, $r \rightarrow \mathcal{R}(\bar{B})$, $g_Y - n \rightarrow \mathcal{R}(\bar{B}) - \rho$, and the wage-growth limit in (41), establishing the fourth case. Step 4 verifies the agent-optimality and transversality requirements of Definition 1 for all these trajectories. $\square$

Proof of Proposition 2. Free entry and the common unit cost $1/B$ imply $p_X = 1/B$. Because improvements are immediately available to every supplier and research receives no separate payment, a supplier's net profit is $-M$. Its optimal research expenditure is zero. The RSI technology (17) then gives $B = B_0$, establishing equation (43). The remaining argument constructs and verifies the allocation at this fixed efficiency.

Net production and normalized dynamics. Write $k = K/(AL)$, $c = C/(AL)$, $u = U/(AL)$, and $y = Y/(AL)$. At $p_X = 1/B_0$, final-good firms' demand condition (13) implies $\partial y/\partial u = 1$. Their choice of AI services therefore maximizes output net of compute at a given k . For the proof, denote this net output per unit of effective labor by

$$f(k) = \max_{u \geq 0} \left\{ k^\alpha \left[\omega_L + \omega_X (B_0 u)^{(\sigma-1)/\sigma} \right]^{\sigma(1-\alpha)/(\sigma-1)} - u \right\}. \quad (97)$$

For $k > 0$, the marginal product of compute, $\partial y/\partial u$, decreases strictly from infinity to zero as u rises from zero to infinity. The derivative of net output is $\partial y/\partial u - 1$, so there is a unique positive maximizer. Its first-order condition is $u = (1 - \alpha)s_X y$. Differentiating this condition, using (15), gives

$$\frac{d \log u}{d \log k} = \frac{\alpha}{e_X}, \quad f'(k) = \alpha \frac{y}{k}, \quad \frac{k f''(k)}{f'(k)} = -\frac{(1 - \alpha)(1 - s_X)}{\sigma e_X} < 0.$$

Thus f is smooth, increasing, and strictly concave, with $f(0) = 0$ and $f'(0+) = \infty$. As $k \rightarrow \infty$, the optimal u tends to infinity and $s_X \rightarrow 1$. The pricing condition gives $X/Y \rightarrow B_0(1 - \alpha)$, while (33) gives $Z/X \rightarrow \omega_X^{\sigma/(\sigma-1)}$. It follows that

$$\lim_{k \rightarrow \infty} f'(k) = \lim_{k \rightarrow \infty} \frac{f(k)}{k} = \mathcal{R}^{\text{mc}}(B_0) + \delta, \quad f(k) - k f'(k) = (1 - \alpha)(1 - s_X)y > 0. \quad (98)$$

This also derives equation (42).

Set $d = \rho - n > 0$ and $G(k) = f(k) - (\delta + n + \gamma)k$. The household's objective, up to a positive factor and an additive constant, is $\int_0^\infty e^{-dt} \log c_t dt$. The resource constraint (26) with $M = 0$, the

Euler equation (5), and the transversality condition (6) become

$$\dot{k} = G(k) - c, \quad \frac{\dot{c}}{c} = G'(k) - d, \quad \lim_{t \rightarrow \infty} e^{-dt} \frac{k_t}{c_t} = 0. \quad (99)$$

An identity useful for constructing the trajectory is

$$\frac{d}{dt} \log \frac{c}{k} = \frac{c}{k} - d - \frac{f(k) - kf'(k)}{k}. \quad (100)$$

Efficiency below the threshold. If $\mathcal{R}^{\text{mc}}(B_0) < \rho + \gamma$, there is a unique $\bar{k} > 0$ satisfying $G'(\bar{k}) = d$. The stationary consumption level is $\bar{c} = G(\bar{k}) > d\bar{k} > 0$, by (98). The Jacobian at this point is

$$\begin{pmatrix} d & -1 \\ \bar{c}f''(\bar{k}) & 0 \end{pmatrix},$$

whose determinant is negative. Its stable curve has positive slope, with $\dot{k}, \dot{c} > 0$ on its left branch and $\dot{k}, \dot{c} < 0$ on its right branch.

Together, the two branches cover every $k > 0$. On the left branch, $dk < c < G(k)$; on the right, $c > \max\{dk, G(k)\}$. To verify these restrictions, at $c = dk$ equation (100) points toward $c/k < d$, from which the path cannot return to $\bar{c}/\bar{k} > d$. At $c = G(k)$, the derivative of $\dot{k} = G(k) - c$ is $-\dot{c}$. Below $\bar{k}$, crossing this curve makes capital decrease while consumption increases, precluding convergence to $\bar{k}$. Above $\bar{k}$, it makes capital increase while consumption decreases, again precluding convergence. The stable branches cannot make either crossing.

Continuing the left branch backward therefore takes k to zero: otherwise its bounded, monotone coordinates would approach a positive stationary point different from $(\bar{k}, \bar{c})$, which does not exist. The right branch extends to arbitrarily large k . It cannot approach another finite stationary point or diverge in consumption at a finite positive k : along this branch $dc/dk = c(G'(k) - d)/(G(k) - c)$ remains bounded when c becomes large on a compact interval of k . For each $k_0 > 0$, the stable curve consequently supplies a positive trajectory converging to $(\bar{k}, \bar{c})$ and satisfying the transversality condition in (99).

Efficiency at or above the threshold. If $\mathcal{R}^{\text{mc}}(B_0) \geq \rho + \gamma$, strict concavity implies $G'(k) > d$ and $G(k) > dk$ at every finite $k > 0$. For any $k_0 > 0$, consider initial consumption in $[dk_0, G(k_0)]$ and the region

$$k > 0, \quad dk < c < G(k).$$

At its lower consumption boundary, (100) points strictly out of the region. At its upper boundary, $\dot{k} = 0$ and $\dot{c} > 0$, so the vector field also points strictly out. Inside, both k and c increase. There is no finite-time escape to infinity because concavity bounds $G(k)$ above by an affine function. Continuous dependence on initial consumption makes the sets of initial choices whose paths first exit through the two boundaries open and disjoint. Both sets are nonempty, as shown by the endpoint initial choices. They cannot exhaust the connected interval of initial choices. Hence at least one trajectory remains in the region forever.

Along this trajectory $k \rightarrow \infty$. A finite limit would also bound c and require a positive stationary point, but $G'(k) > d$ excludes one. Moreover, $d < c/k < G(k)/k$, and the upper bound satisfies

$$\frac{G(k)}{k} \rightarrow \mathcal{R}^{\text{mc}}(B_0) - n - \gamma.$$

If $\mathcal{R}^{\text{mc}}(B_0) = \rho + \gamma$, these bounds imply $c/k \rightarrow d$. If the inequality is strict, the same limit follows from (100): $(f - kf')/k \rightarrow 0$, and any persistent excess of c/k over d would make the ratio increase until it violated its bounded upper limit. More precisely, for any $\varepsilon > 0$ and sufficiently large t , reaching $c/k \geq d + \varepsilon$ implies $d \log(c/k)/dt \geq \varepsilon/2$ thereafter until that contradiction occurs. Thus

$$\frac{c}{k} \rightarrow d, \quad \frac{\dot{k}}{k} \rightarrow \mathcal{R}^{\text{mc}}(B_0) - \rho - \gamma. \quad (101)$$

The transversality condition holds because $c/k > d$. At equality, capital per effective worker diverges but its growth rate tends to zero.

Optimality, uniqueness, and decentralization. The preceding constructions satisfy global optimality, not only the Euler equation. Let $\lambda_t = e^{-dt}/c_t$ along a constructed path; then $\dot{\lambda} = -G'(k)\lambda$. For any alternative feasible path $(\tilde{k}, \tilde{c})$ with the same k_0 , concavity gives

$$\int_0^T e^{-dt} (\log \tilde{c} - \log c) dt \leq -\lambda_T (\tilde{k}_T - k_T) \leq \lambda_T k_T \rightarrow 0.$$

The constructed path has finite utility: in the first case c converges to a positive constant; in the other cases it is positive, increasing, and at most exponentially growing. Thus the inequality verifies optimality in the fixed- B_0 allocation problem. Strict concavity of log utility and concavity of the resource constraint imply uniqueness of consumption; the state equation then gives uniqueness of capital. This verification does not allow a planner to improve B : it concerns the allocation with the technology fixed by the private research choices.

The maximizing u in (97) determines $U = (AL)u$. This compute use, the factor prices in (11) and (12), and $p_X = 1/B_0$ decentralize this allocation. In particular, $wN = (AL)[f(k) - kf'(k)]$, so the household budget coincides with the resource constraint. Its Euler and transversality conditions verify its optimum at these prices. Strict concavity also implies that any other competitive equilibrium would solve the same fixed-technology allocation problem and hence have the same aggregate allocation.

Growth rates and income shares. Below the threshold, convergence to $(\bar{k}, \bar{c})$ gives $r \rightarrow \rho + \gamma$, $g_{Y/N}, g_w \rightarrow \gamma$, and a positive limiting labor share, because optimal u and s_X have finite interior limits. At or above the threshold, $k, u \rightarrow \infty$ and $s_X \rightarrow 1$. Differentiating the static allocation gives

$$\frac{d \log y}{d \log k} = 1 - \frac{(1 - \alpha)(1 - s_X)}{\sigma e_X} \rightarrow 1, \quad \frac{d \log(w/A)}{d \log k} = \frac{\alpha}{\sigma e_X} \rightarrow \frac{1}{\sigma}.$$

Combining these elasticities with (101) proves the growth limits in part (ii). Equation (98) gives the interest-rate limit, and $wL/Y = (1 - \alpha)(1 - s_X) \rightarrow 0$ gives the labor-share limit.

At the boundary $B_0 = \mathcal{B}^{\text{mc}}(\sigma)$, the same calculations give $g_{Y/N}, g_w \rightarrow \gamma$, $r \rightarrow \rho + \gamma$, and $wL/Y \rightarrow 0$. The labor share therefore vanishes without a permanent growth premium. $\square$

Proof of Proposition 3. Fix $B > 0$ and $\sigma > 1$. Combining marginal-cost pricing, $p_X = 1/B$, with the final-good firm's demand in equation (13) gives

$$\frac{X}{Y} = B(1 - \alpha)s_X \longrightarrow B(1 - \alpha) \quad \text{as } s_X \rightarrow 1. \quad (102)$$

The CES limit in equation (33) is unchanged. Substituting that limit and equation (102) into $Y/K = [(Z/X)(X/Y)]^{(1-\alpha)/\alpha}$, from equation (34), yields

$$\frac{Y}{K} \longrightarrow \left[(1 - \alpha)B \omega_X^{\sigma/(\sigma-1)} \right]^{(1-\alpha)/\alpha}.$$

Equation (11) then gives $\mathcal{R}^{\text{mc}}(B)$ in equation (42). It exceeds the monopoly return in equation (35), since $0 < 1 - \alpha < 1$.

The marginal-cost threshold $\mathcal{B}^{\text{mc}}(\sigma)$ is defined by $\mathcal{R}^{\text{mc}}(\mathcal{B}^{\text{mc}}(\sigma)) = \rho + \gamma$. Both returns are strictly increasing in B , and their formulas imply $\mathcal{R}^{\text{mc}}(B) = \mathcal{R}(B/(1 - \alpha))$. Thus equation (36) gives

$$\mathcal{R}^{\text{mc}}((1 - \alpha)\mathcal{B}(\sigma)) = \mathcal{R}(\mathcal{B}(\sigma)) = \rho + \gamma.$$

Uniqueness of the threshold establishes equation (45). $\square$

Proof of Corollary 1. Proposition 3 ensures that the interval in equation (46) is nonempty. Fix $\bar{B}$ in this interval. The labor-bottleneck construction of Proposition 1(iii) uses initial values satisfying equation (91), with $B_0 = \bar{B} - d_0\tau_0$, d_0 near $\bar{d} > 0$, and $\tau_0 = 1/(A_0N_0) > 0$ arbitrarily small. Since $\bar{B} > \mathcal{B}^{\text{mc}}(\sigma)$, I can restrict this construction to $B_0 > \mathcal{B}^{\text{mc}}(\sigma)$ while preserving a nonempty open set of initial states.

From each state in this set, the constructed monopoly equilibrium has $B \rightarrow \bar{B}$ and $g_{Y/N}, g_w \rightarrow \gamma$. Proposition 2 supplies a competitive equilibrium from the same state. Since $B_0 > \mathcal{B}^{\text{mc}}(\sigma)$, this equilibrium is AI-dominated, with $B = B_0$ and both limiting growth rates strictly above γ . This proves the comparison without requiring monopoly equilibrium existence from arbitrary initial states or global uniqueness under monopoly.

The accompanying comparison of AI services follows from the CES share identity in equation (82) and $L = N$. Under monopoly, s_X converges to an interior value, so $X/(AL)$ converges to a positive constant. Under competition, $s_X \rightarrow 1$; since $\sigma > 1$, the same identity gives $X/(AL) \rightarrow \infty$. The common paths of A and L therefore imply that AI services eventually exceed their monopoly level under competition, without establishing their ordering at every date. $\square$

Proof of Corollary 2. Equations (35) and (42) imply

$$\mathcal{R}(\bar{B}) = \mathcal{R}^{\text{mc}}((1 - \alpha)\bar{B}).$$

Since $\mathcal{R}^{\text{mc}}$ is strictly increasing, it follows that

$$\mathcal{R}(\bar{B}) \geq \mathcal{R}^{\text{mc}}(B_0) \iff (1 - \alpha)\bar{B} \geq B_0,$$

with equality on one side if and only if equality holds on the other. Under the two AI-dominated equilibria, these returns are the limiting interest rates. Equations (40), (41), and (44) express limiting output-per-worker and wage growth as strictly increasing functions of the limiting interest rate, with slopes one and $1/\sigma$, respectively. Since the parameters are common to both economies, the same ordering applies to both growth rates, including the equality case. $\square$

Proof of Proposition 4. Equations (49) and (50) follow because $\sigma/(\sigma - 1) < 0$. Define $k = K/(AL)$. The resource constraint (29a), nonnegative consumption and compute expenditure, and $L = N$ imply

$$\dot{k} \leq \omega_L^{\sigma(1-\alpha)/(\sigma-1)} k^\alpha - (\delta + n + \gamma)k. \quad (103)$$

Since $\delta + n + \gamma > 0$ and $0 < \alpha < 1$, the right-hand side is negative above a finite positive level. In particular,

$$k_t \leq \max \left\{ k_0, \left[\frac{\omega_L^{\sigma(1-\alpha)/(\sigma-1)}}{\delta + n + \gamma} \right]^{1/(1-\alpha)} \right\}.$$

Equation (50) then bounds $Y/(AL)$ uniformly. Since $A_t = A_0 e^{\gamma t}$ and $L = N$, Y/N is bounded above by a constant times $e^{\gamma t}$. Taking logarithms, dividing by t , and taking the upper limit gives (51). $\square$

Proof of Proposition 5. All auxiliary variables below are confined to this proof. Write $x = X/(AL)$, $k = K/(AL)$, $c = C/(AL)$, $y = Y/(AL)$, $u = U/(AL)$, $m = M/(AL)$, $d = \rho - n > 0$, and $\nu = \delta + n + \gamma$. No convergence of a growth rate or allocation ratio is assumed.

Step 1. Static optimality gives a limiting production function. Put $\varphi = (\sigma - 1)/\sigma < 0$ and $f(x) = [\omega_L + \omega_X x^\varphi]^{(1-\alpha)/\varphi}$, so $y = k^\alpha f(x)$. The monopoly condition (19) requires $e_X = (1 - s_X)/\sigma + \alpha s_X < 1$. Thus

$$s_X > s_\infty := \frac{1 - \sigma}{1 - \alpha\sigma}, \quad 0 < x < x_\infty := \left[\frac{\omega_L s_\infty}{\omega_X (1 - s_\infty)} \right]^{1/\varphi}. \quad (104)$$

Define $f_\infty = f(x_\infty) > 0$. In particular, $y \leq f_\infty k^\alpha$ and $0 < u \leq x_\infty/B$.

The monopoly condition can be written

$$k^\alpha v(x) = 1/B, \quad v(x) = \frac{(1 - \alpha)s_X f(x)(1 - e_X)}{x}.$$

On $(0, x_\infty)$, v is strictly decreasing, from infinity to zero. Indeed, with a prime here denoting differentiation with respect to $\log x$, $e'_X = (\alpha - \sigma^{-1})\varphi s_X(1 - s_X) > 0$ and $-\mathrm{d} \log v / \mathrm{d} \log x = e_X + e'_X / (1 - e_X) > \alpha s_X$. Consequently x increases with B and k , while

$$\frac{\partial \log y}{\partial \log k} = \alpha + \frac{\alpha(1 - \alpha)s_X}{e_X + e'_X / (1 - e_X)} < 1.$$

Thus y/k decreases with k and increases with B . For k in any compact interval bounded away from zero, as $B \rightarrow \infty$,

$$x \rightarrow x_\infty, \quad y \rightarrow f_\infty k^\alpha, \quad u \rightarrow 0$$

uniformly. These are properties of the monopoly allocation at given k, B , not assumptions about the equilibrium path. They also imply that, for any prescribed finite return, sufficiently small k and sufficiently large B make r exceed that return, uniformly over those small values of k .

Step 2. Optimality bounds consumption per unit of effective labor. Proposition 4 bounds k and y above by finite constants $k_{\max}, y_{\max}$. The household Euler equation and $r \geq -\delta$ give $\dot{c}/c \geq -(\delta + \rho + \gamma)$. Integrating the resource constraint over $[t, t + 1]$ therefore gives

$$c_t \int_0^1 e^{-(\delta + \rho + \gamma)v} \mathrm{d}v \leq \int_t^{t+1} c_s \mathrm{d}s \leq k_{\max} + y_{\max}.$$

Hence c has a finite uniform upper bound $c_{\max}$.

Write $D_{t,s} = \exp(-\int_t^s r_v \mathrm{d}v)$. Euler implies $\int_t^\infty D_{t,s} C_s \mathrm{d}s = C_t/d$. Integrating the household budget (4) and using its TVC (6) shows that the discounted integral of labor income plus developer net profit converges. Since labor income is nonnegative, the discounted integral of developer net profit has a limit that is either finite or minus infinity. The latter is impossible: the developer can stop research and obtain nonnegative operating profits. By continuation optimality, the developer's continuation value $V_t = \int_t^\infty D_{t,s} \Pi_s \mathrm{d}s$ is therefore finite and nonnegative. It follows that

$$\frac{C_t}{d} = K_t + V_t + \int_t^\infty D_{t,s} w_s L_s \mathrm{d}s, \quad c_t \geq dk_t, \quad \int_t^\infty D_{t,s} w_s L_s \mathrm{d}s \leq \frac{C_t}{d}. \quad (105)$$

Here V_t is an auxiliary continuation value, not an additional traded asset or a change to the household budget.

Step 3. Discounted research spending vanishes relative to effective labor. Fix a positive rate a . Step 1 permits a small $k_a > 0$ and a sufficiently late date such that $r - (n + \gamma) \geq a$ whenever $k \leq k_a$. On $[k_a, k_{\max}]$, the wage equation (12) and the static monopoly allocation give a uniform positive lower bound ℓ_a on w/A . Since $r \geq -\delta$, there is a finite constant $H_a > 0$ such that

$$a \leq r - (n + \gamma) + H_a w/A$$

at every sufficiently late date. Multiplication by $D_{t,s}A_sL_s$, integration, and (105) give

$$\int_t^\infty D_{t,s}A_sL_s \, ds \leq \frac{A_tL_t + H_aC_t/d}{a} \leq JA_tL_t \quad (106)$$

for a fixed finite J . The omitted terminal contribution from discounted effective labor is nonpositive; no discount-rate limit is used.

The research condition (22) and costate equation (23) imply

$$\frac{d(qB)}{dt} = rqB - U + \frac{1-\eta}{\eta}M. \quad (107)$$

Since B is nondecreasing, $U_s \leq x_\infty A_s L_s / B_t$ for $s \geq t$. Integrating (107) first over finite intervals and then using the developer TVC (25) gives

$$q_t B_t + \frac{1-\eta}{\eta} \int_t^\infty D_{t,s} M_s \, ds = \int_t^\infty D_{t,s} U_s \, ds \leq \frac{x_\infty J}{B_t} A_t L_t. \quad (108)$$

The right side is finite by (106); nonnegativity then ensures that the research integral is finite as well. Its value divided by $A_t L_t$ tends to zero. This argument requires $0 < \eta < 1$, but neither $\eta \leq 1/2$ nor $\eta < \alpha$.

Step 4. Capital per unit of effective labor stays away from zero. Let $F(k) = f_\infty k^\alpha - \nu k$. Choose $\varepsilon > 0$ small enough that F is positive and increasing on $(0, \varepsilon]$. Step 1 also permits a sufficiently late date and $a > 0$ such that $\dot{c}/c = r - \rho - \gamma \geq a$ whenever $k \leq \varepsilon$. For any $j \leq \varepsilon$, the equilibrium path cannot enter $\{k \leq j, c \geq F(j)\}$: there $\dot{k} \leq F(k) - c \leq 0$, whereas c continues to grow at least at rate a , contradicting its upper bound.

If k stayed below ε forever, consumption would likewise exceed its upper bound. Hence the path eventually reaches ε . Suppose arbitrarily late crossings from ε to $\varepsilon/2$ were possible. Let t_0 be the last crossing of ε before the first subsequent crossing t_1 of $\varepsilon/2$. Throughout this interval $k \in [\varepsilon/2, \varepsilon]$ and c increases. Equation (105) gives $c_{t_0} \geq d\varepsilon$, so

$$t_1 - t_0 \leq T_\varepsilon := a^{-1} \log\left(\frac{c_{\max}}{d\varepsilon}\right) < \infty.$$

At t_1 , the preceding invariant-region argument requires $c_{t_1} < F(\varepsilon/2)$. Thus $c_s < F(\varepsilon/2)$ throughout the crossing. The uniform convergence in Step 1, applied on the crossing interval, yields numbers $\epsilon_{t_0} \rightarrow 0$ such that

$$\dot{k}_s = y_s - u_s - \nu k_s - c_s - m_s \geq -\epsilon_{t_0} - m_s.$$

On this interval r is bounded above by a fixed positive R_ε . Since $n + \gamma > 0$, equation (108) then implies

$$\frac{\varepsilon}{2} \leq T_\varepsilon \epsilon_{t_0} + \int_{t_0}^{t_1} m_s \, ds \leq T_\varepsilon \epsilon_{t_0} + e^{R_\varepsilon T_\varepsilon} \frac{\eta x_\infty J}{(1-\eta)B_{t_0}} \rightarrow 0,$$

a contradiction. Therefore k is eventually bounded away from zero, and so is $c \geq dk$. In particular,

r is now bounded above.

Step 5. Research per unit of effective labor vanishes and Ramsey dynamics determine the limit. Differentiating the research condition and using the costate equation gives

$$(1 - \eta)g_M = r - \frac{\eta\chi UB^{\eta-1}}{M^{1-\eta}} \leq r. \quad (109)$$

Choose a positive R above all eventual values of r and put $H = R/(1 - \eta)$. Then $M_s \geq M_t e^{-H(t-s)}$ on $[t - 1, t]$ at sufficiently late dates. Consequently

$$\frac{1}{A_{t-1}L_{t-1}} \int_{t-1}^t D_{t-1,s} M_s \, ds \geq e^{n+\gamma-R-H} m_t.$$

The left side tends to zero by (108). Hence $m \rightarrow 0$, and Step 1 already gives $u \rightarrow 0$.

The normalized equilibrium equations therefore approach, uniformly on the compact positive region containing the path, the Ramsey system

$$\dot{k} = F(k) - c, \quad \dot{c} = c[\alpha f_\infty k^{\alpha-1} - \delta - \rho - \gamma]. \quad (110)$$

For clarity, the following argument verifies convergence rather than inferring it solely from convergence of the equations. This limiting system has one positive stationary point,

$$k_\infty = \left[\frac{\alpha f_\infty}{\rho + \gamma + \delta} \right]^{1/(1-\alpha)}, \quad c_\infty = F(k_\infty) > 0.$$

Its Jacobian has negative determinant, so it is a saddle. The standard Ramsey phase portrait is discussed by [Groth \(2013\)](#). There is no nonconstant complete orbit confined to a compact subset of the positive quadrant. To see this, the multiplier $1/c^2$ gives

$$\frac{\partial}{\partial k} \frac{F(k) - c}{c^2} + \frac{\partial}{\partial c} \frac{\alpha f_\infty k^{\alpha-1} - \delta - \rho - \gamma}{c} = \frac{\rho - n}{c^2} > 0.$$

The Bendixson–Dulac argument excludes both periodic orbits and homoclinic loops. The planar limit-set theorem, the unique stationary point, and its saddle character then exclude a nonconstant complete orbit confined to such a compact set: its forward and backward limits would otherwise require a periodic orbit or a connection back to that same saddle.

For any sequence of dates tending to infinity, time translations of our bounded positive path have a subsequence converging on every bounded time interval to a complete orbit of (110). This follows from bounded derivatives, uniform convergence of the vector fields, and the integral form of the differential equations. The preceding paragraph forces that orbit to be the stationary point. Thus every accumulation point is (k_∞, c_∞) , proving convergence of the original path.

It follows that $y \rightarrow y_\infty = f_\infty k_\infty^\alpha > 0$ and

$$r \rightarrow \rho + \gamma, \quad \frac{K}{Y} \rightarrow \frac{\alpha}{\rho + \gamma + \delta}, \quad \frac{C}{Y} \rightarrow 1 - \frac{\alpha(\delta + n + \gamma)}{\rho + \gamma + \delta} > 0. \quad (111)$$

Positivity follows because the numerator of the final expression is $\rho - n + (1 - \alpha)(\delta + n + \gamma) > 0$. This proves (52). Since $u, m \rightarrow 0$ and $y \rightarrow y_\infty > 0$, it follows that $U/Y = u/y \rightarrow 0$ and $M/Y = m/y \rightarrow 0$, proving (53). It also gives $\dot{k} \rightarrow 0$ and $g_C \rightarrow n + \gamma$.

Step 6. Output and wage growth converge as well. A convergent level need not have a convergent derivative. The bound on g_M from Step 5 and the RSI technology give

$$B_t^{1-\eta} \geq (1 - \eta)\chi M_t^\eta \int_0^1 e^{-\eta H v} dv.$$

Thus $g_B = \chi M^\eta / B^{1-\eta}$ is bounded above. At x_∞ , the marginal-revenue function v defined in Step 1 has a simple zero: $v(x_\infty) = 0$ and $v'(x_\infty) < 0$, where the prime now denotes differentiation with respect to x . Differentiating $k^\alpha v(x) = 1/B$ yields

$$\dot{x} = -\frac{v(x)}{v'(x)} \left(g_B + \alpha \frac{\dot{k}}{k} \right) \rightarrow 0.$$

Hence $\dot{y} \rightarrow 0$ and $\dot{s}_X \rightarrow 0$. Since $y_\infty > 0$ and $1 - s_\infty > 0$, $g_{Y/N} \rightarrow \gamma$ and, by (12), $g_w \rightarrow \gamma$. The limiting income shares follow by substituting s_∞ into (12) and (13). The result is conditional on equilibrium existence and $B \rightarrow \infty$, not an equilibrium-existence theorem or an assumption of balanced growth. $\square$

Proof of Proposition 6. I use the reference BGP rates and allocation ratios derived in Appendix C, denoted by a superscript $*$. Here $k^* = K/Y$ is a capital-output ratio, rather than capital per unit of effective labor. I first verify the reference BGP and then establish convergence from nearby initial stocks.

The reference BGP. First, all candidate ratios are positive. The denominator of the research share m^* , $\rho - n + \eta g_Y^*$, is positive. Moreover,

$$i^* = \frac{\alpha(g_Y^* + \delta)}{\rho - n + g_Y^* + \delta} < \alpha, \quad m^* < \frac{u^* \eta}{1 - \eta}.$$

Since $\Delta = (1 - \alpha)(1 - \eta - \omega_X) > 0$, $\omega_X < 1 - \eta$ and $\omega_X < 1$. Consequently,

$$u^* + m^* < \frac{u^*}{1 - \eta} < (1 - \alpha)\omega_X < 1 - \alpha, \quad c^* = 1 - u^* - i^* - m^* > 0.$$

Thus positive consumption follows from the stated conditions; it is not an additional assumption.

For given $A_0, N_0 > 0$, choose the reference initial stocks as

$$B_0^* = \left[A_0 N_0 (k^*)^{\alpha/[(1-\alpha)\omega_L]} (u^*)^{\omega_X/\omega_L} m^* \left(\frac{\chi}{g_B^*} \right)^{1/\eta} \right]^{\eta\omega_L/(1-\eta-\omega_X)}, \quad (112)$$

$$Y_0^* = A_0 N_0 (k^*)^{\alpha/[(1-\alpha)\omega_L]} (u^* B_0^*)^{\omega_X/\omega_L}, \quad K_0^* = k^* Y_0^*. \quad (113)$$

The exponent is positive. These choices solve simultaneously the initial production equation and $g_B^* B_0^* = \chi(B_0^* m^* Y_0^*)^\eta$. Set $U_0^* = u^* Y_0^*$, $M_0^* = m^* Y_0^*$, $C_0^* = c^* Y_0^*$, $X_0^* = B_0^* U_0^*$, and

$$q_0^* = \frac{M_0^*}{\eta g_B^* B_0^*}. \quad (114)$$

Let $Y_t^* = Y_0^* e^{g_Y^* t}$, $B_t^* = B_0^* e^{g_B^* t}$, $K_t^* = k^* Y_t^*$, $C_t^* = c^* Y_t^*$, $U_t^* = u^* Y_t^*$, $M_t^* = m^* Y_t^*$, and $q_t^* = q_0^* e^{(g_Y^* - g_B^*)t}$. Use the exogenous A_t, N_t and set $L_t = N_t$ and $X_t^* = B_t^* U_t^*$. The static equilibrium equations determine all remaining quantities and prices.

The two growth identities (58) and (59) ensure that the production equation and the RSI technology hold at every date once they hold initially. The static monopoly condition gives $p_X X = (1 - \alpha)\omega_X Y$ and $U = u^* Y$. Equation (114) gives $M = \eta q \dot{B}$, which is the research first-order condition (28k) with $\psi = 1$. The definitions of k^* , r^* , and c^* verify the capital-return, Euler, and resource equations. The costate equation reduces to

$$g_Y^* - g_B^* = r^* - \frac{u^* \eta g_B^*}{m^*} - \eta g_B^*.$$

This holds by equation (143) and $(1 - \eta)g_B^* = \eta g_Y^*$. All allocations are positive and finite at every finite date.

Both transversality conditions are explicit:

$$e^{-\rho t} \frac{N_t K_t^*}{C_t^*} = \frac{N_0 k^*}{c^*} e^{-(\rho-n)t} \longrightarrow 0, \quad (115)$$

$$e^{-r^* t} q_t^* B_t^* = q_0^* B_0^* e^{-(\rho-n)t} \longrightarrow 0. \quad (116)$$

Discounted developer profit is a constant multiple of $e^{-(\rho-n)t}$. Discounted household utility is $N_0 e^{-(\rho-n)t} [\log(C_0^*/N_0) + (g_Y^* - n)t]$. Both improper objective integrals are finite.

It remains to establish global optimality, rather than only satisfaction of the necessary conditions. At given paths of K, A, L , static optimization gives

$$\begin{aligned} \pi_t(B) &= (1 - \alpha)\omega_X [1 - (1 - \alpha)\omega_X] [K_t^\alpha (A_t L_t)^{(1-\alpha)\omega_L}]^{1/[1-(1-\alpha)\omega_X]} \\ &\quad \times [(1 - \alpha)^2 \omega_X^2 B]^{(1-\alpha)\omega_X/[1-(1-\alpha)\omega_X]}. \end{aligned} \quad (117)$$

To verify developer optimality by concavity, define the invertible state transformation and its

current-value costate by

$$\nu = \frac{1-\eta}{\eta}, \quad S = B^\nu, \quad p = \frac{q}{\nu B^{\nu-1}}. \quad (118)$$

The RSI technology becomes

$$\dot{S} = \nu \chi S^{1-\eta} M^\eta,$$

which is jointly concave in S, M for every $0 < \eta < 1$. At given K, A, L , equation (117) is a positive time-dependent coefficient times

$$S^{\frac{(1-\alpha)\omega_X}{\nu[1-(1-\alpha)\omega_X]}}.$$

Since $\eta + \omega_X < 1$ implies $(1-\alpha)\omega_X < 1-\eta$, this exponent lies strictly between zero and one. Operating profit is therefore concave in S . With $p > 0$, the transformed Hamiltonian $\pi_t(S^{1/\nu}) - M + p_t \nu \chi S^{1-\eta} M^\eta$ is jointly concave. The research condition (22) and costate equation (23), with $\psi = 1$, give its control first-order condition and costate equation by the chain rule.

Write $D_t = \exp(-\int_0^t r_s ds)$ and $J_T = \int_0^T D_t [\pi_t(B_t) - M_t] dt$. For any admissible alternative with the same initial B , concavity, the research condition, and the costate equation give, after integration by parts,

$$\tilde{J}_T - J_T \leq -D_T p_T (\tilde{S}_T - S_T) \leq D_T p_T S_T = \frac{D_T q_T B_T}{\nu} \longrightarrow 0. \quad (119)$$

The last limit follows from the developer TVC. Since the transformation preserves feasible research paths and their payoffs, no admissible alternative has a greater limiting payoff, or a greater upper limit of truncated payoff, than this finite-valued candidate. The same verification applies to nearby candidates satisfying the uncapped equations and the developer TVC; it does not require exact balanced growth.

For the household, set $\Lambda_t = e^{-\rho t} N_t / C_t$. Euler implies $\dot{\Lambda} = -r\Lambda$. Concavity of log utility and the linear budget give, for any feasible alternative with the same K_0 and the same given prices and developer net profit,

$$\int_0^T e^{-\rho t} N_t \left[\log(\tilde{C}_t / N_t) - \log(C_t / N_t) \right] dt \leq -\Lambda_T (\tilde{K}_T - K_T) \leq \Lambda_T K_T \longrightarrow 0.$$

This establishes global household optimality.

Competitive final-good firms face a concave constant-returns technology, and Lemma 1 establishes global static monopoly optimality. The constructed path satisfies Definition 1 with the uncapped RSI technology (48).

Local convergence from nearby initial stocks. Use the ordering $\xi = (\xi_K, \xi_B, \xi_C, \xi_q)'$ in (145),

and let e_K, e_B, e_C, e_q be the corresponding unit row vectors. Define the row vectors

$$y_\xi = \left(\frac{\alpha}{1 - (1 - \alpha)\omega_X}, \frac{(1 - \alpha)\omega_X}{1 - (1 - \alpha)\omega_X}, 0, 0 \right), \quad (120)$$

$$m_\xi = \frac{\eta e_B + e_q}{1 - \eta}, \quad (121)$$

$$b_\xi = g_B^*[(\eta - 1)e_B + \eta m_\xi]. \quad (122)$$

The reduced production equation gives the log deviation of output as $y_\xi \xi$. The research FOC implies $M = (q\chi\eta B^\eta)^{1/(1-\eta)}$, so the log deviation of M is $m_\xi \xi$. Differentiating $g_B = \chi B^{\eta-1} M^\eta$ at the BGP gives the level deviation $b_\xi \xi$.

The four rows of J^* , derived from capital accumulation, AI research, the household Euler equation, and the developer costate equation, are

$$J_K^* = \frac{1 - u^*}{k^*} y_\xi - (g_Y^* + \delta) e_K - \frac{c^*}{k^*} e_C - \frac{m^*}{k^*} m_\xi, \quad (123)$$

$$J_B^* = b_\xi, \quad (124)$$

$$J_C^* = (r^* + \delta)(y_\xi - e_K), \quad (125)$$

$$J_q^* = (r^* + \delta)(y_\xi - e_K) - (\rho - n + \eta g_Y^*)(y_\xi - e_q - e_B) - \eta b_\xi. \quad (126)$$

The last row uses $U/(qB) = \rho - n + \eta g_Y^*$ at the BGP. All entries depend on growth rates and ratios, which are independent of χ . No terminal restrictions were used to derive this Jacobian.

I next establish the two local properties (146) and (147) from the parameter restrictions, rather than impose them. Retain ν from (118) and, for this linearization, define

$$\begin{aligned} a_K &= \frac{\alpha}{1 - (1 - \alpha)\omega_X}, & a_T &= \frac{(1 - \alpha)\omega_X \eta}{[1 - (1 - \alpha)\omega_X](1 - \eta)}, \\ d &= \rho - n, & P &= r^* + \delta, & h &= \rho - n + \eta g_Y^*. \end{aligned}$$

All six quantities are positive, and the restriction $\eta + \omega_X < 1$ gives

$$1 - a_K - a_T = \frac{(1 - \alpha)(1 - \eta - \omega_X)}{[1 - (1 - \alpha)\omega_X](1 - \eta)} > 0. \quad (127)$$

Make the invertible linear change of deviations

$$\widehat{\xi} = (\xi_K, \nu \xi_B, \xi_C, \xi_q + (1 - \nu)\xi_B)'$$

The second and fourth coordinates are the log deviations of the transformed state S and its shadow price p . This changes neither the model nor the eigenvalues. Applying it to (123)–(126) gives a

similar matrix

$$\hat{J} = \begin{pmatrix} A & B & -C & -V \\ 0 & 0 & 0 & D \\ -G & H & 0 & 0 \\ -E & F & 0 & d \end{pmatrix}, \quad (128)$$

where the following letters abbreviate matrix coefficients only:

$$\begin{aligned} A &= d + (1 - \alpha)\omega_X P, & B &= \frac{(1 - u^*)a_T - m^*}{k^*}, & C &= \frac{c^*}{k^*}, \\ V &= \frac{m^*}{k^*(1 - \eta)}, & D &= \frac{\eta g_Y^*}{1 - \eta}, & G &= P(1 - a_K), \\ H &= P a_T, & E &= G + h a_K, & F &= H + h(1 - a_T). \end{aligned}$$

All coefficients other than B are positive by their definitions. The BGP allocation ratios imply the identities

$$L \equiv B - \frac{Vd}{D} = \frac{(1 - \alpha)\omega_X \eta}{(1 - \eta)k^*} > 0, \quad (129)$$

$$GF - HE = hP(1 - a_K - a_T) > 0. \quad (130)$$

Set $N = CG + VE > 0$ and $Q = CH + VF > 0$. Expanding the determinant gives the characteristic polynomial

$$\begin{aligned} p(\ell) &= \det(\ell I - \hat{J}) \\ &= (\ell^2 - A\ell - N)(\ell^2 - d\ell - DF) - DE(Q - L\ell). \end{aligned} \quad (131)$$

In particular,

$$p(0) = DChP(1 - a_K - a_T) > 0.$$

Let $\mu = (d - \sqrt{d^2 + 4DF})/2 < 0$, the negative root of the second quadratic factor. Then $p(\mu) = -DE(Q - L\mu) < 0$. Since $p(\ell) \rightarrow +\infty$ as $\ell \rightarrow -\infty$, there is one negative root below μ and another between μ and zero. These are the only negative roots: the cubic coefficient of p is $-(A + d) < 0$, its linear coefficient is $ADF + dN + LDE > 0$, and its constant coefficient is positive. Thus the coefficient signs of $p(-x)$, in descending powers, are $(+, +, \text{arbitrary}, -, +)$. Descartes' rule of signs gives exactly two sign changes, regardless of the quadratic coefficient. The two negative roots are therefore real, distinct, and simple. Write them as $\ell_1 < \mu < \ell_2 < 0$. The remaining two roots have positive product and positive sum, because $p(0) > 0$ and the sum of all four roots is $A + d > 0$. If real, both are positive; if complex, their real parts are positive. This proves (146).

To establish nonsingularity of the projection onto the two stocks, write a stable eigenvector in transformed coordinates as $(x, y, z, u)'$. Its second component cannot be zero: the second, fourth, and first rows would successively imply $u = x = z = 0$. Normalize $y = 1$. The second and fourth

rows then give

$$x(\ell) = \frac{\text{DF} + d\ell - \ell^2}{\text{DE}}.$$

The ordering of the roots implies $x(\ell_1) < 0 < x(\ell_2)$. Hence the two projected stable eigenvectors $(x(\ell_1), 1)'$ and $(x(\ell_2), 1)'$ are linearly independent. Returning to (ξ_K, ξ_B) only rescales their second component by $1/\nu > 0$, so (147) follows as well.

The exact normalized vector field is continuously differentiable near $\xi = 0$. The spectral property just proved and the stable-manifold theorem supply a two-dimensional local manifold whose trajectories remain near zero and converge exponentially. The derivative of its projection onto (ξ_K, ξ_B) is V_{KB} . The nonsingularity just proved and the inverse-function theorem therefore make this manifold locally the graph of a continuously differentiable function selecting (ξ_C, ξ_q) from the two initial stock deviations. Since exponentiation is a local diffeomorphism, the same holds for (C_0, q_0) as functions of (K_0, B_0) . More explicitly, for some constants $a, H > 0$, trajectories on a sufficiently small stable manifold satisfy

$$\|\xi(t)\| \leq H e^{-at} \|\xi(0)\|, \quad t \geq 0.$$

For any sufficiently small open neighborhood $\mathcal{V}$ of zero, choose a ball contained in $\mathcal{V}$. Continuity of the graph selecting the two jump variables permits an open neighborhood $\mathcal{U}$ of the initial stock deviations for which $H\|\xi(0)\|$ is smaller than that ball's radius. Every initial stock pair in $\mathcal{U}$ therefore has a trajectory staying in $\mathcal{V}$ for all $t \geq 0$ and converging to zero. Exponentiation maps $\mathcal{U}$ to an open neighborhood of (K_0^*, B_0^*) . Shrinking these neighborhoods preserves positivity, feasibility, and existence for every $t \geq 0$. Local uniqueness follows because every trajectory remaining in this local region and converging to zero belongs to the same stable manifold, which is a graph over stocks. This does not assert convergence from arbitrary initial C_0, q_0 or exclude equilibrium trajectories that leave the region.

The statement also permits nearby A_0, N_0 . Equations (112) and (113) make the reference B_0^*, K_0^* smooth functions of these two positive initial values. After normalization by the corresponding BGP, the exact vector field depends only on the parameters and BGP ratios, not on A_0, N_0 . The set of (A_0, N_0, K_0, B_0) whose stock deviations belong to $\mathcal{U}$ is therefore open and contains the reference initial condition. From each such initial condition, the construction converges to the BGP associated with its own A_0, N_0 , not necessarily to the same levels as the original reference path.

Since $f(\xi(t)) \rightarrow 0$, growth rates converge to their BGP values, not only normalized levels. The static equations give the same convergence for the interest rate, wage growth, and allocation ratios. The unit-elastic labor income share remains $(1 - \alpha)\omega_L$ at every date.

Exponential convergence implies that $r_t - r^*$ is integrable. Thus D_t/e^{-r^*t} tends to a finite positive constant, and both TVC expressions equal their BGP counterparts in (115) and (116) times factors tending to finite positive limits. Both TVCs hold. Output and all expenditure flows are bounded multiples of $e^{g_Y^*t}$, while log consumption per person is a linear function of time plus a bounded deviation. Both objective integrals therefore remain finite. The global concavity verifica-

tion established above applies at these new price and allocation paths, proving agent optimality. Static optimality and market clearing then establish equilibrium, not merely a solution of the necessary conditions. $\square$

Proof of Proposition 7. I construct a sequence of research trajectories whose discounted net profit grows without bound, although each trajectory has finite expenditure and AI efficiency. Write $\theta = (1 - \alpha)/\alpha$. For a research trajectory $\{M_t\}_{t \in [0, T]}$, discounted net profit on the fixed horizon is

$$J_T(M) = \int_0^T \exp \left\{ - \int_0^t r_s ds \right\} [\pi(B_t) - M_t] dt.$$

Integrability of r bounds the discount factor above and away from zero on $[0, T]$. At the optimal service choice, equations (13) and (19) imply

$$\pi(B) = p_X X - \frac{X}{B} = e_X p_X X = (1 - \alpha) e_X s_X Y.$$

For $\sigma > 1$, marginal revenue is positive at every finite X and approaches zero as $X \rightarrow \infty$, as established in Lemma 1. The monopoly condition therefore implies $X \rightarrow \infty$ as $B \rightarrow \infty$, holding K, A, L fixed. Consequently, $s_X \rightarrow 1$ and $e_X \rightarrow \alpha$. Substituting these limits into equation (66) gives

$$\pi(B) \sim \kappa_X K B^\theta, \quad \kappa_X = \alpha(1 - \alpha) \left[(1 - \alpha)^2 \omega_X^{\sigma/(\sigma-1)} \right]^\theta > 0.$$

The operating-profit asymptotic is uniform over dates in $[0, T]$ because K, A, L are continuous, bounded, and bounded away from zero on $[0, T]$.

Fix $\Delta \in (0, T)$, choose $M = \mathcal{M}$ on $[0, \Delta]$, and choose $M = 0$ afterward. The uncapped RSI technology gives

$$B_{\mathcal{M}}(t)^{1-\eta} = B_0^{1-\eta} + (1 - \eta) \chi \mathcal{M}^\eta \min\{t, \Delta\}.$$

Since optimized operating profit is nonnegative, a lower bound on discounted net profit counts only operating profits earned after research stops, while subtracting all research expenditure. For all sufficiently large $\mathcal{M}$,

$$J_T(\mathcal{M}) \geq c_\pi \mathcal{M}^{\eta(1-\alpha)/[\alpha(1-\eta)]} - c_M \mathcal{M}, \quad (132)$$

where $c_\pi, c_M > 0$ are independent of $\mathcal{M}$. The exponent exceeds one because $\eta > \alpha$, so $J_T(\mathcal{M}) \rightarrow +\infty$ as $\mathcal{M} \rightarrow \infty$. Each finite $\mathcal{M}$ keeps AI efficiency and discounted net profit finite on $[0, T]$; it is the sequence of payoffs that is unbounded.

To compare with an infinite-horizon equilibrium candidate, extend the deviation after T using the candidate's subsequent research expenditure. Once the initial research phase leaves AI efficiency above the candidate's level at T , subsequent operating profits cannot be lower than along that candidate. $\square$

Proposition 8 (An AI bottleneck under complementarity). *Suppose $0 < \sigma < 1$ and $0 < B_0 < \bar{B} < \mathcal{B}(\sigma)$. Any equilibrium satisfies*

$$\limsup_{t \rightarrow \infty} \frac{1}{t} \log \left(\frac{Y_t/N_t}{Y_0/N_0} \right) \leq \mathcal{R}(\bar{B}) - \rho < \gamma,$$

$$\frac{X}{AL} \rightarrow 0, \quad s_X \rightarrow 1, \quad \frac{wL}{Y} \rightarrow 0.$$

Proof of Proposition 8. Consider any equilibrium with $0 < \sigma < 1$ and $0 < B_0 < \bar{B} < \mathcal{B}(\sigma)$. The increasing share map in equation (83), established in Step 1 of the proof of Proposition 1, and $B_t \leq \bar{B}$ imply

$$r_t \leq \mathcal{R}(B_t) \leq \mathcal{R}(\bar{B}) < \rho + \gamma, \quad \frac{Y_t}{K_t} \leq \frac{\mathcal{R}(\bar{B}) + \delta}{\alpha}. \quad (133)$$

The present value of the developer's net profit from any date onward is nonnegative: it can stop research and earn nonnegative operating profit. Wage income is also nonnegative. Integrating the household budget (4), using its transversality condition (6) and the Euler equation (5), therefore gives the first inequality below; integrating the Euler equation with the return bound gives the second:

$$C_t \geq (\rho - n)K_t, \quad \frac{C_t}{N_t} \leq \frac{C_0}{N_0} e^{[\mathcal{R}(\bar{B}) - \rho]t}. \quad (134)$$

Combining these inequalities yields

$$\frac{Y_t}{N_t} \leq \frac{\mathcal{R}(\bar{B}) + \delta}{\alpha(\rho - n)} \frac{C_0}{N_0} e^{[\mathcal{R}(\bar{B}) - \rho]t}, \quad \limsup_{t \rightarrow \infty} \frac{1}{t} \log \left(\frac{Y_t/N_t}{Y_0/N_0} \right) \leq \mathcal{R}(\bar{B}) - \rho < \gamma. \quad (135)$$

Consequently $Y/(AL) \rightarrow 0$. Since equation (32) gives $X/Y \leq \bar{B}(1 - \alpha)^2$ under complementarity, $X/(AL) \rightarrow 0$: AI production services become scarce relative to effective labor. Equation (10) then implies $s_X \rightarrow 1$ and hence $wL/Y \rightarrow 0$. The growth bound concerns the long-run average, not convergence of the instantaneous growth rate. Neither convergence of C/Y nor $B_t \rightarrow \bar{B}$ is assumed. The growth bound and vanishing labor share are necessary properties of any equilibrium in this region, not a proof of equilibrium existence. $\square$

B Numerical algorithm and equilibrium audit

B.1 Economic intuition and initial conditions

For the RSI-activation comparisons in Sections 7.2 and 7.3, I compute numerical approximations to equilibrium trajectories by solving a boundary-value problem (BVP). Initial capital K_0 and AI efficiency B_0 are given, whereas initial consumption C_0 and the developer's shadow price q_0 are determined jointly with the transition. I solve the equilibrium differential equations over a finite horizon, using the given stocks as initial conditions and the long-run characterization in Proposition 1 to construct terminal conditions. The checks described below assess feasibility, optimality,

and sensitivity to the computational horizon.

For each σ , I first solve the existing-AI economy with efficiency fixed at B_0 and research unavailable. A scalar root search in $\log K_0$ solves Equation (67) using the production and monopoly equations. The resource constraint (26), with $M = 0$ and $\dot{K} = (n + \gamma)K$, gives pre-event consumption. Homogeneity extends these values to a BGP: K, Y, C, U, X grow at $n + \gamma$, wages grow at γ , and B, p_X, r and income shares are constant. Since $r - (n + \gamma) = \rho - n > 0$, pre-event developer profit has finite present value and the household's transversality expression decays.

At unexpected RSI activation, capital and AI efficiency are inherited, as are the exogenous levels of population and labor-augmenting technology. The positive- χ problem selects consumption, research and the shadow value anew. A separate audit checks continuity of production quantities and prices, verifies the pre-event Euler, resource and monopoly conditions, and records the endogenous jumps in consumption and research.

B.2 Boundary-value solution and reported outcomes

I use `scipy.integrate.solve_bvp` from the open-source Python library SciPy (Virtanen et al., 2020). The solver implements fourth-order collocation with residual control similar to that of Kierzenka and Shampine (2001). I use the computational coordinates

$$\left(\log K, \log \frac{B}{\bar{B} - B}, \log C, \log q \right).$$

The second coordinate expresses AI efficiency relative to the remaining gap below its upper bound; it can grow without bound while AI efficiency remains bounded. Subtracting the analytical long-run aggregate growth rate times t from each coordinate removes its asymptotic linear trend. These transformations change the numerical representation, not the economic variables or calendar time. At each trial state the code solves the static monopoly problem and recovers research from its first-order condition. The left boundary fixes K_0, B_0 ; two right-boundary projections remove unstable components of the normalized equilibrium dynamics near the long-run limit, using Appendix A.

For each of the eight parameter configurations, I start near its analytical limiting point and move the initial stocks to their prescribed values in 32 continuation stages. Neither $\omega_X = 0$ nor $\chi = 0$ is used as the starting point for continuation. These stages change the initial conditions of the numerical problem; they are not dates on the economic trajectory.

At unit elasticity the static block uses its exact Cobb–Douglas limit; near one, `log1p` and `expm1` reduce the loss of precision from subtracting nearly equal numbers. Under complementarity, marginal revenue is evaluated directly and the static root is polished when needed near zero marginal revenue; the first-order condition is unchanged.

The values $\chi = 7.5$ and $\chi = 1.5$ are supplied directly: there is no outer calibration search. Research shares and price declines are outputs. Annual research expenditure relative to output is measured as $\int_0^1 M_t dt / \int_0^1 Y_t dt$, not M_0/Y_0 . Gauss–Legendre quadrature with 64 and 128 nodes must agree within 10^{-8} in logs. The annual ratio must also change by less than 2×10^{-5} in logs

across the horizon extensions.

B.3 Equilibrium checks

I extend each computational horizon twice by 500 years and tighten the collocation tolerance from 2×10^{-6} to 10^{-8} and 10^{-9} . The final boundary tolerance is 10^{-11} . Initial meshes contain 221, 401 and 601 points and adapt as needed. The displayed window lies inside each solved horizon; no spline is extrapolated.

The original equations are reconstructed independently with five-point differences at 1,001 dates and steps of 0.003 and 0.001 years. An additional 801-date check over the first ten years uses steps of 0.0003 and 0.0001. Acceptance of a computed trajectory requires dynamic residuals below 10^{-6} , first-order residuals below 10^{-9} , final-horizon changes below 2×10^{-5} in detrended logs, and terminal-coordinate gaps below 10^{-4} . The horizon comparison measures changes in the computed trajectory when the terminal date is extended; the terminal-coordinate check measures its distance from the analytical limiting point at that date. These are the existing numerical tolerances, not relaxed economic conditions.

The audit checks feasibility and developer global optimality. When the concavity conditions of Lemma 3 fail, it checks the Hamiltonian support condition in Lemma 2, using counterfactual-state minimization and an analytical bound beyond the grid. At each checked date, the support test compares the maximized Hamiltonian at alternative AI-efficiency levels with its tangent at the candidate state, holding the transformed shadow price and the dated values of capital and effective labor fixed. The two grids contain 81 dates and 101 alternative states, and 321 dates and 241 alternative states, respectively, supplemented by dense first-decade checks when needed. The analytical support bound and the asymptotic rate $n - \rho < 0$ of both transversality expressions support the infinite-horizon continuation. These are numerically verified equilibrium approximations, not interval-arithmetic existence certificates for arbitrary initial stocks.

Table 4: Numerical accuracy of the illustrative RSI comparisons

χ	σ	Final horizon	Mesh points	Dynamic residual	Horizon change
7.5	0.90	5,428.0	1,193	1.18×10^{-9}	3.06×10^{-8}
7.5	1.00	4,944.5	1,290	1.38×10^{-9}	3.46×10^{-8}
7.5	1.10	4,828.8	1,326	1.66×10^{-9}	3.73×10^{-8}
7.5	1.50	3,013.0	1,586	1.69×10^{-9}	1.56×10^{-8}
1.5	0.90	6,047.0	841	1.02×10^{-9}	4.30×10^{-8}
1.5	1.00	5,563.5	893	1.11×10^{-9}	3.80×10^{-8}
1.5	1.10	5,447.8	911	1.19×10^{-9}	4.08×10^{-8}
1.5	1.50	3,632.0	1,104	1.23×10^{-9}	1.80×10^{-8}

Horizons are measured in years. Dynamic residuals are the largest independently reconstructed errors across the full-horizon and dense first-decade checks. Horizon change compares the last two solutions in detrended logarithmic coordinates over years 0–500. Error bounds are rounded upward. All eight paths pass the equation, event-continuity, developer-optimality and long-run-continuation checks. The $\sigma = 1.50$ cases use the global Hamiltonian-support test and its analytical continuation bound; the other cases pass the concavity test.

B.4 Analytical checks on developer optimality

At unit elasticity, the operating-profit concavity condition in Lemma 3 has a simple primitive form. Define $\beta = (1 - \alpha)\omega_X$. The static monopoly solution implies $\pi_t(B) = a_t B^{\beta/(1-\beta)}$ for some $a_t > 0$, conditional on the dated values of capital and effective labor. Here R_B is the transformed AI-efficiency state in equation (71). Because $B(R_B)$ is increasing and concave, $R_B \mapsto \pi_t(B(R_B))$ is globally concave whenever $\beta \leq 1/2$. In the benchmark calibration in Table 2, $\beta = 0.067$, so this sufficient condition holds with substantial slack. The research-term condition (76) also holds because $\eta = 0.20 < 1/2$. These are sufficient conditions, not additional maintained assumptions; when they fail, the support condition can instead be checked directly.

The AI-dominated continuation supplies an analytical long-run check on the support inequality. For $1 < \sigma \leq 1/\alpha$, the service elasticity defined in equation (77) satisfies $\varepsilon_{B,t}(b) \leq 1/\alpha$ for every t and b . Suppressing these arguments, subtracting $\alpha(1 - e_X)$ from the denominator in that equation gives

$$(\sigma^{-1} - \alpha)(1 - s_X) [(1 - \sigma^{-1})(1 - s_X) + (\sigma^{-1} - \alpha)s_X] \geq 0.$$

Choose $B_* \in (B_0, \bar{B})$ with $B_*/\bar{B} > \max\{0, (1 - 2\alpha)/(1 - \alpha)\}$. Operating profit is then concave in the transformed state R_B for $B \geq B_*$ by equation (79). If $\eta \leq 1/2$, the research term in (73) is concave there as well. For a candidate with $B_t \geq B_*$, it consequently suffices to check

$$\hat{\mathcal{H}}_t(R_{B,t}) - \hat{\mathcal{H}}'_t(R_{B,t})(R_{B,t} - R_{B,0}) \geq \hat{\mathcal{H}}_t(R_B(B_*)). \quad (136)$$

Above B_* the tangent bound follows from concavity. Below B_* the Hamiltonian is increasing in B , so its value is at most the right-hand side, whereas the tangent is minimized at $R_{B,0}$.

Along an AI-dominated continuation, the difference between the two sides of (136), divided by Y_t , converges to

$$\alpha(1 - \alpha) \left[1 - \left(\frac{B_*}{\bar{B}} \right)^{(1-\alpha)/\alpha} \right] > 0.$$

To obtain this limit, use $M/Y \rightarrow 0$, $\psi(B)R_B \rightarrow 0$, and $\pi(B_*)/Y \rightarrow \alpha(1 - \alpha)(B_*/\bar{B})^{(1-\alpha)/\alpha}$. Thus the global support condition holds sufficiently late on this continuation even when the stronger curvature condition fails at low counterfactual AI-efficiency levels. The finite transition still requires its own support check.

B.5 Replication files

I use NumPy for array operations and linear algebra (Harris et al., 2020) and Matplotlib to generate the figures (Hunter, 2007). The package versions are recorded in `requirements-rewrite.txt`.

The code, simulation data, and replication instructions are available at <https://github.com/oliverpardo1979/ai-growth-and-labor>. The [replication guide](#) explains how to install the dependencies, check the published results, and recompute the eight trajectories. From the repository root, run:

`python scripts/simulate_rewrite_illustrative_rsi.py`

This command solves and audits all four elasticities in both scenarios. Use `--chi 7.5` or `--chi 1.5` for one scenario; `--sigma 1.5`, for example, solves one elasticity. With all four checkpoints available, `--finish` repeats the equilibrium checks and exports the figures. The full replication entry point is `scripts/reproduce_rewrite_results.py`.

The [higher-productivity results folder](#) and [lower-productivity results folder](#) contain the scenario inputs, validated CSV files, outcome summaries, equilibrium checks and figure metadata. Checkpoint and CSV hashes bind the audits to the plotted data. No comparison is exported unless the equilibrium and activation-continuity checks pass for all four elasticities in that comparison.

The repository includes the two RSI-activation comparisons reported here and the competitive-to-monopoly comparisons documented in Appendix D. Earlier experiments are preserved separately and are not needed for replication.

C The uncapped economy with unit elasticity

This appendix derives the rates and allocation ratios supporting Proposition 6 in Section 6.2 and defines the normalized variables used to prove local convergence. I use the uncapped RSI technology (48) and the production function (57). In this appendix and its proofs, a superscript $*$ denotes the reference BGP characterized in Section 6.2, distinguishing it from nearby equilibrium trajectories.

C.1 An explicit balanced-growth equilibrium

At unit elasticity, $s_X = \omega_X$ and $e_X = 1 - (1 - \alpha)\omega_X$. The AI-demand and monopoly conditions, equations (13) and (19), give

$$\frac{p_X X}{Y} = (1 - \alpha)\omega_X, \quad \frac{U}{Y} = (1 - \alpha)^2 \omega_X^2. \quad (137)$$

Substituting $X = BU = (1 - \alpha)^2 \omega_X^2 BY$ into production yields

$$Y^{1-(1-\alpha)\omega_X} = K^\alpha (AL)^{(1-\alpha)\omega_L} [(1 - \alpha)^2 \omega_X^2 B]^{(1-\alpha)\omega_X}. \quad (138)$$

On an exact balanced-growth path with positive constant K/Y , C/Y , and M/Y , log differentiation of (138) and (48) gives the two growth restrictions (58) and (59). Define

$$\Delta \equiv (1 - \alpha)(1 - \eta - \omega_X). \quad (139)$$

For $\Delta > 0$, their unique solution has

$$g_Y^* = \frac{(1 - \eta)(1 - \omega_X)}{1 - \eta - \omega_X} (n + \gamma), \quad (140)$$

and g_B^* as in (64). The equality $g_K = g_C = g_Y^*$ follows from constant capital and consumption ratios; equations (12) and (5) then give (61) and (62).

Define the candidate ratios $k^* = K/Y$, $u^* = U/Y$, $i^* = (\dot{K} + \delta K)/Y$, $m^* = M/Y$, and $c^* = C/Y$ by

$$k^* = \frac{\alpha}{r^* + \delta}, \quad u^* = (1 - \alpha)^2 \omega_X^2, \quad (141)$$

$$i^* = (g_Y^* + \delta)k^*, \quad (142)$$

$$m^* = \frac{u^* \eta g_B^*}{\rho - n + \eta g_Y^*}, \quad (143)$$

$$c^* = 1 - u^* - i^* - m^*. \quad (144)$$

In this appendix, k^* is a capital–output ratio, not capital per unit of effective labor, and i^* is gross physical investment as a share of output. The research share comes from the developer’s first-order and costate conditions; the consumption share comes from market clearing. These ratios are endogenous, not fixed saving or research rules. The proof in Appendix A verifies their positivity and constructs the reference initial levels from the production and research equations, for given initial population and labor-augmenting technology. Along the BGP, $B_t = B_0^* e^{g_B^* t} \rightarrow \infty$, $g_K = g_C = g_U = g_M = g_Y^*$, $g_X = g_Y^* + g_B^*$, and $g_q = g_Y^* - g_B^*$.

Since $n + \gamma > 0$, equations (58) and (59) require $\Delta > 0$ for this positive finite-rate BGP. If $\Delta = 0$, they are inconsistent; if $\Delta < 0$, they imply $g_B < 0$, contrary to the RSI technology. Thus $\eta + \omega_X \geq 1$ excludes this BGP, not every possible equilibrium trajectory. The same restriction $\eta + \omega_X < 1$ also suffices for global developer optimality: the proof uses the invertible state transformation $S = B^{(1-\eta)/\eta}$, in which the RSI technology is jointly concave in S, M and operating profit is concave in S . Finally, $\rho > n$ makes both transversality expressions decline and the objective integrals converge.

C.2 Equilibrium trajectories near the balanced-growth path

The BGP supplies a reference for proving existence from nearby initial stocks that need not be on that path. For the given exogenous paths of population and labor-augmenting technology, compare each variable with its value on the corresponding reference BGP at the same date. Define log deviations

$$\xi_K = \log \frac{K}{K^*}, \quad \xi_B = \log \frac{B}{B^*}, \quad \xi_C = \log \frac{C}{C^*}, \quad \xi_q = \log \frac{q}{q^*}. \quad (145)$$

Time subscripts are suppressed: the reference quantities in the denominators follow the BGP rather than remain fixed at their initial values. Substitution into the four equilibrium dynamics gives a smooth autonomous system $\dot{\xi} = f(\xi)$ with $f(0) = 0$, where $\xi = (\xi_K, \xi_B, \xi_C, \xi_q)'$. Let $J^* = Df(0)$ denote its Jacobian. The proof in Appendix A derives its entries directly from equations (29a)–(29d), with $\psi = 1$ and $\psi' = 0$. The normalized Jacobian is independent of χ , A_0 , and N_0 ; changing research productivity changes the reference levels, not the eigenvalues governing local convergence.

The parameter restrictions in Proposition 6 imply two local properties. First, the linearized dynamics have exactly two stable directions and no eigenvalue with zero real part. Writing E^s for the real invariant subspace associated with eigenvalues having negative real parts, the first property is

$$\text{spec}(J^*) \cap i\mathbb{R} = \emptyset, \quad \dim E^s = 2. \quad (146)$$

Second, these stable directions allow independent changes in the two predetermined stocks. Let $V_S \in \mathbb{R}^{4 \times 2}$ be any real basis matrix for E^s , and let V_{KB} consist of its first two rows. Then

$$\det V_{KB} \neq 0. \quad (147)$$

Nonsingularity does not depend on the choice of basis. It allows the stable manifold to select C_0, q_0 for every sufficiently close pair K_0, B_0 , rather than only for stocks on a curve. Both properties follow analytically from the proposition's parameter restrictions; neither is an additional assumption or a calibration-specific test. The complete argument, including both transversality conditions and global agent optimality, is in Appendix A, [Proof of Proposition 6](#).

No constant growth rates or allocation shares are imposed along these transitions. The result is local in initial stocks, not in time. In particular, it concerns equilibrium choices of consumption and research, not convergence from arbitrary nearby values of all four variables: there are also two unstable directions. It does not rule out equilibria that leave the local neighborhood.

C.3 Growth and distribution without an upper bound

The uncapped BGP has

$$\frac{wL}{Y} = (1 - \alpha)\omega_L > 0, \quad \frac{(r + \delta)K}{Y} = \alpha, \quad \frac{\Pi}{Y} = (1 - \alpha)\omega_X - u^* - m^*. \quad (148)$$

Equation (61) shows that wages and output per worker grow together at a rate above γ , even though labor retains a positive, constant income share. Without an upper bound, continued growth in AI efficiency sustains this wage-growth premium. With any upper bound, the unit-elastic equilibria in Proposition 1(ii) instead have $g_{Y/N}, g_w \rightarrow \gamma$. The equivalence between wage growth above γ and a vanishing labor share across the four regimes in Proposition 1 does not extend to the uncapped economy.

Within the restrictions of Proposition 6, increasing the AI weight raises the common growth rate and the interest rate:

$$\frac{\partial g_{Y/N}^*}{\partial \omega_X} = \frac{\partial g_w^*}{\partial \omega_X} = \frac{\partial r^*}{\partial \omega_X} = \frac{\eta(1 - \eta)(n + \gamma)}{(1 - \eta - \omega_X)^2} > 0. \quad (149)$$

Labor's income share falls across these parameterizations, but remains constant over time on each BGP. As $\omega_X \rightarrow 0^+$, the common per-person growth rate converges to γ and $r^* \rightarrow \rho + \gamma$. These statements compare balanced-growth equilibria; they do not describe a transition from a no-AI

economy or establish an uncapped equilibrium for $\sigma > 1$.

D Additional competitive-to-monopoly simulation results

I supplement the $\sigma = 1.5$ comparison in Section 7.4 with the original four-elasticity exercises at both research productivities. No additional equilibria are computed for this presentation. The level figures retain the three windows used in that section; the growth, price, and distributional figures below show two pre-event years and years 0–10 above, with years 10–500 below. Thin horizontal lines show continued competition; black dotted long-run lines show the analytical monopoly limits at $\sigma = 1.5$. The benchmark lines also distinguish the competitive income shares from the monopoly limits: competitive AI revenue pays for inference, whereas monopoly revenue also finances research and developer profit. The date-zero level changes are not finite growth rates: open and filled dots distinguish the values on either side of the event, and growth curves describe the smooth trajectory on each side separately.

Table 5: Competitive initial capital and the impact of exclusive AI rights

σ	K_0	Change in output	Change in wage
0.9	6.235	−32.4%	−36.6%
1.0	6.278	−17.6%	−17.6%
1.1	6.324	−13.7%	−11.9%
1.5	6.535	−10.4%	−5.4%

Notes: The capital stocks are derived from the competitive BGP, not selected independently. All other parameters and initial stocks are in Table 2. Changes compare levels immediately after and before the grant. They are identical for both values of χ , since K and B cannot jump. The ratio $K_0/Y_{0-} = 3.3$ applies before the grant, not after output falls.

Level comparisons across elasticities. Figures 13 and 14 preserve the level comparisons for all four elasticities. Research can produce level gains even where it does not permanently raise growth. With $\chi = 7.5$, output first regains its competitive counterpart around year 18.2 at $\sigma = 1.1$ and year 46.9 at $\sigma = 1$. With $\chi = 1.5$, those recoveries occur around years 87.3 and 212.3. At $\sigma = 0.9$, neither output nor wages regain their competitive counterparts within 500 years under either χ . The initial AI-service price in that case is about 131 times its competitive level, a large distortion implied by the illustrative parameters rather than an empirical estimate. These recovery dates compare contemporaneous levels; they do not rank welfare.

Levels relative to continued competition | $\chi = 7.5$

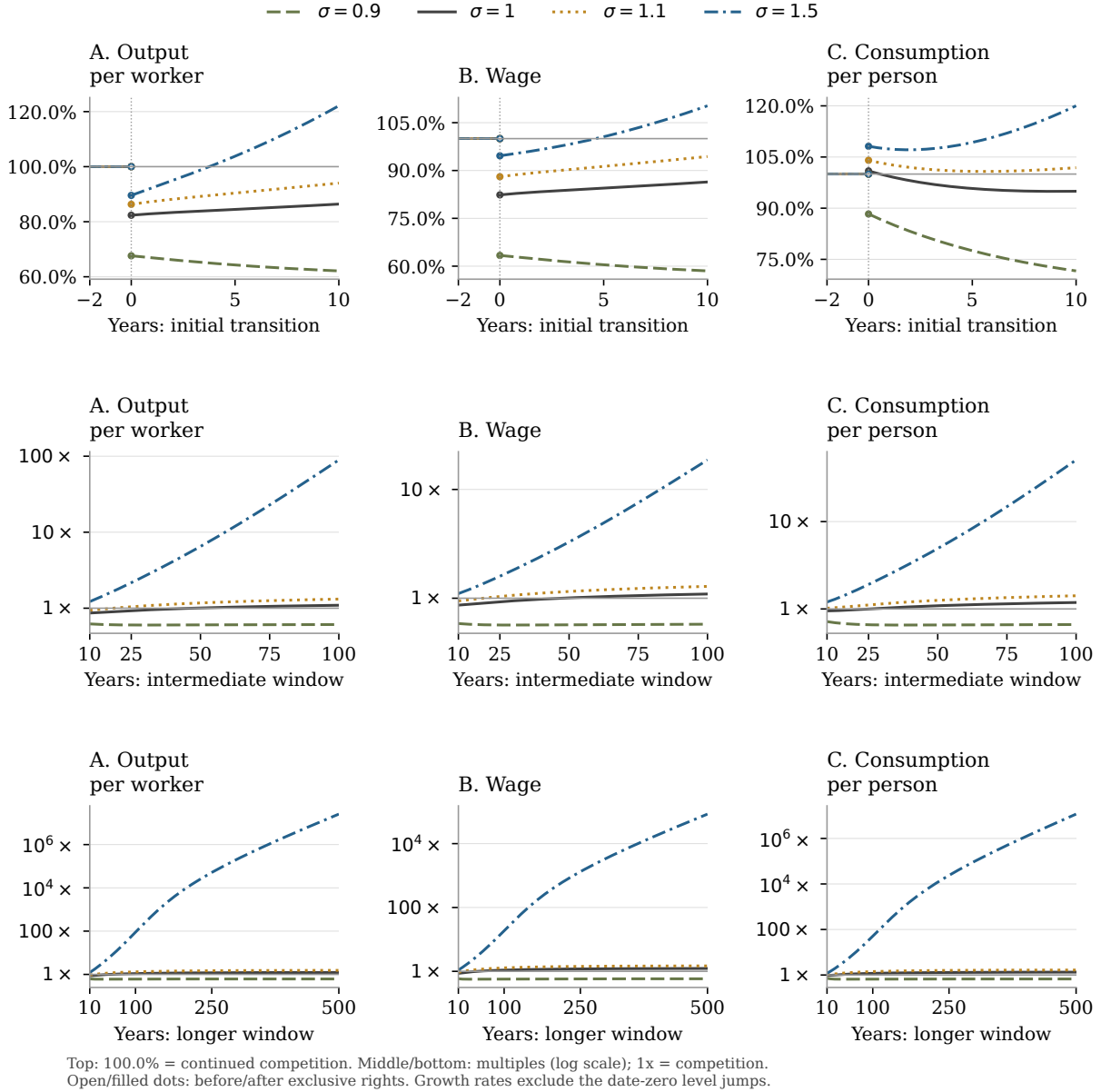


Figure 13: From competition to monopoly with RSI, $\chi = 7.5$: levels relative to continued competition for all four elasticities. The rows cover years $-2-10$, $10-100$, and $10-500$. The upper row uses percentages; the middle and lower rows use multiples on logarithmic scales. The horizontal benchmark is 100.0% or $1\times$. Open and filled dots mark levels before and after the grant. Vertical scales differ across windows and between the two research-productivity exercises.

Levels relative to continued competition | $\chi = 1.5$

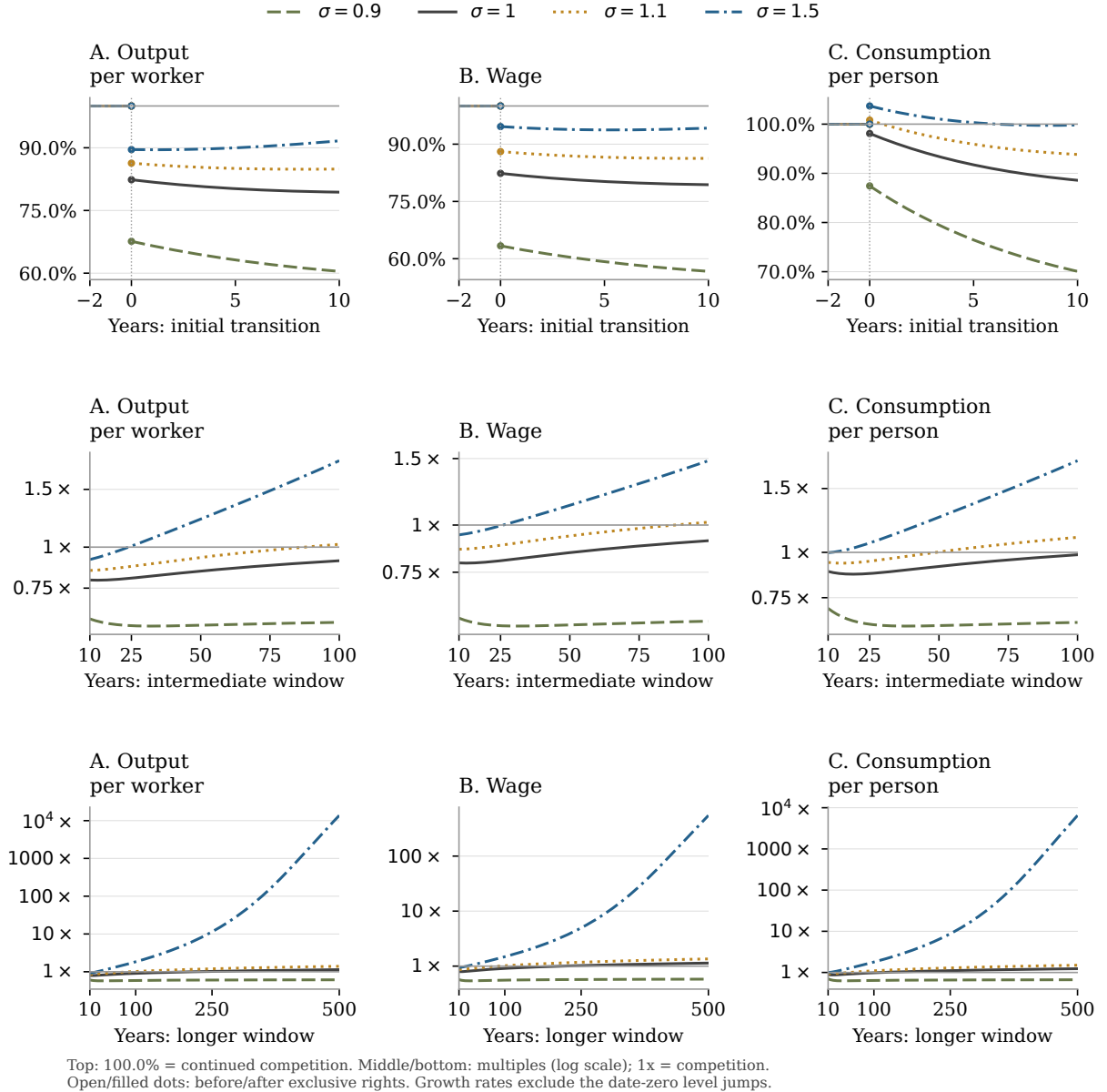


Figure 14: From competition to monopoly with RSI, $\chi = 1.5$: output per worker, wages, and consumption per person relative to continued competition for all four elasticities. Windows, normalizations, and line styles match Figure 13. Vertical scales are fitted to this exercise.

Initial conditions and numerical checks. I obtain each competitive initial stock by imposing $r = \rho + \gamma$ with the marginal-cost pricing condition $p_X = 1/B_0$. Pre-event consumption follows from the resource constraint with $M = 0$ and $\dot{K} = (n + \gamma)K$. The post-event calculation uses the same monopoly BVP and equilibrium checks as Appendix B. Previously computed RSI-activation trajectories supply numerical initial guesses, but I solve again at the competitive K_0 , imposing

neither their consumption nor their research choices. All eight paths pass the same residual, horizon-refinement, transversality, and developer-optimality checks; no tolerance is relaxed.

Higher research productivity. Figures 15–17 report $\chi = 7.5$. Unlike the pure RSI-activation experiment, the grant immediately reduces the return on capital: output falls while capital cannot jump, so equation (11) implies a lower interest rate. The Euler equation (5) then gives lower consumption growth immediately after the grant, even where the level of consumption jumps upward. Subsequent RSI and capital adjustment determine recovery. Research absorbs resources before its gains are fully realized, and the distribution panels separate developer net profit from expenditure on inference and research.

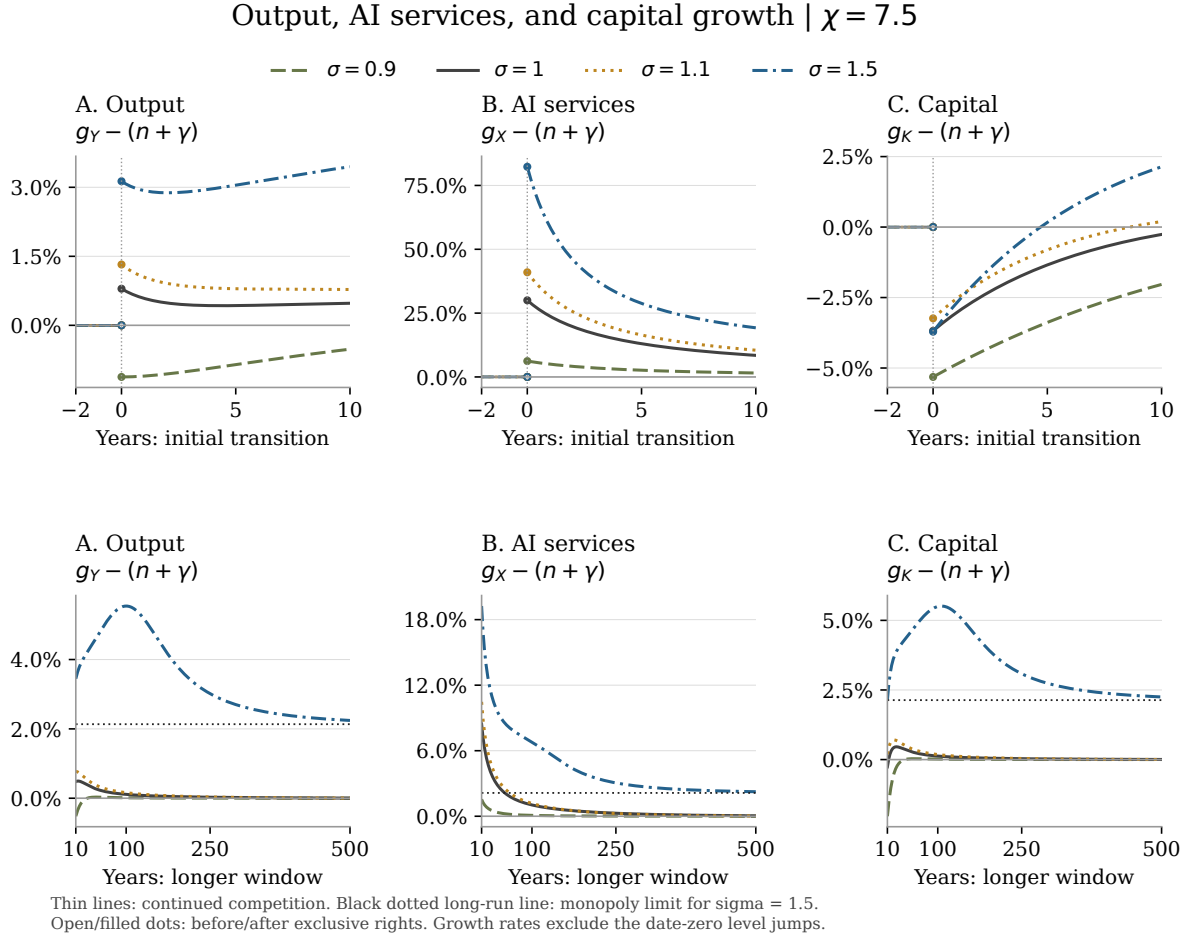


Figure 15: Competition to monopoly, $\chi = 7.5$: annual growth of output, AI services, and capital relative to effective labor. Zero denotes growth at $n + \gamma$. Growth rates exclude the discrete level changes at date zero.

Wages, interest rates, and AI prices | $\chi = 7.5$

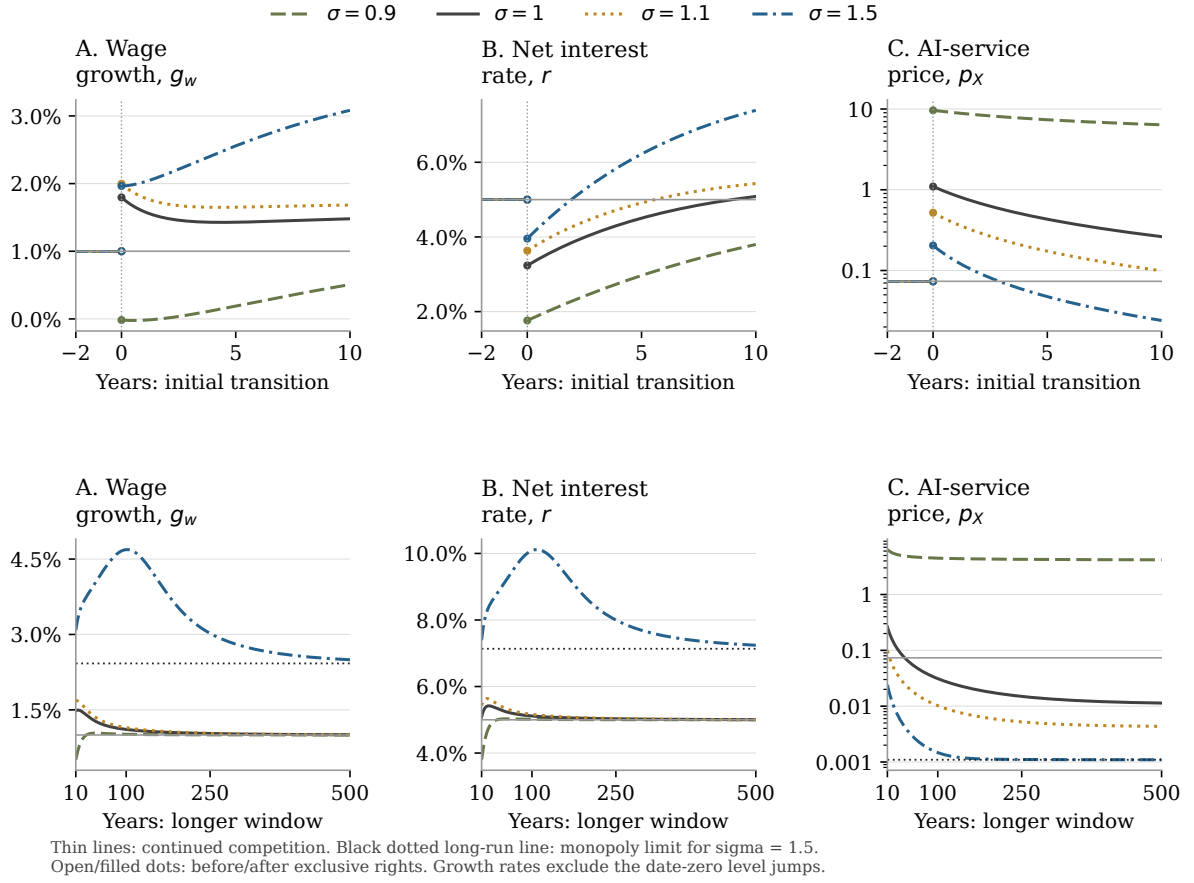


Figure 16: Competition to monopoly, $\chi = 7.5$: annual wage growth, net interest, and AI-service prices. Prices are measured in final-good units on a logarithmic scale. The wage level falls at the grant; panel A reports wage growth on either side, not that level change.

Income and compute expenditure shares | $\chi = 7.5$

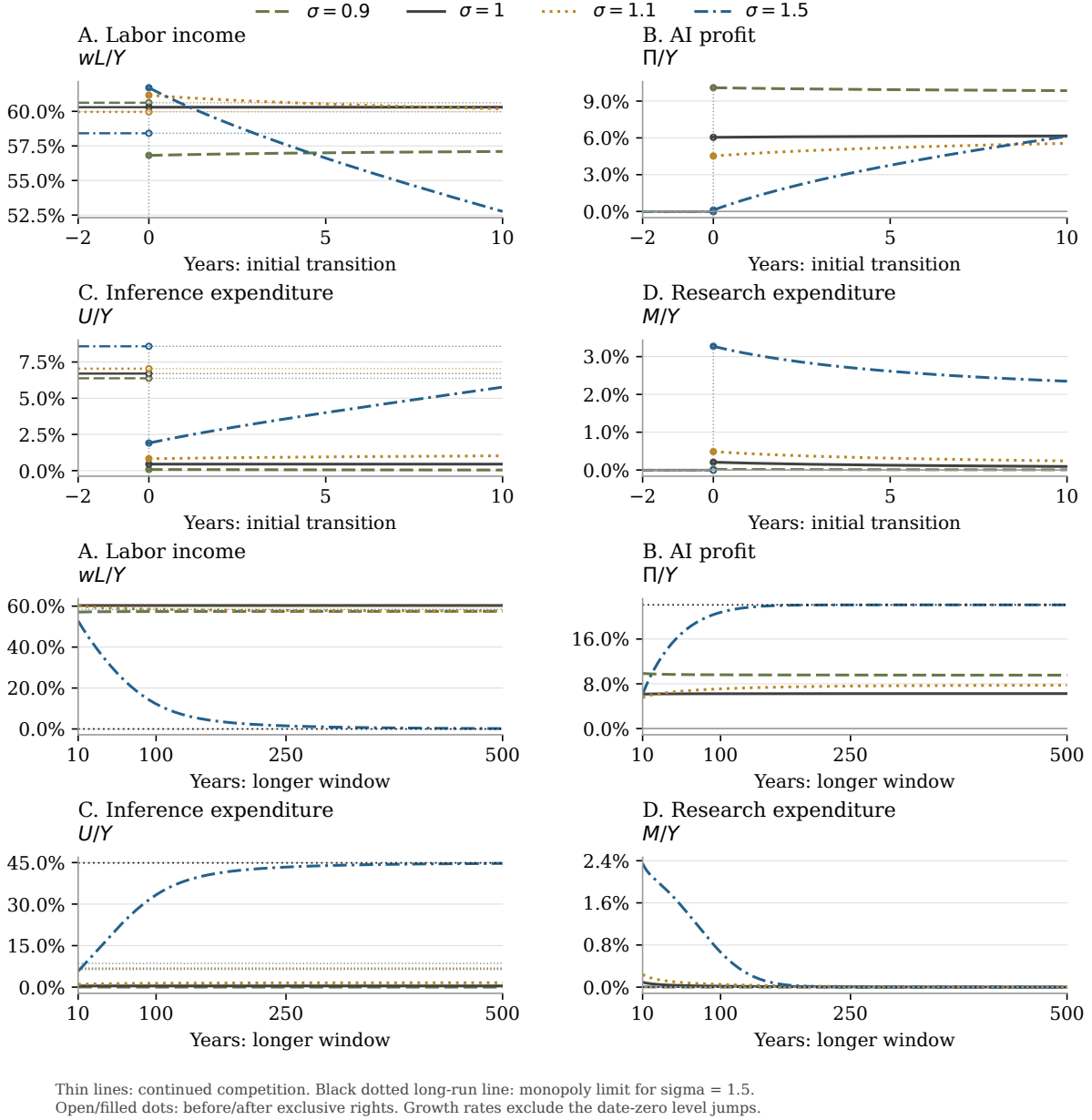


Figure 17: Competition to monopoly, $\chi = 7.5$: labor income, developer net profit, inference expenditure, and research expenditure as shares of output. The first two rows cover years -2 – 10 ; the last two cover years 10 – 500 . Gross capital income remains 33.0% of output.

Lower research productivity. Figures 18–20 report $\chi = 1.5$. The impact pricing distortion is unchanged, but the initial research allocation and efficiency growth are smaller. At $\sigma = 1.5$, the eventual rise in growth and decline in labor’s share emerge later. At year 500, growth and returns still exceed their monopoly limits, so the final displayed values should not be read as those limits.

Output, AI services, and capital growth | $\chi = 1.5$

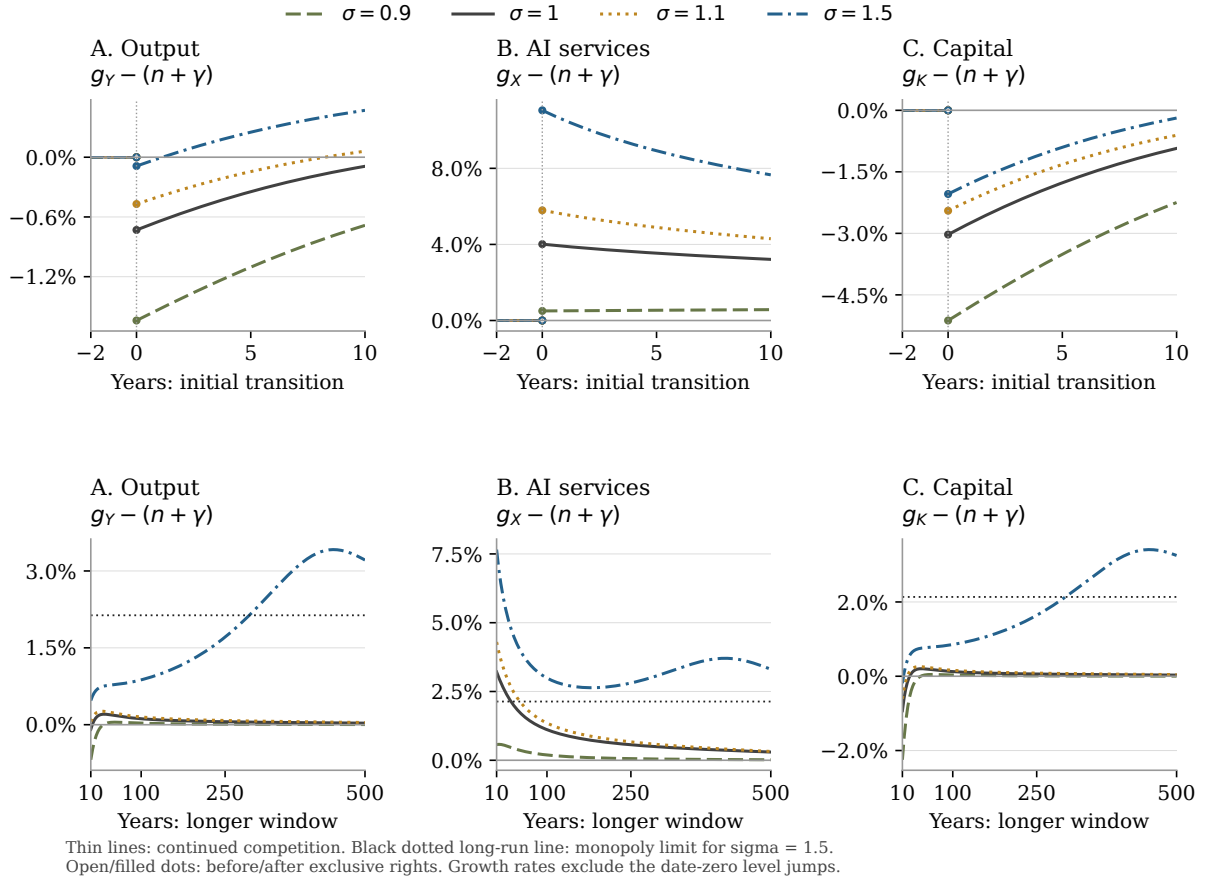


Figure 18: Competition to monopoly, $\chi = 1.5$: annual growth of output, AI services, and capital relative to effective labor. Windows and benchmarks match Figure 15.

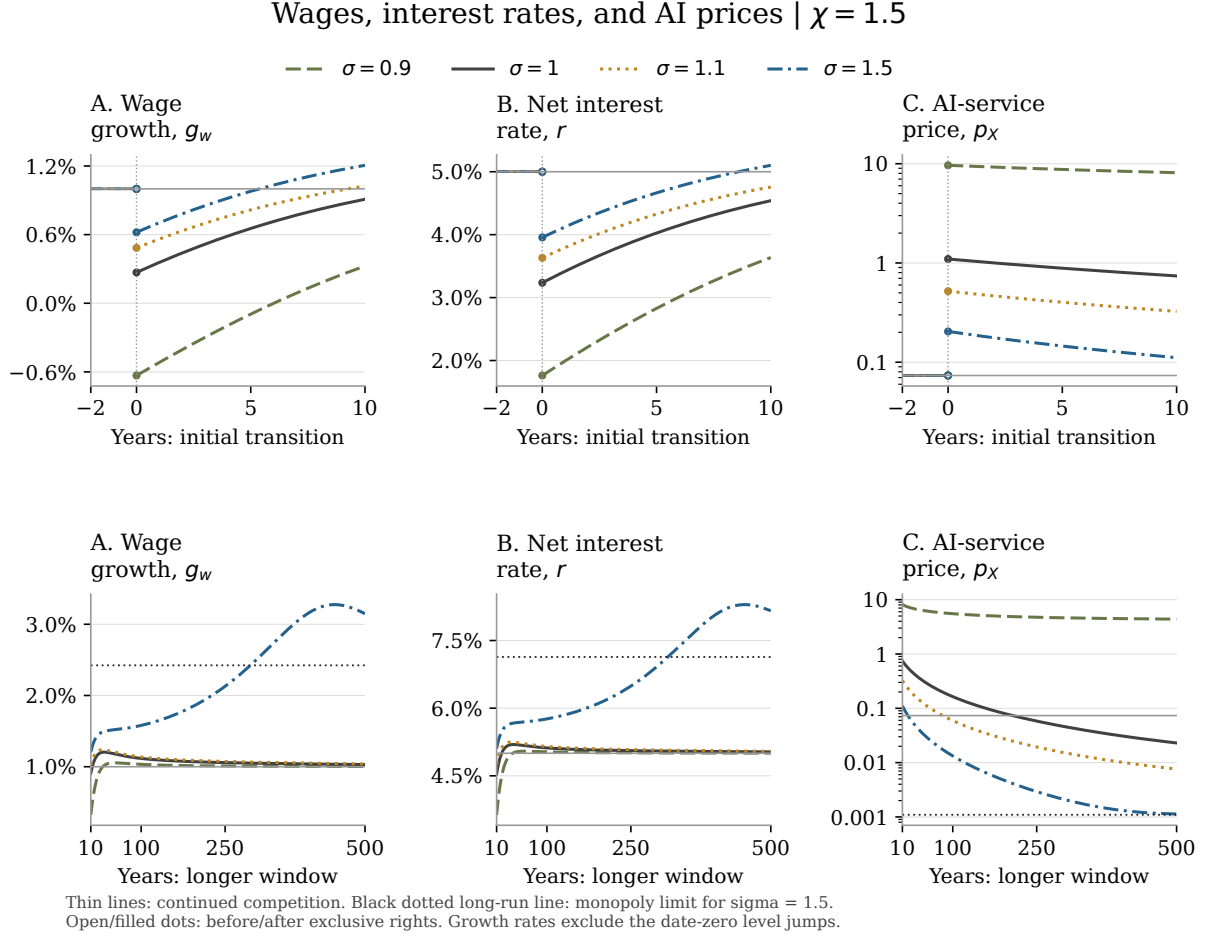
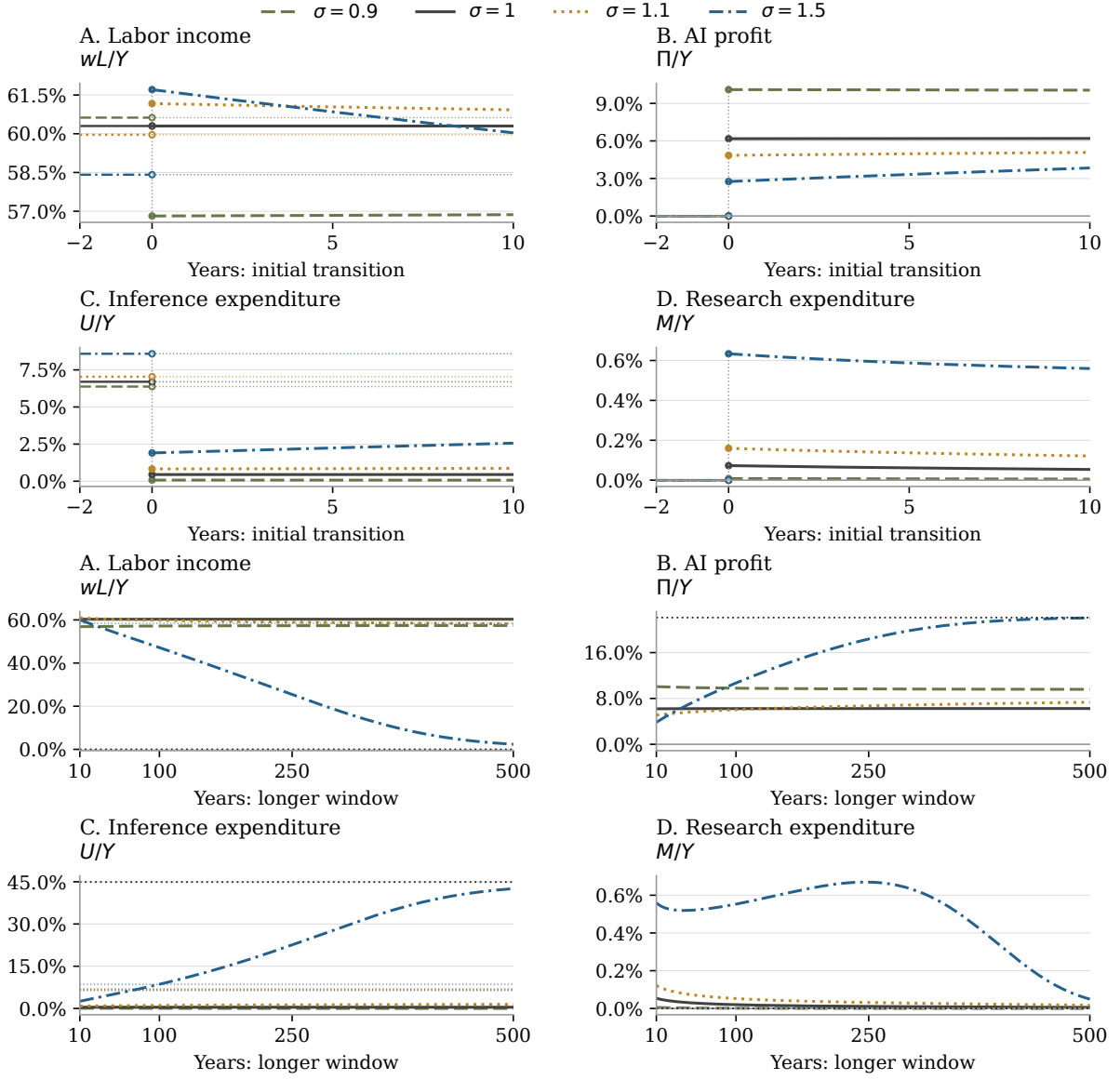


Figure 19: Competition to monopoly, $\chi = 1.5$: annual wage growth, net interest, and AI-service prices in final-good units. Windows, benchmarks, and the logarithmic price scale match Figure 16.

Income and compute expenditure shares | $\chi = 1.5$



Thin lines: continued competition. Black dotted long-run line: monopoly limit for $\sigma = 1.5$.
Open/filled dots: before/after exclusive rights. Growth rates exclude the date-zero level jumps.

Figure 20: Competition to monopoly, $\chi = 1.5$: labor income, developer net profit, inference expenditure, and research expenditure relative to output. Layout and benchmarks match Figure 17.

Replication. The repository’s [competitive-transition guide](#) records the parameters, the institutional experiment, and the replication commands. The driver is `scripts/simulate_competitive_to_monopoly.py`; the renderer is `scripts/report_competitive_to_monopoly.py`. The guide links to the separate results folders for each research productivity, containing the original paths, competitive benchmarks, impact changes, and numerical audits. The main-text $\sigma = 1.5$ compari-

son is rendered by `scripts/report_competitive_main_comparison.py`, which reads these same audited paths without changing them. Figure windows only select dates from those stored paths; they do not change the simulated economy or the numerical solution horizon.

The repository’s [growth-reversal replication guide](#) records the separate input choices, data, audit reports, and commands. The original growth-reversal experiment, initialized at $A_0 = N_0 = 1$, remains archived in separate files; the other simulation exercises are unchanged.

Growth reversal under exclusive rights. For Section 7.5, I first construct a competitive prehistory with $B = B_0$ above the competitive threshold. At the start of this prehistory, I normalize $A = N = 1$ and set $K = 14,047.3$, the limiting monopoly capital–effective-labor ratio at the chosen upper bound. These seed values are illustrative; they are not the stocks at the monopoly grant. I advance the competitive path to year 190, the first whole year at which the three rates in (44) are each within 0.1 percentage point of their limits; all subsequent annual observations through year 1,000 also satisfy this criterion. I then set that date to zero and inherit A , N , and K without changing units. Re-solving the competitive equilibrium from those stocks reproduces the continued prehistory path to within 10^{-9} in log levels over the next 500 years. The figures include its last ten pre-grant years.

I solve the competitive resource constraint and Euler equation in (99), with $M = 0$ and $B = B_0$, using the inherited capital stock and the terminal condition $C/K \rightarrow \rho - n$ from (101). The numerical coordinates are $\log(C/K)$ and $\log(K/(AL)) - [\mathcal{R}^{\text{mc}}(B_0) - \rho - \gamma]t$. Removing the limiting trend avoids extremely large levels without imposing balanced growth on the transition. I use the same `scipy.integrate.solve_bvp` routine described in the numerical appendix. Extending its horizon from 1,200 to 1,800 years changes these coordinates by less than 10^{-9} over the displayed 500 years; independent finite-difference residuals are below 10^{-9} . The static supply condition, positive allocation, limiting growth rates, and household transversality diagnostics also pass the checks.

I solve the two monopoly paths using the existing continuation method, without changing its equilibrium-admission tolerances. Their final solution horizon is about 4,592 years, although the figures display only the first 500 years. Both pass the equation, horizon-extension, transversality, and developer-optimality checks, including denser checks over the first decade.

Declaration of AI use

During the preparation of this work, I used OpenAI Codex to assist with mathematical derivations, consistency checks, manuscript editing, and the preparation and testing of computational code. I take full responsibility for the content of the paper.

References

- Acemoglu, D. and Restrepo, P. (2018). The race between man and machine: Implications of technology for growth, factor shares, and employment. *American Economic Review*, 108(6):1488–1542.
- Acemoglu, D. and Restrepo, P. (2019). Automation and new tasks: How technology displaces and reinstates labor. *Journal of Economic Perspectives*, 33(2):3–30.
- Aghion, P., Jones, B. F., and Jones, C. I. (2019). Artificial intelligence and economic growth. In Agrawal, A., Gans, J., and Goldfarb, A., editors, *The Economics of Artificial Intelligence: An Agenda*, pages 237–282. University of Chicago Press.
- Autor, D. H. (2013). The “task approach” to labor markets: An overview. *Journal for Labour Market Research*, 46(3):185–199.
- Autor, D. H. and Kausik, B. N. (2026). Resolving the automation paradox: Falling labor share, rising wages. arXiv:2601.06343.
- Beaudry, P. and Portier, F. (2014). News-driven business cycles: Insights and challenges. *Journal of Economic Literature*, 52(4):993–1074.
- Berg, A., Buffie, E. F., and Zanna, L.-F. (2018). Should we fear the robot revolution? (The correct answer is yes). *Journal of Monetary Economics*, 97:117–148.
- Bloom, N., Jones, C. I., Van Reenen, J., and Webb, M. (2020). Are ideas getting harder to find? *American Economic Review*, 110(4):1104–1144.
- Brynjolfsson, E., Korinek, A., and Agrawal, A. K. (2025). A research agenda for the economics of transformative AI. NBER Working Paper 34256, National Bureau of Economic Research.
- Caselli, F. and Manning, A. (2019). Robot arithmetic: New technology and wages. *American Economic Review: Insights*, 1(1):1–12.
- Chow, T., Halperin, B., and Mazlish, J. Z. (2026). Transformative AI, existential risk, and real interest rates. Working paper; first posted January 2023.
- Davidson, T., Halperin, B., Houlden, T., and Korinek, A. (2026). When does automating AI research produce explosive growth? Feedback loops in innovation networks. NBER Working Paper 35155, National Bureau of Economic Research.
- Demirer, M., Fradkin, A., Tadelis, N., and Peng, S. (2025). The emerging market for intelligence: Pricing, supply, and demand for LLMs. NBER Working Paper 34608, National Bureau of Economic Research.

- Duffy, J. and Papageorgiou, C. (2000). A cross-country empirical investigation of the aggregate production function specification. *Journal of Economic Growth*, 5(1):87–120.
- Dyatlov, S. (2018). Notes on hyperbolic dynamics.
- Erdil, E. and Besiroglu, T. (2022). Algorithmic progress in computer vision. *arXiv preprint arXiv:2212.05153*.
- Erdil, E., Potlogea, A., Besiroglu, T., Roldan, E., Ho, A., Sevilla, J., Barnett, M., Vrzla, M., and Sandler, R. (2025). GATE: An integrated assessment model for AI automation. arXiv:2503.04941.
- Feldman, W. M. (2021). Stable manifold theorem. MATH 6410: Ordinary Differential Equations, University of Utah. Fall lecture slides.
- Gans, J. S. (2024). Market power in artificial intelligence. NBER Working Paper 32270, National Bureau of Economic Research.
- Groth, C. (2013). Lecture notes in macroeconomics: The Ramsey model. University of Copenhagen. Chapter 10, especially Section 10.8, Appendices A and E.
- Harris, C. R., Millman, K. J., van der Walt, S. J., et al. (2020). Array programming with NumPy. *Nature*, 585(7825):357–362.
- Hémous, D. and Olsen, M. (2022). The rise of the machines: Automation, horizontal innovation, and income inequality. *American Economic Journal: Macroeconomics*, 14(1):179–223.
- Hernandez, D. and Brown, T. B. (2020). Measuring the algorithmic efficiency of neural networks. *arXiv preprint arXiv:2005.04305*.
- Ho, A., Besiroglu, T., Erdil, E., Owen, D., Rahman, R., Guo, Z. C., Atkinson, D., Thompson, N., and Sevilla, J. (2024). Algorithmic progress in language models. Technical Report 2403.05812, arXiv.
- Hunter, J. D. (2007). Matplotlib: A 2D graphics environment. *Computing in Science & Engineering*, 9(3):90–95.
- Jones, C. I. (1995). R&D-based models of economic growth. *Journal of Political Economy*, 103(4):759–784.
- Jones, C. I. (2026). AI and our economic future. *Journal of Economic Perspectives*, 40(3):3–22.
- Jones, C. I. and Tonetti, C. (2026). Past automation and future A.I.: How weak links tame the growth explosion. Working paper, Version 0.5, May 2026.
- Jones, L. E. and Manuelli, R. E. (1990). A convex model of equilibrium growth: Theory and policy implications. *Journal of Political Economy*, 98(5, Part 1):1008–1038.

- Kierzenka, J. and Shampine, L. F. (2001). A BVP solver based on residual control and the MATLAB PSE. *ACM Transactions on Mathematical Software*, 27(3):299–316.
- Korinek, A. and McKelvey, P. (2026). Measuring the AI economy. Staff Working Paper 2026-20, Bank of Canada.
- Korinek, A. and Ng, D. X. (2019). Digitization and the macro-economics of superstars. Working paper, December 2019 version.
- Korinek, A. and Stiglitz, J. E. (2019). Artificial intelligence and its implications for income distribution and unemployment. In Agrawal, A., Gans, J., and Goldfarb, A., editors, *The Economics of Artificial Intelligence: An Agenda*, pages 349–390. University of Chicago Press.
- Korinek, A. and Vipra, J. (2025). Concentrating intelligence: Scaling and market structure in artificial intelligence. *Economic Policy*, 40(121):225–256.
- Kremer, M. (1998). Patent buyouts: A mechanism for encouraging innovation. *The Quarterly Journal of Economics*, 113(4):1137–1167.
- Kydland, F. E. and Prescott, E. C. (1977). Rules rather than discretion: The inconsistency of optimal plans. *Journal of Political Economy*, 85(3):473–491.
- Liu, T. and Wan, R. (2026). AGI, ANI and economic growth. Working paper, SSRN 6776339.
- Moll, B., Rachel, L., and Restrepo, P. (2022). Uneven growth: Automation’s impact on income and wealth inequality. *Econometrica*, 90(6):2645–2683.
- OECD (2025). OECD Global Long-Run Economic Scenarios: 2025 Update. Technical Report 36, OECD Publishing, Paris.
- Ray, D. and Mookherjee, D. (2022). Growth, automation, and the long-run share of labor. *Review of Economic Dynamics*, 46:1–26.
- Rebelo, S. (1991). Long-run policy analysis and long-run growth. *Journal of Political Economy*, 99(3):500–521.
- Restrepo, P. (2025). We won’t be missed: Work and growth in the AGI world. NBER Working Paper 34423, National Bureau of Economic Research.
- Romer, P. M. (1990). Endogenous technological change. *Journal of Political Economy*, 98(5, Part 2):S71–S102.
- Sastry, G., Heim, L., Belfield, H., et al. (2024). Computing power and the governance of artificial intelligence. *arXiv preprint arXiv:2402.08797*.
- Trammell, P. and Korinek, A. (2023). Economic growth under transformative AI. NBER Working Paper 31815, National Bureau of Economic Research. Revised April 2026.

- United Nations (2024). World Population Prospects 2024: Summary of Results. Technical Report UN DESA/POP/2024/TR/NO. 9, United Nations, Department of Economic and Social Affairs, Population Division, New York.
- Virtanen, P., Gommers, R., Oliphant, T. E., et al. (2020). SciPy 1.0: Fundamental algorithms for scientific computing in Python. *Nature Methods*, 17(3):261–272.
- Zeira, J. (1998). Workers, machines, and economic growth. *Quarterly Journal of Economics*, 113(4):1091–1117.